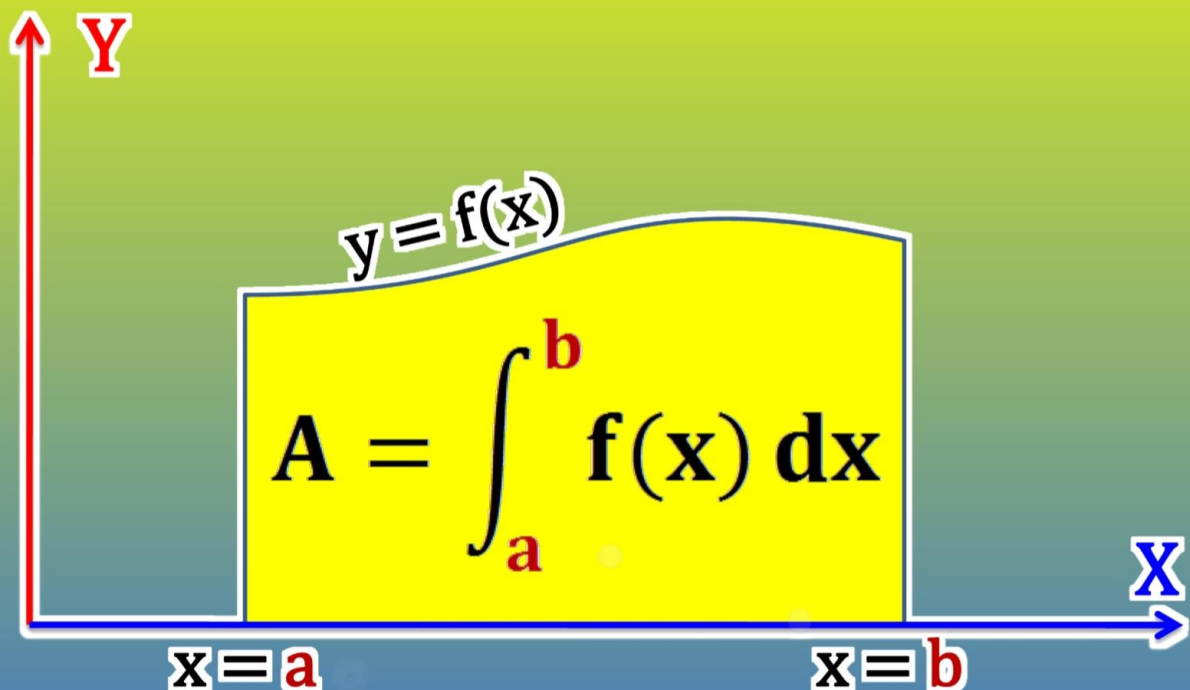


Mathematics-II Suggestion (2024-2025)

PDF

For 2nd Semester of Polytechnic

Top 400 Questions with
Step by Step solution
From All Chapters



Prepared by **NatiTute** YouTube Channel

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1. If $A = \begin{bmatrix} 2 & 0 \\ -1 & 2 \\ 0 & 4 \end{bmatrix}$ then $A^T = ?$

Solution:

Given, $A = \begin{bmatrix} 2 & 0 \\ -1 & 2 \\ 0 & 4 \end{bmatrix}$

$\therefore A^T = \begin{bmatrix} 2 & -1 & 0 \\ 0 & 2 & 4 \end{bmatrix}$ (Ans)

2. For a matrix $A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 0 & 5 \end{pmatrix}$, $A^T = \underline{\hspace{2cm}}$.

Solution:

$A = \begin{pmatrix} 3 & 2 & 1 \\ 2 & 0 & 5 \end{pmatrix}$

$\Rightarrow A^T = \begin{pmatrix} 3 & 2 \\ 2 & 0 \\ 1 & 5 \end{pmatrix}$ (Ans)

3. A square matrix is singular if-

(a) $\det A = 0$

(b) $\det A \neq 0$

(c) $\det A = 1$

(d) $\det A \neq 1$

Answer: (a)

4. Given that matrix A has size 3×4 . Then $A^T B$ and BA^T are possible when B has size-

(a) 4×4

(b) 4×3

(c) 3×4

(d) 3×3

Solution: Given that matrix A is 3×4 .

Let the B matrix be $P \times Q$.

Therefore, A^T is 4×3 .

Since $A^T B$ is defined, so number of columns of A^T must be equal to number of rows of B , therefore, $P = 3$.

Also, BA^T is defined, so the number of columns of B must be equal to number of rows of A^T , then $Q = 4$.

Therefore, matrix B is 3×4 .

Answer: (c)

5. The value of $\begin{vmatrix} 1 & -2 & 3 \\ 2 & -1 & 4 \\ -2 & 3 & 1 \end{vmatrix}$ is-

(a) 0 (b) 13 (c) 19 (d) none of these

Solution:

$\begin{vmatrix} 1 & -2 & 3 \\ 2 & -1 & 4 \\ -2 & 3 & 1 \end{vmatrix}$

$= 1(-1 - 12) + 2(2 + 8) + 3(6 - 2)$

$= -13 + 20 + 12 = 19$

Answer: (c)

6. The value of the determinant $\begin{vmatrix} 1 & \omega & -\omega^2 \\ -\omega & 0 & \omega^3 \\ \omega^2 & -\omega^3 & 0 \end{vmatrix}$ is

_____.

Solution:

$\begin{vmatrix} 1 & \omega & -\omega^2 \\ -\omega & 0 & \omega^3 \\ \omega^2 & -\omega^3 & 0 \end{vmatrix}$

$= \begin{vmatrix} 1 & \omega & -\omega^2 \\ -\omega & 0 & 1 \\ \omega^2 & -1 & 0 \end{vmatrix}$

$= 1(0 + 1) - \omega(0 - \omega^2) - \omega^2(\omega - 0)$

$= 1 + \omega^3 - \omega^3 = 1$ (Ans.)

7. The value of $\begin{vmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 1 & 2 & 3 \end{vmatrix}$ is-

(a) 0

(b) 13

(c) -13

(d) NOTA

Solution:

$\begin{vmatrix} 2 & 3 & -1 \\ 1 & 0 & 2 \\ 1 & 2 & 3 \end{vmatrix}$

$= 2(0 - 4) - 3(3 - 2) - 1(2 - 0)$

$= -8 - 3 - 2$

$= -13$

Answer: (c)

8. The value of the determinant, $\begin{vmatrix} 1 & -1 & 3 \\ 1 & -1 & 3 \\ 1 & 5 & 2 \end{vmatrix}$ is - (a) 3

(b) 0 (c) 6 (d) None of these

Solution: As, two rows of given determinant are identical.

$\therefore \begin{vmatrix} 1 & -1 & 3 \\ 1 & -1 & 3 \\ 1 & 5 & 2 \end{vmatrix} = 0$

Answer: (b)

9. The value of $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & \omega \\ \omega^2 & 1 & 1 \end{vmatrix}$ (where ω is a cube root of unity) is-

(a) 0

(b) 1

(c) 2

(d) None of these

Solution:

$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & \omega \\ \omega^2 & 1 & 1 \end{vmatrix}$

$= 1(\omega^2 - \omega) - \omega(\omega - \omega^3) + \omega^2(\omega - \omega^4)$

$= \omega^2 - \omega - \omega^2 + \omega + 1 - 1 [\omega^3 = 1]$

$= 0$

Answer: (a)

10. The value of $\begin{vmatrix} 1 & \omega^3 & \omega^2 \\ \omega^3 & 1 & \omega \\ \omega^2 & \omega & 1 \end{vmatrix}$ (where ω is a cube root of unity) is- (a) 0 (b) 3 (c) 1 (d) none of these

Solution:

$$\begin{vmatrix} 1 & \omega^3 & \omega^2 \\ \omega^3 & 1 & \omega \\ \omega^2 & \omega & 1 \end{vmatrix} \\ = 1(1 - \omega^2) - \omega^3(\omega^3 - \omega^3) + \omega^2(\omega^4 - \omega^2) \\ = 1 - \omega^2 + \omega^6 - \omega^4 \\ = 1 - \omega^2 + 1 - \omega \\ = 2 - (\omega^2 + \omega) = 2 + 1 = 3$$

Answer: (b)

11. The value of $\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix}$ (where ω is a cube root of unity) is- (a) -1 (b) 1 (c) 0 (d) Ω

Solution:

$$\begin{vmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{vmatrix} \\ = 1(\omega^3 - 1) - \omega(\omega^2 - \omega^2) + \omega^2(\omega - \omega^4) \\ = \omega^3 - 1 + \omega^3 - \omega^6 \\ = 1 - 1 + 1 - 1 = 0$$

Answer: (c)

12. Show that $(a + b + c)$ is a factor of $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix}$.

Solution:

$$\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} \\ = \begin{vmatrix} a+b+c & b & c \\ b+c+a & c & a \\ c+a+b & a & b \end{vmatrix} \\ [\text{doing } C'_1 = C_1 + C_2 + C_3] \\ = (a + b + c) \begin{vmatrix} 1 & b & c \\ 1 & c & a \\ 1 & a & b \end{vmatrix} \quad (\text{Proved})$$

13. Without expanding prove that $\begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} = 0$.

Solution:

$$\text{LHS} = \begin{vmatrix} 1 & a & b+c \\ 1 & b & c+a \\ 1 & c & a+b \end{vmatrix} \\ = \begin{vmatrix} 1 & a & a+b+c \\ 1 & b & b+c+a \\ 1 & c & c+a+b \end{vmatrix} \quad [C'_3 = C_2 + C_3] \\ = (a + b + c) \begin{vmatrix} 1 & a & 1 \\ 1 & b & 1 \\ 1 & c & 1 \end{vmatrix} \\ = (a + b + c) \times 0 \quad [\text{As two columns are identical}] \\ = 0 \quad (\text{Proved})$$

14. In the matrix $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$ minor of (-1) is ____.

Solution: Given matrix is $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ -1 & 2 & 1 \end{bmatrix}$

$\therefore$ The minor of (-1) is $\begin{vmatrix} 1 & 1 \\ 2 & 0 \end{vmatrix} = 0 - 2 = -2$ (Ans.)

15. If $\Delta = \begin{vmatrix} 0 & 1 & 4 \\ -3 & 0 & -1 \\ 5 & -2 & 7 \end{vmatrix}$ then the minor of the element 4 is - (a) 6 (b) 8 (c) 16 (d) none of these

Solution:

Minor of 4 = $\begin{vmatrix} -3 & 0 \\ 5 & -2 \end{vmatrix} = 6 - 0 = 6$

Answer: (a)

16. The cofactor of 6 in the determinant $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$ is ____.

Solution:

Given determinant $\begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$

$\therefore$ The cofactor of 6 is $(-1)^{2+3} \begin{vmatrix} 1 & 2 \\ 7 & 8 \end{vmatrix}$

$= (-1)^5 \times (8 - 14)$

$= (-1) \times (-6) = 6$ (Ans)

17. The co-factor of element 2 of the determinant

$\begin{vmatrix} -1 & 3 & -2 \\ 5 & 4 & 0 \\ 2 & -3 & 5 \end{vmatrix}$ is- (a) 8 (b) -8 (c) -1 (d) none

Solution:

Now $M_{13} = \begin{vmatrix} 5 & 4 \\ 2 & -3 \end{vmatrix} = 0 - (-8) = 8$

Then $C_{13} = (-1)^{1+3} \times M_{13} \\ = 1 \times 8 = 8$

Answer: (a)

18. The co-factor of element '3' of the determinant

$\begin{vmatrix} 1 & -1 & 3 \\ 0 & 2 & 6 \\ 1 & 5 & 2 \end{vmatrix}$ is- (a) 2 (b) -2 (c) 0 (d) 1

Solution: The co-factor of element '3' of the given determinant

$= (-1)^{3+1} \begin{vmatrix} 0 & 2 \\ 1 & 5 \end{vmatrix} = (-1)^4(0 - 2) = -2$

Answer: (b)

19. The co-factor of element '5' of the determinant

$$\begin{vmatrix} 1 & 3 & 4 \\ -1 & 0 & 2 \\ 3 & 5 & 8 \end{vmatrix} \text{ is- (a) 5 (b) 6 (c) 0 (d) -6}$$

Solution: The co-factor of element '5' of the given determinant

$$= (-1)^{2+3} \begin{vmatrix} 1 & 4 \\ -1 & 2 \end{vmatrix} = (-1)^5 (2 + 4) = -6$$

Answer: (d)

20. Is the matrix $\begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 1 \end{bmatrix}$ non-singular? (Yes/No)

Solution:

$$\text{Given matrix is } \begin{bmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 1 \end{bmatrix}$$

$$\text{Now, } \begin{vmatrix} 1 & -1 & 2 \\ 1 & 0 & 3 \\ -2 & -3 & 1 \end{vmatrix}$$

$$= 1(0 + 9) + 1(1 + 6) + 2(-3 - 0)$$

$$= 9 + 7 - 6 = 10 \neq 0$$

∴ The given matrix is non-singular

Answer: Yes

21. What is symmetric matrix?

Answer:

Symmetric Matrix: A symmetric matrix is a square matrix that is equal to its transpose. That means, for a matrix A, it is symmetric if: $A = A^T$

Example:

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}, \Rightarrow A^T = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix} = A$$

22. Determinant value of any skew symmetric matrix of order 2×2 is

(a) strictly positive

(b) strictly negative

(c) 0

(d) none of these

Solution: Let, a 2×2 ordered skew symmetric matrix

$$A = \begin{bmatrix} 0 & a \\ -a & 0 \end{bmatrix}$$

$$\Rightarrow |A| = \begin{vmatrix} 0 & a \\ -a & 0 \end{vmatrix} = 0 + a^2 = a^2$$

Hence, determinant of a 2×2 order skew symmetric matrix is a perfect square i.e strictly positive.

Answer: (a)

23. The matrix $\begin{bmatrix} 0 & 2 & -1 \\ -2 & 0 & 1 \\ 1 & -1 & 0 \end{bmatrix}$ is -

(a) Triangular Matrix

(b) Symmetric Matrix

(c) Skew Symmetric Matrix

(d) no one is correct

Answer: (c)

24. The matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is -

(a) Unit Matrix

(b) Scalar Matrix

(c) Diagonal Matrix

(d) none of these

Answer: (c)

25. Is the matrix $\begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$ orthogonal? (Yes/No)

Solution:

$$\text{Let, } A = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\therefore A^T = \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$$

$$\begin{aligned} \therefore AA^T &= \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & \sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \\ &= \begin{bmatrix} \cos^2 \theta + \sin^2 \theta & 2 \sin \theta \cos \theta \\ 2 \sin \theta \cos \theta & \cos^2 \theta + \sin^2 \theta \end{bmatrix} \\ &= \begin{bmatrix} 1 & 2 \sin \theta \cos \theta \\ 2 \sin \theta \cos \theta & 1 \end{bmatrix} \\ &\neq \text{Identity matrix.} \end{aligned}$$

So, given matrix is not an orthogonal matrix.

Answer: No

26. $I = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ then $I^2 + 2I$ equals to -

(a) I (b) 2I (c) 3I (d) none is true

Solution:

$$I^2 + 2I$$

$$= I + 2I [\because I \text{ is a unit matrix}]$$

$$= 3I$$

Answer: (c)

27. If $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then $AB = ?$

Solution:

∵ A is an unit matrix

$$AB = IB = B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \text{ (Ans)}$$

28. If $A = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ then $AB = ?$

Solution:

$$\begin{aligned} AB &= \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} \\ &= \begin{bmatrix} 1 \times 1 + (-1) \times 3 & 1 \times 2 + (-1) \times 4 \\ 2 \times 1 + 3 \times 3 & 2 \times 2 + 3 \times 4 \end{bmatrix} \\ &= \begin{bmatrix} -2 & -2 \\ 11 & 16 \end{bmatrix} \quad (\text{Ans.}) \end{aligned}$$

29. Matrix $(1 \ 2) \begin{pmatrix} -2 & 5 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix}$ is equal to ____.

Solution:

$$\begin{aligned} &(1 \ 2) \begin{pmatrix} -2 & 5 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ &= (-2 + 6 \ 5 + 4) \begin{pmatrix} 1 \\ 2 \end{pmatrix} \\ &= (4 \ 9) \begin{pmatrix} 1 \\ 2 \end{pmatrix} = 4 + 18 = 22 \quad (\text{Ans}) \end{aligned}$$

30. If $A = \begin{bmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{bmatrix}$ then $\begin{vmatrix} a & b & c \\ b & c & a \\ 1 & 1 & 1 \end{vmatrix} = ?$

(a) 0 (b) A (c) -A (d) none

Solutions:

$$\begin{aligned} A &= \begin{bmatrix} 1 & a & b \\ 1 & b & c \\ 1 & c & a \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ a & b & c \\ b & c & a \end{bmatrix} = - \begin{bmatrix} a & b & c \\ 1 & 1 & 1 \\ b & c & a \end{bmatrix} \\ &= \begin{bmatrix} a & b & c \\ b & c & a \\ 1 & 1 & 1 \end{bmatrix} = A \end{aligned}$$

Answer: (b)

31. If $\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$ then the value of a, b, c and d respectively are-

(a) (1, -2, 3, -4)

(b) (1, 2, 3, 4)

(c) (1, 2, -3, 4)

(d) none is true

Solution:

$$\begin{bmatrix} a-b & 2a+c \\ 2a-b & 3c+d \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ 0 & 13 \end{bmatrix}$$

Comparing both side we get,

$$2a - b = 0 \Rightarrow b = 2a$$

$$\text{again, } a - b = -1, \Rightarrow a - 2a = -1$$

$$\Rightarrow -a = -1, \Rightarrow a = 1$$

$$\therefore b = 2.1 = 2$$

$$\text{again, } 2a + c = 5, \Rightarrow 2.1 + c = 5 \Rightarrow c = 3$$

$$\text{again, } 3c + d = 13, \Rightarrow 3.3 + d = 13$$

$$\Rightarrow 9 + d = 13, \Rightarrow d = 4$$

$\therefore$ The value of a, b, c and d respectively are 1, 2, 3, 4.

Answer: (b)

32. The value of $\begin{vmatrix} 0 & a & -b \\ -a & 0 & -c \\ b & c & 0 \end{vmatrix}$ is -

(a) abc (b) 0 (c) $a + b + c$ (d) none

Explanation: As all elements across the primary diagonal are zero.

Answer: (b)

33. Find the value of a if $\begin{bmatrix} 2 & a \\ 3 & 6 \end{bmatrix}$ is singular matrix.

Solution:

$\therefore \begin{bmatrix} 2 & a \\ 3 & 6 \end{bmatrix}$ is singular matrix.

$$\text{So, } \begin{vmatrix} 2 & a \\ 3 & 6 \end{vmatrix} = 0$$

$$\Rightarrow 2 \times 6 - 3 \times a = 0$$

$$\Rightarrow 12 - 3a = 0$$

$$\Rightarrow 3a = 12, \Rightarrow a = 4 \quad (\text{Ans.})$$

34. If $A = \begin{bmatrix} 2 & m-3 \\ 3 & 6 \end{bmatrix}$ is singular, then the value of m is- (a) 3 (b) -3 (c) 7 (d) -7

Solution: As, A is singular

$$\therefore \begin{vmatrix} 2 & m-3 \\ 3 & 6 \end{vmatrix} = 0$$

$$\Rightarrow (2 \times 6) - (m-3)3 = 0$$

$$\Rightarrow 12 - 3m + 9 = 0$$

$$\Rightarrow 3m = 21, \Rightarrow m = 7$$

Answer: (c)

35. Find the adjoint of the matrix $\begin{bmatrix} 5 & 0 \\ 1 & -2 \end{bmatrix}$.

Solution:

Cofactor of elements 5, 0, 1, -2, in $\begin{bmatrix} 5 & 0 \\ 1 & -2 \end{bmatrix}$ are -2, -1, 0, 5.

$$\begin{aligned} \therefore \text{The requires adjoint is } &\begin{bmatrix} -2 & -1 \\ 0 & 5 \end{bmatrix}^T \\ &= \begin{bmatrix} -2 & 0 \\ -1 & 5 \end{bmatrix} \quad (\text{Ans}) \end{aligned}$$

36. If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, find $A(\text{adj } A)$.

Solution:

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}, \Rightarrow \text{adj } A = \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$\therefore A(\text{adj } A) = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix} \times \begin{bmatrix} 4 & -2 \\ -1 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 12-2 & -6+6 \\ 4-4 & -2+12 \end{bmatrix}$$

$$= \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = 10 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 10I_2 \quad (\text{Ans})$$

37. Find the inverse of the matrix $\begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$.

Solution: Let $A = \begin{bmatrix} 2 & 3 \\ 1 & 5 \end{bmatrix}$

$$\therefore |A| = \begin{vmatrix} 2 & 3 \\ 1 & 5 \end{vmatrix} = 10 - 3 = 7$$

$$\text{And } Adj(A) = \begin{bmatrix} 5 & -3 \\ -1 & 2 \end{bmatrix}$$

$$\text{So, } A^{-1} = \frac{Adj(A)}{|A|} = \frac{\begin{bmatrix} 5 & -3 \\ -1 & 2 \end{bmatrix}}{7} = \frac{1}{7} \begin{bmatrix} 5 & -3 \\ -1 & 2 \end{bmatrix} \text{ (Ans)}$$

38. If $A = \begin{bmatrix} 1 & 2 \\ -1 & 1 \end{bmatrix}$, $A^{-1} = ?$

Solution:

$$A^{-1} = \frac{AdjA}{|A|} = \frac{\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}}{\begin{vmatrix} 1 & 2 \\ -1 & 1 \end{vmatrix}} = \frac{\begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix}}{3} = \frac{1}{3} \begin{bmatrix} 1 & -2 \\ 1 & 1 \end{bmatrix} \text{ (Ans)}$$

39. Evaluate: $\begin{vmatrix} 41 & 42 & 43 \\ 47 & 48 & 49 \\ 44 & 45 & 46 \end{vmatrix}$

Solution:

$$\begin{vmatrix} 41 & 42 & 43 \\ 47 & 48 & 49 \\ 44 & 45 & 46 \end{vmatrix} = \begin{vmatrix} 41 & 1 & 2 \\ 47 & 1 & 2 \\ 44 & 1 & 2 \end{vmatrix} \quad [C'_2 = C_2 - C_1, C'_3 = C_3 - C_1]$$

$$= 0 \quad \text{(Ans.)}$$

40. The value of $\begin{vmatrix} 100 & 101 & 102 \\ 105 & 106 & 107 \\ 110 & 111 & 112 \end{vmatrix}$ is -

(a) 2 (b) 0 (c) -1 (d) none of these

Solution:

$$\begin{vmatrix} 100 & 101 & 102 \\ 105 & 106 & 107 \\ 110 & 111 & 112 \end{vmatrix} = \begin{vmatrix} 100 & 1 & 2 \\ 105 & 1 & 2 \\ 110 & 1 & 2 \end{vmatrix} \quad [C'_2 = C_2 - C_1, C'_3 = C_3 - C_1]$$

$$= 0 \quad [\text{As two columns are identical}]$$

Answer: (b)

41. If $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, then evaluate

$2A - B$.

Solution:

$$2A - B = 2 \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 4 \\ 6 & 8 \end{bmatrix} - \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 3 \\ 2 & 5 \end{bmatrix} \quad \text{(Ans.)}$$

42. If $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$, then show that $A^2 - 4A - I_2 = O$ and hence find A^{-1} .

Solution: Given $A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$

$$\therefore A^2 = A \times A = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} \times \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9+4 & 6+2 \\ 6+2 & 4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix}$$

$$\therefore A^2 - 4A - I_2$$

$$= \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix} - 4 \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 13 & 8 \\ 8 & 5 \end{bmatrix} - \begin{bmatrix} 12 & 8 \\ 8 & 4 \end{bmatrix} - \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \quad \text{(Proved)}$$

$$\text{Now, } A^2 - 4A - I_2 = O$$

$$\Rightarrow A^2 - 4A = I_2$$

$$\Rightarrow I_2 = A(A - 4I_2)$$

$$\Rightarrow \frac{I_2}{A} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \begin{bmatrix} 3 & 2 \\ 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} = \begin{bmatrix} -1 & 2 \\ 2 & -3 \end{bmatrix} \quad \text{(Ans)}$$

43. If $A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$, then show that $A^2 - 4A + 7I_2 = O$ and hence find A^{-1} .

Solution:

$$\text{Given } A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

$$\therefore A^2 = A \times A = \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

$$= \begin{bmatrix} 4-3 & -2-2 \\ 6+6 & -3+4 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix}$$

$$\therefore A^2 - 4A + 7I_2$$

$$= \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} - 4 \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} + 7 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -4 \\ 12 & 1 \end{bmatrix} - \begin{bmatrix} 8 & -4 \\ 12 & 8 \end{bmatrix} + \begin{bmatrix} 7 & 0 \\ 0 & 7 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} = O \quad \text{(Proved)}$$

$$\text{Now, } A^2 - 4A + 7I_2 = O$$

$$\Rightarrow A^2 - 4A = -7I_2$$

$$\Rightarrow 4A - A^2 = 7I_2$$

$$\Rightarrow A \left(\frac{4}{7} I_2 - \frac{1}{7} A \right) = I_2$$

$$\therefore A^{-1} = \frac{4}{7} I_2 - \frac{1}{7} A \quad [\because AA^{-1} = I]$$

$$= \frac{4}{7} \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \frac{1}{7} \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$$

$$= \frac{1}{7} \left\{ \begin{bmatrix} 4 & 0 \\ 0 & 4 \end{bmatrix} - \begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \right\}$$

$$= \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix} \quad \text{(Ans)}$$

44. Show that the matrix $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ satisfies the equation $A^2 - 4A - 5I = O$ and hence find A^{-1} .
Solution:

$$\text{Given, } A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$\therefore A^2 = A \times A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1+4+4 & 2+2+4 & 2+4+2 \\ 2+2+4 & 4+1+4 & 4+2+2 \\ 2+4+2 & 4+2+2 & 4+4+1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix}$$

$$\therefore A^2 - 4A - 5I$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - 4 \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 5 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{bmatrix} - \begin{bmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{bmatrix} - \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = O \text{ (Proved)}$$

$$\text{Now, } A^2 - 4A - 5I = O$$

$$\Rightarrow A^2 - 4A = 5I$$

$$\Rightarrow \frac{A^2}{5} - \frac{4A}{5} = I$$

$$\Rightarrow A \cdot \frac{1}{5} (A - 4I) = I$$

$$\therefore A^{-1} = \frac{1}{5} (A - 4I) [\because AA^{-1} = I]$$

$$= \frac{1}{5} \left\{ \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

$$= \frac{1}{5} \left\{ \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} - \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix} \right\}$$

$$= \frac{1}{5} \begin{bmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{bmatrix} \text{ (Ans)}$$

45. If A is symmetric matrix, then prove that $(A + A^T)$ is a symmetric and $(A - A^T)$ is skew-symmetric.

Solution:

$$(A + A^T)^T = A^T + (A^T)^T = A^T + A = A + A^T$$

So, $A + A^T$ is symmetric matrix. (Proved)

and

$$(A - A^T)^T = A^T - (A^T)^T = A^T - A = -(A - A^T)$$

So, $A - A^T$ is skew-symmetric matrix. (Proved)

46. Find A^{-1} , if $A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 1 & 4 \\ 0 & -1 & 1 \end{bmatrix}$ and show that

$$AA^{-1} = I_3.$$

Solution:

$$\text{Given, } A = \begin{bmatrix} 0 & -1 & 0 \\ 1 & 1 & 4 \\ 0 & -1 & 1 \end{bmatrix}$$

$$\therefore |A| = \begin{vmatrix} 0 & -1 & 0 \\ 1 & 1 & 4 \\ 0 & -1 & 1 \end{vmatrix} = -(-1)(1-0) = 1 \neq 0$$

Elements with respective cofactor are-

Element	0	-1	0	1	1	4	0	-1	1
Cofactor	5	-1	-1	1	0	0	-4	0	1

$$\therefore \text{Adj}A = \begin{bmatrix} 5 & -1 & -1 \\ 1 & 0 & 0 \\ -4 & 0 & 1 \end{bmatrix}^T = \begin{bmatrix} 5 & 1 & -4 \\ -1 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix}$$

$$\therefore A^{-1} = \frac{\text{Adj}A}{|A|} = \begin{bmatrix} 5 & 1 & -4 \\ -1 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \text{ (Ans)}$$

$$\begin{aligned} \text{Now, } AA^{-1} &= \begin{bmatrix} 0 & -1 & 0 \\ 1 & 1 & 4 \\ 0 & -1 & 1 \end{bmatrix} \times \begin{bmatrix} 5 & 1 & -4 \\ -1 & 0 & 0 \\ -1 & 0 & 1 \end{bmatrix} \\ &= \begin{bmatrix} 0+1+0 & 0+0+0 & 0+0+0 \\ 5-1-4 & 1+0+0 & -4+0+4 \\ 0+1-1 & 0+0+0 & 0+0+1 \end{bmatrix} \\ &= \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \text{ (Proved)} \end{aligned}$$

47. Find the inverse of the matrix $\begin{pmatrix} 2 & -3 & 4 \\ 1 & 0 & 1 \\ 0 & -1 & 4 \end{pmatrix}$

Solution:

$$\text{Let, } A = \begin{pmatrix} 2 & -3 & 4 \\ 1 & 0 & 1 \\ 0 & -1 & 4 \end{pmatrix}$$

$$\therefore |A| = \begin{vmatrix} 2 & -3 & 4 \\ 1 & 0 & 1 \\ 0 & -1 & 4 \end{vmatrix}$$

$$= 2(0+1) + 3(4-0) + 4(-1-0)$$

$$= 2 + 12 - 4$$

$$= 10$$

$$\therefore A^T = \begin{pmatrix} 2 & 1 & 0 \\ -3 & 0 & -1 \\ 4 & 1 & 4 \end{pmatrix}$$

$$\therefore \text{Adj}(A) = \begin{pmatrix} 1 & 8 & -3 \\ -4 & 8 & 2 \\ -1 & 2 & 3 \end{pmatrix}$$

$$\therefore A^{-1} = \frac{\text{Adj}(A)}{|A|} = \frac{1}{10} \begin{pmatrix} 1 & 8 & -3 \\ -4 & 8 & 2 \\ -1 & 2 & 3 \end{pmatrix} \text{ (Ans)}$$

48. If the matrix $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$, Show that

$$AA^T = I_3.$$

Solution:

$$\text{Given, } A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$\therefore A^T = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

$$\therefore AA^T = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 1+4+4 & 2+2-4 & -2+4-2 \\ 2+2-4 & 4+1+4 & -4+2+2 \\ -2+4-2 & -4+2+2 & 4+4+1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \frac{1}{9} \times 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3 \text{ (Proved)}$$

49. If $A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$ show that $AA^T = A^T A = I_3$.

Hence find A^{-1} .

Solution:

$$A = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix}$$

$$\Rightarrow A^T = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

$$\therefore AA^T = \frac{1}{3} \begin{bmatrix} -1 & 2 & -2 \\ -2 & 1 & 2 \\ 2 & 2 & 1 \end{bmatrix} \times \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 1+4+4 & 2+2-4 & -2+4-2 \\ 2+2-4 & 4+1+4 & -4+2+2 \\ -2+4-2 & -4+2+2 & 4+4+1 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= \frac{1}{9} \times 9 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I_3$$

Similarly we can prove-

$$A^T A = I_3$$

Since $AA^T = A^T A = I_3$, A is an orthogonal matrix, and for an orthogonal matrix-

$$A^{-1} = A^T = \frac{1}{3} \begin{bmatrix} -1 & -2 & 2 \\ 2 & 1 & 2 \\ -2 & 2 & 1 \end{bmatrix} \text{ (Ans)}$$

50. Prove that $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (a-b)(b-c)(c-a)$.

Solution:

$$\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ a & b-c & c-a \\ a^2 & b^2-c^2 & c^2-a^2 \end{vmatrix} \begin{matrix} C'_2 = C_2 - C_3 \\ C'_3 = C_3 - C_1 \end{matrix}$$

$$= \begin{vmatrix} b-c & c-a \\ b^2-c^2 & c^2-a^2 \end{vmatrix} \text{ [Expanding w.r.t. } R_1]$$

$$= \begin{vmatrix} b-c & c-a \\ (b+c)(b-c) & (c+a)(c-a) \end{vmatrix}$$

$$= (b-c)(c-a) \begin{vmatrix} 1 & 1 \\ (b+c) & (c+a) \end{vmatrix}$$

$$= (b-c)(c-a)(c+a-b-c)$$

$$= (a-b)(b-c)(c-a) \text{ (Proved)}$$

51. Show that $\begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix} = - \begin{vmatrix} \alpha^2 & \beta^2 & \gamma^2 \\ \alpha & \beta & \gamma \\ 1 & 1 & 1 \end{vmatrix}$

Solution:

$$\begin{vmatrix} 1 & \alpha & \alpha^2 \\ 1 & \beta & \beta^2 \\ 1 & \gamma & \gamma^2 \end{vmatrix}$$

[Interchanging rows into columns]

$$= \begin{vmatrix} 1 & 1 & 1 \\ \alpha & \beta & \gamma \\ \alpha^2 & \beta^2 & \gamma^2 \end{vmatrix}$$

[Interchanging R_2 and R_3]

$$= - \begin{vmatrix} 1 & 1 & 1 \\ \alpha^2 & \beta^2 & \gamma^2 \\ \alpha & \beta & \gamma \end{vmatrix}$$

[Interchanging R_1 and R_2]

$$= \begin{vmatrix} \alpha^2 & \beta^2 & \gamma^2 \\ 1 & 1 & 1 \\ \alpha & \beta & \gamma \end{vmatrix}$$

[Interchanging R_2 and R_3]

$$= - \begin{vmatrix} \alpha^2 & \beta^2 & \gamma^2 \\ \alpha & \beta & \gamma \\ 1 & 1 & 1 \end{vmatrix} \text{ (Proved)}$$

52. If A is symmetric matrix then show that $P^T A P$ is symmetric.

Solution:

As A is symmetric, $\therefore A^T = A$

$$\text{Now, } (P^T A P)^T = P^T A^T (P^T)^T$$

$$= P^T A^T P$$

$$= P^T A P \text{ } [\because A^T = A]$$

So $P^T A P$ also symmetric matrix. (Proved)

53. If $x + y + z = 0$, then show that $\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = 0$

Solution:

$$\begin{vmatrix} 1 & 1 & 1 \\ x & y & z \\ x^3 & y^3 & z^3 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 \\ x & x-y & y-z \\ x^3 & x^3-y^3 & y^3-z^3 \end{vmatrix} \begin{matrix} C'_2 = C_1 - C_2 \\ C'_3 = C_2 - C_3 \end{matrix}$$

$$= (x-y)(y^3-z^3) - (y-z)(x^3-y^3)$$

$$= (x-y)(y-z)(y^2+yz+z^2) - (y-z)(x-y)(x^2+xy+y^2)$$

$$= (x-y)(y-z)(y^2+yz+z^2-x^2-xy-y^2)$$

$$= (x-y)(y-z)(yz+z^2-x^2-xy)$$

$$= (x-y)(y-z)\{(z+x)(z-x)+y(z-x)\}$$

$$= (x-y)(y-z)(z-x)(x+y+z)$$

$$= 0 [\because x+y+z=0] \text{ (Proved)}$$

54. Show that

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = abc + ab + bc + ca$$

Solution:

$$\begin{vmatrix} 1+a & 1 & 1 \\ 1 & 1+b & 1 \\ 1 & 1 & 1+c \end{vmatrix} = (1+a)\{(1+b)(1+c)-1\} - (1+c-1) + (1-1-b)$$

$$= (1+a)\{1+c+b+bc-1\} - c - b$$

$$= (1+a)\{c+b+bc\} - c - b$$

$$= c + b + bc + ca + ab + abc - c - b$$

$$= abc + ab + bc + ca \text{ (Proved)}$$

55. Find the value of determinant

$$\begin{vmatrix} 1 & \cos 60^\circ & \cos 30^\circ \\ \frac{1}{2} & \sin 30^\circ & \sin 60^\circ \\ 1 & 0 & 2 \end{vmatrix}$$

Solution:

$$\begin{vmatrix} 1 & \cos 60^\circ & \cos 30^\circ \\ \frac{1}{2} & \sin 30^\circ & \sin 60^\circ \\ 1 & 0 & 2 \end{vmatrix} = \begin{vmatrix} 1 & \frac{1}{2} & \frac{\sqrt{3}}{2} \\ \frac{1}{2} & \frac{1}{2} & \frac{\sqrt{3}}{2} \\ 1 & 0 & 2 \end{vmatrix}$$

$$= 1(1-0) - \frac{1}{2}\left(1 - \frac{\sqrt{3}}{2}\right) + \frac{\sqrt{3}}{2}\left(0 - \frac{1}{2}\right)$$

$$= 1 - \frac{1}{2} + \frac{\sqrt{3}}{4} - \frac{\sqrt{3}}{4} = \frac{1}{2} \text{ (Ans)}$$

56. Express $A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{bmatrix}$ as sum of two matrices such that one of them is symmetric and other is skew-symmetric matrix.

Solution:

$$\text{Given, } A = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{bmatrix} \therefore A^T = \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 3 & 5 & 7 \end{bmatrix}$$

$$\text{Now, } \frac{1}{2}(A + A^T) = \frac{1}{2} \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{bmatrix} + \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 3 & 5 & 7 \end{bmatrix} \right\}$$

$$= \frac{1}{2} \begin{bmatrix} 2 & 5 & 8 \\ 5 & 8 & 11 \\ 8 & 11 & 14 \end{bmatrix} \text{ [symmetric]}$$

Similarly,

$$\frac{1}{2}(A - A^T) = \frac{1}{2} \left\{ \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{bmatrix} - \begin{bmatrix} 1 & 3 & 5 \\ 2 & 4 & 6 \\ 3 & 5 & 7 \end{bmatrix} \right\}$$

$$= \frac{1}{2} \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix} \text{ [skew-symmetric]}$$

$$\therefore A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$\Rightarrow \begin{bmatrix} 1 & 2 & 3 \\ 3 & 4 & 5 \\ 5 & 6 & 7 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 2 & 5 & 8 \\ 5 & 8 & 11 \\ 8 & 11 & 14 \end{bmatrix} + \frac{1}{2} \begin{bmatrix} 0 & -1 & -2 \\ 1 & 0 & -1 \\ 2 & 1 & 0 \end{bmatrix}$$

(Proved)

57. Show that the matrix $A = \begin{bmatrix} 2 & 0 & 2 \\ 4 & 6 & 0 \\ 0 & 2 & 4 \end{bmatrix}$ satisfies the equation $A = \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$.

Solution:

$$A = \begin{bmatrix} 2 & 0 & 2 \\ 4 & 6 & 0 \\ 0 & 2 & 4 \end{bmatrix} \therefore A^T = \begin{bmatrix} 2 & 4 & 0 \\ 0 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

$$A + A^T = \begin{bmatrix} 2 & 0 & 2 \\ 4 & 6 & 0 \\ 0 & 2 & 4 \end{bmatrix} + \begin{bmatrix} 2 & 4 & 0 \\ 0 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 2 \\ 4 & 12 & 2 \\ 2 & 2 & 8 \end{bmatrix}$$

$$\Rightarrow \frac{1}{2}(A + A^T) = \begin{bmatrix} 2 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{bmatrix}$$

$$A - A^T = \begin{bmatrix} 2 & 0 & 2 \\ 4 & 6 & 0 \\ 0 & 2 & 4 \end{bmatrix} - \begin{bmatrix} 2 & 4 & 0 \\ 0 & 6 & 2 \\ 2 & 0 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & -4 & 2 \\ 4 & 0 & -2 \\ -2 & 2 & 0 \end{bmatrix}$$

$$\Rightarrow \frac{1}{2}(A - A^T) = \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$\therefore \frac{1}{2}(A + A^T) + \frac{1}{2}(A - A^T)$$

$$= \begin{bmatrix} 2 & 2 & 1 \\ 2 & 6 & 1 \\ 1 & 1 & 4 \end{bmatrix} + \begin{bmatrix} 0 & -2 & 1 \\ 2 & 0 & -1 \\ -1 & 1 & 0 \end{bmatrix}$$

$$= \begin{bmatrix} 2 & 0 & 2 \\ 4 & 6 & 0 \\ 0 & 2 & 4 \end{bmatrix} = A \text{ (Proved)}$$

58. Evaluate by Chio's Method: $\begin{vmatrix} 1 & 2 & -2 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 0 \end{vmatrix}$

Solution:

$$\begin{vmatrix} 1 & 2 & -2 & 0 \\ 0 & 1 & 0 & 0 \\ -2 & 0 & 1 & 1 \\ 1 & 0 & -1 & 0 \end{vmatrix}$$

$$= \frac{1}{1^{(4-2)}} \begin{vmatrix} \begin{vmatrix} 1 & 2 \\ 0 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 0 & 0 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 0 & 0 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ -2 & 0 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ -2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ -2 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ 1 & 0 \end{vmatrix} & \begin{vmatrix} 1 & -2 \\ 1 & -1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 1 & 0 \end{vmatrix} \end{vmatrix}$$

$$= \begin{vmatrix} 1 & 0 & 0 \\ 4 & -3 & 1 \\ -2 & 1 & 0 \end{vmatrix}$$

$$= \frac{1}{1^{(3-2)}} \begin{vmatrix} \begin{vmatrix} 1 & 0 \\ 4 & -3 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ 4 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 0 \\ -2 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 0 \\ -2 & 0 \end{vmatrix} \end{vmatrix}$$

$$= \begin{vmatrix} -3 & 1 \\ 1 & 0 \end{vmatrix} = 0 - 1 = -1 \quad (\text{Ans})$$

59. Evaluate by Chio's method $\begin{vmatrix} 0 & -1 & 2 & 3 \\ 1 & 2 & -1 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \end{vmatrix}$

Solution:

$$\begin{vmatrix} 0 & -1 & 2 & 3 \\ 1 & 2 & -1 & 1 \\ 1 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \end{vmatrix}$$

$$= - \begin{vmatrix} \begin{vmatrix} 1 & 2 & -1 & 1 \\ 0 & -1 & 2 & 3 \\ 1 & 1 & 2 & 3 \\ 3 & 2 & 1 & 0 \end{vmatrix} \end{vmatrix} \quad [\text{Interchanging 1st and 2nd Row}]$$

$$= -(1)^{4-2} \begin{vmatrix} \begin{vmatrix} 1 & 2 \\ 0 & -1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 0 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 0 & 3 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ 1 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 1 & 2 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 1 & 3 \end{vmatrix} \\ \begin{vmatrix} 1 & 2 \\ 3 & 2 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 3 & 1 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 3 & 0 \end{vmatrix} \end{vmatrix}$$

$$= - \begin{vmatrix} -1 & 2 & 3 \\ -1 & 3 & 2 \\ -4 & 4 & -3 \end{vmatrix}$$

$$= -(-1)^{3-2} \begin{vmatrix} \begin{vmatrix} -1 & 2 \\ -1 & 3 \end{vmatrix} & \begin{vmatrix} -1 & 3 \\ -1 & 2 \end{vmatrix} \\ \begin{vmatrix} -1 & 2 \\ -4 & 4 \end{vmatrix} & \begin{vmatrix} -1 & 3 \\ -4 & -3 \end{vmatrix} \end{vmatrix}$$

$$= \begin{vmatrix} -1 & 1 \\ 4 & 15 \end{vmatrix} = -15 - 4 = -19 \quad (\text{Ans})$$

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60. Evaluate by Chio's method $\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ 4 & 4 & 4 & 1 \end{vmatrix}$

Solution:

$$\begin{vmatrix} 0 & -1 & -1 & 1 \\ 4 & 5 & 1 & 1 \\ 3 & 9 & 4 & 1 \\ 4 & 4 & 4 & 1 \end{vmatrix}$$

$$= - \begin{vmatrix} \begin{vmatrix} -1 & 0 & -1 & 1 \\ 5 & 4 & 1 & 1 \\ 9 & 3 & 4 & 1 \\ 4 & 4 & 4 & 1 \end{vmatrix} \end{vmatrix} \quad [\text{Interchanging 1st and 2nd Column}]$$

$$= \begin{vmatrix} 1 & 0 & 1 & -1 \\ 5 & 4 & 1 & 1 \\ 9 & 3 & 4 & 1 \\ 4 & 4 & 4 & 1 \end{vmatrix}$$

$$= \frac{1}{1^{4-2}} \begin{vmatrix} \begin{vmatrix} 1 & 0 \\ 5 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 5 & 1 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 5 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 0 \\ 9 & 3 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 9 & 4 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 9 & 1 \end{vmatrix} \\ \begin{vmatrix} 1 & 0 \\ 4 & 4 \end{vmatrix} & \begin{vmatrix} 1 & 1 \\ 4 & 4 \end{vmatrix} & \begin{vmatrix} 1 & -1 \\ 4 & 1 \end{vmatrix} \end{vmatrix}$$

$$= \begin{vmatrix} 4 & -4 & 6 \\ 3 & -5 & 10 \\ 4 & 0 & 5 \end{vmatrix}$$

$$= \frac{1}{4^{3-2}} \begin{vmatrix} \begin{vmatrix} 4 & -4 \\ 3 & -5 \end{vmatrix} & \begin{vmatrix} 4 & 6 \\ 3 & 10 \end{vmatrix} \\ \begin{vmatrix} 4 & -4 \\ 4 & 0 \end{vmatrix} & \begin{vmatrix} 4 & 6 \\ 4 & 5 \end{vmatrix} \end{vmatrix}$$

$$= \frac{1}{4} \begin{vmatrix} -8 & 22 \\ 16 & -4 \end{vmatrix}$$

$$= \frac{1}{4} \times (32 - 352) = \frac{1}{4} \times (-320) = -80 \quad (\text{Ans})$$

61. If $I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ then $I^5 + 3I$ equals to-

- (a) $8I$
- (b) $5I$
- (c) $3I$
- (d) $4I$

Explanation:

$$I^5 + 3I$$

$$= I + 3I \quad [\because I^5 = I]$$

$$= 4I$$

Answer: (d)

62. Multiply $\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix}$ and $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.

Solution:

$$\begin{bmatrix} 2 & -1 \\ 3 & 2 \end{bmatrix} \times \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

$$= \begin{bmatrix} 2 \times 1 + (-1) \times 3 & 2 \times 2 + (-1) \times 4 \\ 3 \times 1 + 2 \times 3 & 3 \times 2 + 2 \times 4 \end{bmatrix}$$

$$= \begin{bmatrix} -1 & 0 \\ 9 & 14 \end{bmatrix} \quad (\text{Ans})$$

63. Solve the equations by Cramer's rule:

$$x + y + z = 3$$

$$2x + z = 5$$

$$x + 2y = -1$$

Solution: Given equation can be written as

$$x + y + z = 3$$

$$2x + 0.y + z = 5$$

$$x + 2y + 0.z = -1$$

$$\text{Here, } \Delta = \begin{vmatrix} 1 & 1 & 1 \\ 2 & 0 & 1 \\ 1 & 2 & 0 \end{vmatrix}$$

$$\Rightarrow \Delta = 1(0 - 2) - 1(0 - 1) + 1(4 - 0)$$

$$\Rightarrow \Delta = -2 + 1 + 4 = 3$$

$$\Delta_x = \begin{vmatrix} 3 & 1 & 1 \\ 5 & 0 & 1 \\ -1 & 2 & 0 \end{vmatrix}$$

$$\Rightarrow \Delta_x = 3(0 - 2) - 1(0 + 1) + 1(10 - 0)$$

$$\Rightarrow \Delta_x = -6 - 1 + 10 = 3$$

$$\Delta_y = \begin{vmatrix} 1 & 3 & 1 \\ 2 & 5 & 1 \\ 1 & -1 & 0 \end{vmatrix}$$

$$\Rightarrow \Delta_y = 1(0 + 1) - 3(0 - 1) + 1(-2 - 5)$$

$$\Rightarrow \Delta_y = 1 + 3 - 7 = -3$$

$$\Delta_z = \begin{vmatrix} 1 & 1 & 3 \\ 2 & 0 & 5 \\ 1 & 2 & -1 \end{vmatrix}$$

$$\Rightarrow \Delta_z = 1(0 - 10) - 1(-2 - 5) + 3(4 - 0)$$

$$\Rightarrow \Delta_z = -10 + 7 + 12 = 9$$

∴ The required solutions are:

$$x = \frac{\Delta_x}{\Delta} = \frac{3}{3} = 1 \text{ (Ans)}$$

$$y = \frac{\Delta_y}{\Delta} = \frac{-3}{3} = -1 \text{ (Ans)}$$

$$z = \frac{\Delta_z}{\Delta} = \frac{9}{3} = 3 \text{ (Ans)}$$

64. Solve the equations by Cramer's Rule:

$$2x + 5y + 3z = 5$$

$$3x + y + 2z = 5$$

$$x + 2y - z = 0$$

Solution:

$$2x + 5y + 3z = 5$$

$$3x + y + 2z = 5$$

$$x + 2y - z = 0$$

$$\therefore \Delta = \begin{vmatrix} 2 & 5 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & -1 \end{vmatrix}$$

$$= 2(-1 - 4) - 5(-3 - 2) + 3(6 - 1)$$

$$= -10 + 25 + 15$$

$$= 30$$

$$\Delta_x = \begin{vmatrix} 5 & 5 & 3 \\ 5 & 1 & 2 \\ 0 & 2 & -1 \end{vmatrix}$$

$$= 5(-1 - 4) - 5(-5 - 0) + 3(10 - 0)$$

$$= -25 + 25 + 30$$

$$= 30$$

$$\Delta_y = \begin{vmatrix} 2 & 5 & 3 \\ 3 & 5 & 2 \\ 1 & 0 & -1 \end{vmatrix}$$

$$= 2(-5 - 0) - 5(-3 - 2) + 3(0 - 5)$$

$$= -10 + 25 - 15 = 0$$

$$\Delta_z = \begin{vmatrix} 2 & 5 & 5 \\ 3 & 1 & 5 \\ 1 & 2 & 0 \end{vmatrix}$$

$$= 2(0 - 10) - 5(0 - 5) + 5(6 - 1)$$

$$= -20 + 25 + 25 = 30$$

$$\therefore x = \frac{\Delta_x}{\Delta} = \frac{30}{30} = 1 \text{ (Ans)}$$

$$y = \frac{\Delta_y}{\Delta} = \frac{0}{30} = 0 \text{ (Ans)}$$

$$z = \frac{\Delta_z}{\Delta} = \frac{30}{30} = 1 \text{ (Ans)}$$

65. Solve the following system of linear equations by

Cramer's Rule:

$$2x + y + z = -2$$

$$x + 2y + z = 1$$

$$x - 2y - 3z = 1$$

Solution:

$$2x + y + z = -2$$

$$x + 2y + z = 1$$

$$x - 2y - 3z = 1$$

$$\therefore \Delta = \begin{vmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & -2 & -3 \end{vmatrix}$$

$$= 2(-6 + 2) - 1(-3 - 1) + 1(-2 - 2)$$

$$= -8 + 4 - 4 = -8$$

$$\Delta_x = \begin{vmatrix} -2 & 1 & 1 \\ 1 & 2 & 1 \\ 1 & -2 & -3 \end{vmatrix}$$

$$= -2(-6 + 2) - 1(-3 - 1) + 1(-2 - 2)$$

$$= 8 + 4 - 4 = 8$$

$$\Delta_y = \begin{vmatrix} 2 & -2 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & -3 \end{vmatrix}$$

$$= 2(-3 - 1) + 2(-3 - 1) + 1(1 - 1)$$

$$= -8 - 8 = -16$$

$$\Delta_z = \begin{vmatrix} 2 & 1 & -2 \\ 1 & 2 & 1 \\ 1 & -2 & 1 \end{vmatrix}$$

$$= 2(2 + 2) - 1(1 - 1) - 2(-2 - 2)$$

$$= 8 + 0 + 8 = 16$$

$$\therefore x = \frac{\Delta_x}{\Delta} = \frac{8}{-8} = -1 \text{ (Ans)}$$

$$y = \frac{\Delta_y}{\Delta} = \frac{-16}{-8} = 2 \text{ (Ans)}$$

$$z = \frac{\Delta_z}{\Delta} = \frac{16}{-8} = -2 \text{ (Ans)}$$

66. Solve the equations by Cramer's Rule:

$$x + 2y + 3z = 5$$

$$3x - 2y + z = -1$$

$$4x + 2y + z = 13$$

Solution:

$$x + 2y + 3z = 5$$

$$3x - 2y + z = -1$$

$$4x + 2y + z = 13$$

$$\therefore \Delta = \begin{vmatrix} 1 & 2 & 3 \\ 3 & -2 & 1 \\ 4 & 2 & 1 \end{vmatrix}$$

$$= 1(-2 - 2) - 2(3 - 4) + 3(6 + 8)$$

$$= -4 + 2 + 42 = 40$$

$$\Delta_x = \begin{vmatrix} 5 & 2 & 3 \\ -1 & -2 & 1 \\ 13 & 2 & 1 \end{vmatrix}$$

$$= 5(-2 - 2) - 2(-1 - 13) + 3(-2 + 26)$$

$$= -20 + 28 + 72 = 80$$

$$\Delta_y = \begin{vmatrix} 1 & 5 & 3 \\ 3 & -1 & 1 \\ 4 & 13 & 1 \end{vmatrix}$$

$$= 1(-1 - 13) - 5(3 - 4) + 3(39 + 4)$$

$$= -14 + 5 + 129 = 120$$

$$\Delta_z = \begin{vmatrix} 1 & 2 & 5 \\ 3 & -2 & -1 \\ 4 & 2 & 13 \end{vmatrix}$$

$$= 1(-26 + 2) - 2(39 + 4) + 5(6 + 8)$$

$$= -24 - 86 + 70 = -40$$

$$\therefore x = \frac{\Delta_x}{\Delta} = \frac{80}{40} = 2 \text{ (Ans)}$$

$$y = \frac{\Delta_y}{\Delta} = \frac{120}{40} = 3 \text{ (Ans)}$$

$$z = \frac{\Delta_z}{\Delta} = \frac{-40}{40} = -1 \text{ (Ans)}$$

67. If A and B be two square matrices then which one is correct?

(a) $\det(AB) = \det(A) \cdot \det(B)$

(b) $\det(A+B) = \det(A) + \det(B)$

(c) $\det(A-B) = \det(A) - \det(B)$

(d) None of these

Answer: (a)

68. If A is a symmetric matrix prove that $A \cdot A^T$ is symmetric.

Solution:

$$\text{Let, } B = A \cdot A^T$$

$$\Rightarrow B^T = (A \cdot A^T)^T$$

$$\Rightarrow B^T = A^T \cdot (A^T)^T \text{ by reversal law}$$

$$\Rightarrow B^T = A^T \cdot A = A \cdot A^T \text{ } [\because A \text{ is symmetric}]$$

$$\Rightarrow B^T = B$$

$$\therefore A \cdot A^T \text{ is symmetric matrix (Proved)}$$

69. If A be a skew symmetric matrix then prove that A^2 is symmetric.

Solution:

A be a skew symmetric matrix it means-

$$A^T = -A$$

We need to prove $(A^2)^T = A^2$

$$\text{Now, } (A^2)^T = (A \cdot A)^T$$

$$\Rightarrow (A^2)^T = A^T \cdot A^T \text{ [Applying rule } (AB)^T = B^T \cdot A^T]$$

$$\Rightarrow (A^2)^T = (-A)(-A) \text{ } [\because A^T = -A]$$

$$\Rightarrow (A^2)^T = A^2$$

$$\therefore A^2 \text{ is symmetric (Proved)}$$

70. If $2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 14 & 14 \end{bmatrix}$ then find the value of x and y.

Solution:

$$\text{Given } 2 \begin{bmatrix} x & 5 \\ 7 & y-3 \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 14 & 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x & 10 \\ 14 & 2(y-3) \end{bmatrix} = \begin{bmatrix} 7 & 10 \\ 14 & 14 \end{bmatrix}$$

Comparing both side we get,

$$2x = 7, \Rightarrow x = \frac{7}{2} \text{ (Ans)}$$

$$2(y-3) = 14$$

$$\Rightarrow y-3 = 7$$

$$\Rightarrow y = 3 + 7 = 10 \text{ (Ans)}$$

71. If $2 \begin{bmatrix} x & 2 \\ 7 & y+5 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 15 & 14 \end{bmatrix}$, then the value of (x, y) is- (a) (0, 0) (b) (1, 0) (c) (1, 1) (d) (0, 1)

Solution:

$$\text{Given, } 2 \begin{bmatrix} x & 2 \\ 7 & y+5 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 15 & 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x & 4 \\ 14 & 2y+10 \end{bmatrix} + \begin{bmatrix} 3 & 4 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 15 & 14 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 2x+3 & 8 \\ 15 & 2y+12 \end{bmatrix} = \begin{bmatrix} 3 & 8 \\ 15 & 14 \end{bmatrix}$$

$$\therefore 2x + 3 = 3, \Rightarrow x = 0$$

$$\text{And, } 2y + 12 = 14, \Rightarrow 2y = 2, \Rightarrow y = 1$$

$$\therefore (x, y) = 0, 1$$

Answer: (d)

72. Find x and y such that $\begin{pmatrix} 3 & x+1 \\ y+2 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix}$.

Solution:

$$\text{Given, } \begin{pmatrix} 3 & x+1 \\ y+2 & 5 \end{pmatrix} = \begin{pmatrix} 3 & 2 \\ 7 & 5 \end{pmatrix}$$

$$\therefore x + 1 = 2, \Rightarrow x = 2 - 1 = 1 \text{ (Ans)}$$

$$y + 2 = 7, \Rightarrow y = 7 - 2 = 5 \text{ (Ans)}$$

{Prepared by [NatiTute](#) YouTube Channel}

1. The point $(2, \sqrt{3} - 2)$ lies in –

- (a) 1st quadrant
- (b) 2nd quadrant
- (c) 3rd quadrant
- (d) 4th quadrant

Explanation: In the point $(2, \sqrt{3} - 2)$

Abscissa = 2 i.e. (+ve)

Ordinate = $\sqrt{3} - 2$ i.e. (-ve)

∴ (Abscissa, Ordinate) → (+, -)

∴ The point lies in 4th quadrant.

Answer: (d)

2. The Cartesian co-ordinate of a point are $(-\sqrt{3}, 1)$, its polar co-ordinates are- (a) $(2, \frac{\pi}{6})$, (b) $(2, -\frac{\pi}{6})$, (c)

$(2, \frac{5\pi}{6})$, (d) $(2, -\frac{5\pi}{6})$.

Solution:

Given $(x, y) \rightarrow (-\sqrt{3}, 1)$

$$\therefore r = \sqrt{(-\sqrt{3})^2 + 1^2} = \sqrt{4} = 2$$

$$\text{and } \theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$= \tan^{-1} \left(\frac{1}{-\sqrt{3}} \right)$$

$$= \pi - \frac{\pi}{6} \quad [\because \theta \text{ lies in 2nd quadrant}]$$

$$= \frac{5\pi}{6}$$

$$\therefore \text{Polar co-ordinates } (r, \theta) = \left(2, \frac{5\pi}{6} \right)$$

Answer: (c)

3. The polar co-ordinates of the point whose Cartesian co-ordinates are $(-1, 1)$ are-

- (a) $(\sqrt{2}, \frac{3\pi}{4})$
- (b) $(\sqrt{2}, -\frac{3\pi}{4})$
- (c) $(\sqrt{2}, \frac{\pi}{4})$
- (d) none of these

Solution: Given $(x, y) \rightarrow (-1, 1)$

$$\therefore r = \sqrt{(-1)^2 + 1^2} = \sqrt{2}$$

$$\text{and } \theta = \tan^{-1} \left(\frac{y}{x} \right) = \tan^{-1} \left(\frac{1}{-1} \right) = \tan^{-1}(-1)$$

$$\Rightarrow \theta = \pi - \frac{\pi}{4} \quad [\because \theta \text{ lies in 2nd quadrant}]$$

$$= \frac{3\pi}{4}$$

$$\therefore \text{The required polar coordinate is } \left(\sqrt{2}, \frac{3\pi}{4} \right)$$

Answer: (a)

4. The Cartesian coordinates of the point whose polar coordinate are $(3, \frac{\pi}{6})$ are-

(a) $(\frac{3}{2}, \frac{3}{2})$

(b) $(\frac{3\sqrt{3}}{2}, \frac{3}{2})$

(c) $(\frac{\sqrt{3}}{2}, \frac{3}{2})$

(d) none of these

Explanation: Given $(r, \theta) \rightarrow (3, \frac{\pi}{6})$

$$\therefore x = r \cos \theta = 3 \cos \frac{\pi}{6} = 3 \times \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

$$\text{and } y = r \sin \theta = 3 \sin \frac{\pi}{6} = 3 \times \frac{1}{2} = \frac{3}{2}$$

$$\therefore \text{The required Cartesian coordinate is } \left(\frac{3\sqrt{3}}{2}, \frac{3}{2} \right)$$

Answer: (b)

5. The area of a triangle with vertices at $(-4, 1)$, $(1, 2)$ and $(4, -3)$ is-

(a) 14 sq unit

(b) 15 sq unit

(c) 16 sq unit

(d) none of these

Solution: Area of triangle

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |(-4)(2 - 3) + 1(-3 - 1) + 4(1 - 2)|$$

$$= \frac{1}{2} |-20 - 4 - 4| = \frac{1}{2} |-28|$$

$$= \frac{1}{2} \times 28 = 14 \text{ sq unit}$$

Answer: (a)

6. The coordinates of centroid of a triangle having vertices $(-1, 2)$, $(-1, -1)$ and $(2, -1)$ is-

(a) $(0, 0)$

(b) $(1, 1)$

(c) $(2, -1)$

(d) $(3, 3)$

Solution: The coordinate of centroid of a triangle is

$$\left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3} \right).$$

$$= \left(\frac{-1 - 1 + 2}{3}, \frac{2 - 1 - 1}{3} \right) = (0, 0)$$

Answer: (a)

7. The X and Y intercept of the straight line

$3x + 4y = 12$ are-

- (a) 3, 3
- (b) 4, 5
- (c) 4, 3
- (d) none of these

Solution: Given equation is $3x + 4y = 12$

$$\Rightarrow \frac{3x}{12} + \frac{4y}{12} = 1$$

$$\Rightarrow \frac{x}{4} + \frac{y}{3} = 1$$

∴ X intercept = 4 and Y intercept = 3

Answer: (c)

8. The area bounded by the curves $4x + 3y = 12$, $y = 0$, $x = 0$, is _____.

Solution:

Given equation: $4x + 3y = 12$

$$\Rightarrow \frac{4x}{12} + \frac{3y}{12} = 1, \Rightarrow \frac{x}{3} + \frac{y}{4} = 1$$

$$\begin{aligned} \therefore \text{Area} &= \frac{1}{2} \times (\text{X intercept}) \times (\text{Y intercept}) \\ &= \frac{1}{2} \times 3 \times 4 = 6 \text{ Sq. unit (Ans)} \end{aligned}$$

9. The area of the triangle formed by the coordinate axes and the straight line $2x + 3y = 6$ is-

- (a) 1 sq unit
- (b) 9 sq unit
- (c) 6 sq unit
- (d) 3 sq unit

Solution:

Given equation is $2x + 3y = 6$

$$\Rightarrow \frac{2x}{6} + \frac{3y}{6} = 1$$

$$\Rightarrow \frac{x}{3} + \frac{y}{2} = 1$$

∴ X intercept = 3 and Y intercept = 2

$$\therefore \text{Area of the triangle} = \frac{1}{2} \times 3 \times 2 = 3 \text{ sq unit}$$

Answer: (d)

10. Length of the perpendicular from (1, 2) to the straight line $3x + 4y = 1$ is-

- (a) 1
- (b) 5
- (c) 2
- (d) none of these

Explanation: Given point (1, 2)

Straight line $3x + 4y = 1, \Rightarrow 3x + 4y - 1 = 0$

∴ Length of perpendicular from given point,

$$p = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}} = \frac{|3 \times 1 + 4 \times 2 - 1|}{\sqrt{3^2 + 4^2}} = \frac{10}{5} = 2$$

Answer: (c)

11. The distance between the lines $3x + 4y = 9$ and $6x + 8y = 15$ is-

- (a) $\frac{3}{2}$
- (b) $\frac{3}{10}$
- (c) 6
- (d) none of these

Solution: Two parallel lines are

$$ax + by = c_1 \text{ and } ax + by = c_2$$

then distance between the lines = $\frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$

Given equations $3x + 4y = 9$

$$\text{and } 6x + 8y = 15, \Rightarrow 3x + 4y = \frac{15}{2}$$

$$\therefore \text{Distance } d = \frac{\left|9 - \frac{15}{2}\right|}{\sqrt{3^2 + 4^2}} = \frac{\frac{3}{2}}{5} = \frac{3}{10}$$

Answer: (b)

12. Check whether the points (-3, 4), (0, -1) and (1, 2) are the co-linear.

Solution: If three points are collinear, the area of the triangle they form is zero.

$$\therefore \frac{1}{2} |-3(-1 - 2) + 0 + 1(4 + 1)|$$

$$= \frac{1}{2} |9 + 5| = \frac{1}{2} \times 14 = 7 \neq 0$$

Since the area is not zero, the points are not collinear.

13. Check the co-linearity of the points (2, 5), (0, 0) and (-1, -2.5).

Solution:

Area of the triangle formed by the given three points (2, 5), (0, 0) and (-1, -2.5)

$$= \frac{1}{2} [2(0 + 2.5) + 0(-2.5 - 5) - 1(5 - 0)]$$

$$= \frac{1}{2} [5 + 0 - 5] = \frac{1}{2} \times 0 = 0$$

Hence, the given three points are collinear. (Ans)

14. The value of p for which the points (1, 2), (3, 6), (p , 4) are collinear is- (a) 2 (b) 3 (c) 1 (d) -1

Solution:

As, given points are collinear.

$$\therefore 1(6 - 4) + 3(4 - 2) + p(2 - 6) = 0$$

$$\Rightarrow 2 + 6 - 4p = 0$$

$$\Rightarrow 4p = 8, \Rightarrow p = 2$$

Answer: (a)

15. Find slope of the line be $2x - 3y = 3$.

Solution: Given equation of line $2x - 3y = 3$

$$\Rightarrow -3y = -2x + 3, \Rightarrow y = \frac{2}{3}x - 1$$

Comparing with $y = mx + c$, we see that:

$$\text{Slope} = m = \frac{2}{3} \quad (\text{Ans})$$

16. Transfer the Cartesian form of the equation $x^2 + y^2 = 2ax$ to its polar form.

Solution: We are given $x^2 + y^2 = 2ax$

The relationships between Cartesian and polar coordinates: $x = r \cos \theta$, $y = r \sin \theta$

Substituting these in the given equation:

$$(r \cos \theta)^2 + (r \sin \theta)^2 = 2a \cdot r \cos \theta$$

$$\Rightarrow r^2 \cos^2 \theta + r^2 \sin^2 \theta = 2ar \cos \theta$$

$$\Rightarrow r^2(\cos^2 \theta + \sin^2 \theta) = 2ar \cos \theta$$

$$\Rightarrow r^2 = 2ar \cos \theta$$

$$\Rightarrow r^2 - 2ar \cos \theta = 0$$

$$\Rightarrow r(r - 2a \cos \theta) = 0$$

[As r can't be 0]

$$\Rightarrow r - 2a \cos \theta = 0$$

$$\Rightarrow r = 2a \cos \theta$$

This is the polar form of given equation.

17. Find the angle between the lines $x + 2y = 5$ and $y = 2x$.

Solution:

$$\text{Now, } x + 2y = 5$$

$$\Rightarrow 2y = -x + 5$$

$$\Rightarrow y = \left(-\frac{1}{2}\right)x + \frac{5}{2}$$

$$\therefore \text{Slope of line } m_1 = -\frac{1}{2}$$

$$\text{Again, } y = 2x$$

$$\therefore \text{Slope of line } m_2 = 2$$

$$\therefore m_1 \times m_2 = -\frac{1}{2} \times 2 = -1$$

$\therefore$ The angle between the line is 90° (Ans)

18. Find the equation of the straight line making 60° angle with positive direction of X-axis and passing through the point of the points (7, 8).

Solution: According to the given condition-

$$\text{Slope of straight line } m = \tan 60^\circ = \sqrt{3}$$

$$\text{A given point on the line } (x_1, y_1) = (7, 8)$$

The equation of a straight line in point-slope form is: $y - y_1 = m(x - x_1)$

Applying the given conditions we get,

$$y - 8 = \sqrt{3}(x - 7)$$

$$\Rightarrow \sqrt{3}x - 7\sqrt{3} = y - 8$$

$$\Rightarrow \sqrt{3}x - y = 7\sqrt{3} - 8 \quad (\text{Ans})$$

19. The value of k for which the line $2x + 3y + 4 + k(6x - y + 11) = 0$ is parallel to the Y axis is - (a) $1/3$ (b) $-1/3$ (c) 3 (d) none of these

Solution:

$$\text{Given, } 2x + 3y + 4 + k(6x - y + 11) = 0$$

$$\Rightarrow 2x + 3y + 4 + 6kx - ky + 11k = 0$$

$$\Rightarrow (2 + 6k)x + (3 - k)y + (4 + 11k) = 0$$

As the line is parallel to Y axis therefore coefficient of y must be zero.

$$\Rightarrow 3 - k = 0, \Rightarrow k = 3$$

Answer: (c)

20. Find the St. Line passes through the point (1, 0) and parallel to the St. Line $2x + 3y = 6$.

Solution: Given equation

$$2x + 3y = 6$$

$$\Rightarrow 3y = -2x + 6$$

$$\Rightarrow y = -\frac{2}{3}x + 2$$

Comparing with the standard form

$$y = mx + c$$

We get, the slope of the given line $m = -\frac{2}{3}$

The point-slope form of a line is

$$y - y_1 = m(x - x_1)$$

Here $m = -\frac{2}{3}$ and the point $(x_1, y_1) = (1, 0)$

$$\therefore y - 0 = -\frac{2}{3}(x - 1)$$

$$\Rightarrow 3y = -2x + 2$$

$$\Rightarrow 2x + 3y = 2$$

$\therefore$ The equation of required line is $2x + 3y = 2$

21. If the lines joining the points (a, -2) and (4, -a) be parallel to the line joining (2, 1) and (3, 4), find the value of a.

Solution:

Slope of the line joining (a, -2) and (4, -a)

$$m_1 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{-a + 2}{4 - a}$$

Slope of the lines joining (2, 1) and (3, 4)

$$m_2 = \frac{y_2 - y_1}{x_2 - x_1} = \frac{4 - 1}{3 - 2} = \frac{3}{1} = 3$$

Since the lines are parallel, their slopes must be equal: $m_1 = m_2$

$$\Rightarrow \frac{-a+2}{4-a} = 3$$

$$\Rightarrow -a + 2 = 12 - 3a$$

$$\Rightarrow 3a - a = 12 - 2$$

$$\Rightarrow 2a = 10, \Rightarrow a = 5 \quad (\text{Ans})$$

22. Prove that the triangle formed by the lines

$x + y = 0$, $3x + y = 4$, $x + 3y = 4$ is an isosceles triangle.

Solution: Given lines are-

$$x + y = 0 \dots (i)$$

$$3x + y = 4 \dots (ii)$$

$$x + 3y = 4 \dots (iii)$$

Solving (i) and (ii) we get,

$$x + y = 0, \Rightarrow y = -x$$

$$3x + y = 4, \Rightarrow 3x - x = 4, \Rightarrow 2x = 4, \Rightarrow x = 2$$

$$\therefore y = -2$$

So, intersection of (i) and (ii) is $(2, -2)$

Similarly, we can find out-

Intersection of (ii) and (iii) is $(-2, 2)$

Intersection of (iii) and (i) is $(1, 1)$

Therefore,

Length of side between $(2, -2)$ and $(-2, 2)$

$$= \sqrt{(2+2)^2 + (-2-2)^2} = \sqrt{16+16} = \sqrt{32}$$

Length of side between $(-2, 2)$ and $(1, 1)$

$$= \sqrt{(-2-1)^2 + (2-1)^2} = \sqrt{9+1} = \sqrt{10}$$

Length of side between $(1, 1)$ and $(2, -2)$

$$= \sqrt{(1-2)^2 + (1+2)^2} = \sqrt{1+9} = \sqrt{10}$$

Since two sides have the same length ($\sqrt{10}$) the triangle is isosceles.

23. A straight line passes through the point $(7, 3)$ and has intercepts on axes in equal magnitude but in opposite sign. Find its equation. Also find the co-ordinates of the point on the line at which the abscissa is triple the ordinate.

Solution: Let the intercepts of the line on the axes be a on the x-axis and $(-a)$ on the y-axis. The equation of a line in intercept form is:

$$\frac{x}{a} + \frac{y}{-a} = 1$$

$$\Rightarrow x - y = a$$

$$\Rightarrow x - y = a$$

Since the line passes through $(7, 3)$

$$\Rightarrow 7 - 3 = a, \Rightarrow a = 4$$

Thus, the equation of the line is-

$$x - y = 4 \text{ (Ans)}$$

If the co-ordinates of the point on the line at which the abscissa is triple the ordinate then we can say-

$$x = 3y$$

Substituting this in the equation $x - y = 4$

$$\Rightarrow 3y - y = 4$$

$$\Rightarrow 2y = 4, \Rightarrow y = 2$$

$$\therefore x = 3 \times 2 = 6$$

So, the required point is $(6, 2)$ (Ans)

24. If p_1 and p_2 be the length of perpendiculars from the origin upon the lines $3x - 4y = 5 \sin \alpha$ and $4x + 3y = 5 \cos \alpha$, prove that $p_1^2 + p_2^2 = 1$.

Solution:

Length of perpendicular from the origin upon the line $3x - 4y = 5 \sin \alpha$ is-

$$p_1 = \frac{5 \sin \alpha}{\sqrt{3^2 + (-4)^2}} = \frac{5 \sin \alpha}{\sqrt{9+16}} = \frac{5 \sin \alpha}{5} = \sin \alpha$$

Length of perpendicular from the origin upon the line $4x + 3y = 5 \cos \alpha$ is-

$$p_2 = \frac{5 \cos \alpha}{\sqrt{4^2 + 3^2}} = \frac{5 \cos \alpha}{\sqrt{16+9}} = \frac{5 \cos \alpha}{5} = \cos \alpha$$

$$\text{Now, } p_1^2 + p_2^2 = \sin^2 \alpha + \cos^2 \alpha = 1 \text{ (Proved)}$$

25. A straight line forms a right angle triangle with the coordinate axis. If the area of triangle is 6 and the hypotenuse is 5, find the equation of the straight line.

Solution:

Hypotenuse of triangle is 5.

Let, other two sides of triangle is a and b .

$$\therefore \text{Area of triangle is } = \frac{1}{2}ab = 6 \text{ (given)}$$

$$\Rightarrow ab = 12$$

$$\text{and, } a^2 + b^2 = 5^2$$

$$\Rightarrow a^2 + b^2 = 25$$

$$\Rightarrow (a+b)^2 - 2ab = 25$$

$$\Rightarrow (a+b)^2 - 2 \times 12 = 25$$

$$\Rightarrow (a+b)^2 - 24 = 25$$

$$\Rightarrow (a+b)^2 = 49$$

$$\Rightarrow a+b = 7$$

($\because$ Side of triangle may not be negative)

$$\Rightarrow a + \frac{12}{a} = 7 \text{ [}\because ab = 12\text{]}$$

$$\Rightarrow a^2 + 12 = 7a$$

$$\Rightarrow a^2 - 7a + 12 = 0$$

$$\Rightarrow a^2 - 3a - 4a + 12 = 0$$

$$\Rightarrow a(a-3) - 4(a-3) = 0$$

$$\Rightarrow (a-3)(a-4) = 0$$

$$\therefore a = 3 \text{ or } 4$$

So, $b = 3$ when $a = 4$

$\therefore$ The required equation of straight line is-

$$\frac{x}{4} + \frac{y}{3} = 1 \text{ or } 3x + 4y = 12 \text{ (Ans)}$$

26. Write down the centre and radius of the circle

$$x^2 + y^2 - 3x + 4y + 4 = 0.$$

Solution:

$$\text{Given equation } x^2 + y^2 - 3x + 4y + 4 = 0$$

$$\Rightarrow x^2 + y^2 + 2\left(-\frac{3}{2}\right)x + 2.2y + 4 = 0$$

Comparing with general equation,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{we get, } g = -\frac{3}{2}, f = 2 \text{ and } c = 4$$

$$\therefore \text{Centre} = (-g, -f) = \left(\frac{3}{2}, 2\right) \text{ (Ans)}$$

$$\text{Radius} = \sqrt{g^2 + f^2 - c}$$

$$= \sqrt{\left(-\frac{3}{2}\right)^2 + 2^2 - 4}$$

$$= \sqrt{\left(\frac{3}{2}\right)^2} = \frac{3}{2} \text{ (Ans)}$$

27. If the length of the diameter of the circle

$$x^2 + y^2 + 4x - 6y - k = 0 \text{ is 4 then the value of } k \text{ is } \underline{\hspace{2cm}}.$$

Solution:

$$x^2 + y^2 + 4x - 6y - k = 0$$

$$\Rightarrow x^2 + y^2 + 2.2x + 2(-3)y + (-k) = 0$$

$$\text{Now, } 2\sqrt{g^2 + f^2 - c} = \text{Diameter}$$

$$\Rightarrow 2\sqrt{2^2 + (-3)^2 - (-k)} = 4$$

$$\Rightarrow \sqrt{4 + 9 + k} = 2$$

$$\Rightarrow 13 + k = 2^2 = 4$$

$$\Rightarrow k = -9 \text{ (Ans)}$$

28. Coordinates of centre of the circle

$$2x^2 + 2y^2 - 8x - 5 = 0 \text{ are-}$$

$$\text{(a) (0, 2) (b) (2, 0) (c) (4, 0) (-4, 0)}$$

Solution:

$$\text{Given equation } 2x^2 + 2y^2 - 8x - 5 = 0$$

$$\Rightarrow x^2 + y^2 - 4x - \frac{5}{2} = 0$$

$$\Rightarrow x^2 + y^2 + 2.(-2)x + 2.0.y - \frac{5}{2} = 0$$

Comparing with general equation,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{we get, } g = -2, f = 0$$

$$\therefore \text{Centre} = (-g, -f) = (2, 0)$$

Answer: (b)

29. The centre of the circle

$$ax^2 + (2a - 3)y^2 - 4x - 1 = 0 \text{ is -}$$

$$\text{(a) } \left(\frac{2}{3}, 0\right)$$

$$\text{(b) } \left(-\frac{2}{3}, 0\right)$$

$$\text{(c) } (2, 0)$$

$$\text{(d) none of these}$$

Solution: Given equation of circle is

$$ax^2 + (2a - 3)y^2 - 4x - 1 = 0$$

$$\text{From circle } 2a - 3 = a$$

$$\Rightarrow 2a - a = 3, \Rightarrow a = 3$$

$\therefore$ The equation of the circle is

$$3x^2 + 3y^2 - 4x - 1 = 0$$

$$\Rightarrow x^2 + y^2 - \frac{4}{3}x - \frac{1}{3} = 0$$

$$\Rightarrow x^2 + y^2 + 2\left(-\frac{2}{3}\right)x + 2 \times 0 \times y + \left(-\frac{1}{3}\right) = 0$$

Comparing with general equation,

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

$$\text{we get, } g = -\frac{2}{3} \text{ and } f = 0$$

$$\text{Thus centre, } (-g, -f) = \left(\frac{2}{3}, 0\right)$$

Answer: (a)

30. A circle has radius 4 unit and its centre lies on $x + y = 0$ and it passes through (1, 1). Find its equation.

Solution: Let (a, b) be the center of the circle.

Then by the problem, $a + b = 0, \Rightarrow b = -a$

$\therefore$ Center of the circle becomes $(a, -a)$

Given radius of the circle is 4.

$\therefore$ The equation of the circle is

$$(x - a)^2 + (y + a)^2 = 4^2$$

$$\Rightarrow (x - a)^2 + (y + a)^2 = 16$$

$\therefore$ It passes through (1, 1)

$$\Rightarrow (1 - a)^2 + (1 + a)^2 = 16$$

$$\Rightarrow 2(1^2 + a^2) = 16$$

$$\Rightarrow 1 + a^2 = 8$$

$$\Rightarrow a^2 = 7, \Rightarrow a = \pm\sqrt{7}$$

$\therefore$ The required equation is

$$\Rightarrow (x \mp \sqrt{7})^2 + (y \pm \sqrt{7})^2 = 16$$

$$\Rightarrow x^2 \mp 2\sqrt{7}x + 7 + y^2 \pm 2\sqrt{7}y + 7 = 16$$

$$\Rightarrow x^2 + y^2 \mp 2\sqrt{7}x \pm 2\sqrt{7}y - 2 = 0 \text{ (Ans)}$$

31. Find the equation of the circle passing through (4, 5) having the centre at (2, 2).

Solution:

Let the equation of the circle is

$$x^2 + y^2 + 2gx + 2fy + c = 0$$

Given centre of circle (2, 2)

$$\therefore g = -2 \text{ and } f = -2$$

Putting the given point (4, 5)

$$4^2 + 5^2 + 2(-2)4 + 2(-2)5 + c = 0$$

$$\Rightarrow 16 + 25 - 16 - 20 + c = 0$$

$$\Rightarrow c = -5$$

$\therefore$ The required equation is

$$x^2 + y^2 + 2(-2)x + 2(-2)y - 5 = 0$$

$$\Rightarrow x^2 + y^2 - 4x - 4y - 5 = 0 \text{ (Ans)}$$

32. Find the equation of the circle whose radius is 5 unit and which touches the circle

$$x^2 + y^2 - 2x - 4y - 20 = 0 \text{ at a point (5, 5).}$$

Solution:

The given circle is $x^2 + y^2 - 2x - 4y - 20 = 0$

$$\Rightarrow x^2 + y^2 + 2(-1)x + 2(-2)y + (-20) = 0$$

$\therefore$ Centre = (1, 2)

$$\text{and radius} = \sqrt{(-1)^2 + (-2)^2 + 20}$$

$$= \sqrt{1 + 4 + 20} = 5$$

And the required circle has radius 5 hence circles touch each other externally.

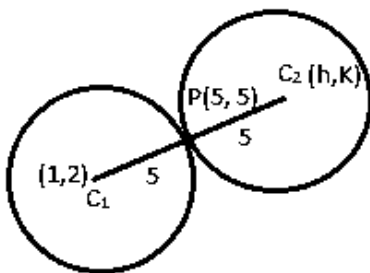
Since, point of contact is (5, 5)

P is the mid point of C_1 and C_2 .

Let coordinate of C_2 is (h, k) then

$$\frac{1+h}{2} = 5, \Rightarrow 1 + h = 10, \Rightarrow h = 9$$

$$\frac{2+k}{2} = 5, \Rightarrow 2 + k = 10, \Rightarrow k = 8$$



$\therefore$ The equation of required circle is

$$(x - 9)^2 + (y - 8)^2 = 5^2$$

$$\Rightarrow x^2 - 18x + 81 + y^2 - 16y + 64 = 25$$

$$\Rightarrow x^2 + y^2 - 18x - 16y + 120 = 0 \text{ (Ans)}$$

33. Find the equations to the circles which touch both the axes and passes through (3, 6).

Solution: Since circle touches both axis

$\Rightarrow$ equations of circle is of the form

$$(x - a)^2 + (y - a)^2 = a^2 \text{ -----(1)}$$

Since circle passes through (3, 6)

$$\Rightarrow (3 - a)^2 + (6 - a)^2 = a^2$$

$$\Rightarrow 9 - 6a + a^2 + 36 - 12a + a^2 = a^2$$

$$\Rightarrow a^2 - 18a + 45 = 0$$

$$\Rightarrow a^2 - 15a - 3a + 45 = 0$$

$$\Rightarrow a(a - 15) - 3(a - 15) = 0$$

$$\Rightarrow (a - 15)(a - 3) = 0$$

$$\Rightarrow a = 15 \text{ and } a = 3$$

By putting $a = 15$ in equation (1)

$$(x - 15)^2 + (y - 15)^2 = 15^2$$

$$\Rightarrow x^2 - 30x + 225 + y^2 - 30y + 225 = 225$$

$$\Rightarrow x^2 + y^2 - 30x - 30y + 225 = 0 \text{ (Ans)}$$

By putting $a = 3$ in equation (1)

$$(x - 3)^2 + (y - 3)^2 = 3^2$$

$$\Rightarrow x^2 - 6x + 9 + y^2 - 6y + 9 = 9$$

$$\Rightarrow x^2 + y^2 - 6x - 6y + 9 = 0 \text{ (Ans)}$$

34. The latus rectum of the parabola $y^2 = -48x$ is-

(a) -48

(b) -12

(c) 48

(d) 12

Solution: Given equation $y^2 = -48x$

$$\Rightarrow y^2 = -4.12. x$$

Comparing with standard equation $y^2 = -4ax$

We get, $a = 12$

$\therefore$ Length of latus rectum = $4a$

$$= 4 \times 12$$

$$= 48$$

Answer: (c)

35. The vertex of the parabola $y^2 = 8x + 4y + 4$ is-

(a) (-1, -2)

(b) (2, 1)

(c) (1, -2)

(d) (-1, 2)

Solution: Given equation $y^2 = 8x + 4y + 4$

$$\Rightarrow y^2 - 4y = 8x + 4$$

$$\Rightarrow y^2 - 2.y.2 + 2^2 = 8x + 8$$

$$\Rightarrow (y - 2)^2 = 4.2(x + 1)$$

Comparing with standard equation

$$(y - \beta)^2 = 4a(x - \alpha)$$

$$\alpha = -1 \text{ and } \beta = 2$$

$\therefore$ The vertex $(\alpha, \beta) = (-1, 2)$

Answer: (d)

36. The parabola $y^2 = 4ax$ passes through the centre of the circle $4x^2 + 4y^2 - 8x + 12y - 7 = 0$; find the focus, latus rectum and equation to the directrix of the parabola.

Solution:

$$\text{Circle: } 4x^2 + 4y^2 - 8x + 12y - 7 = 0$$

$$\Rightarrow x^2 + y^2 - 2x + 3y - \frac{7}{4} = 0$$

$$\Rightarrow x^2 + y^2 + 2(-1)x + 2 \times \frac{3}{2}y - \frac{7}{4} = 0$$

$$\therefore \text{Centre of the circle } \left(1, -\frac{3}{2}\right)$$

Given parabola $y^2 = 4ax$ passes through this centre so $\left(-\frac{3}{2}\right)^2 = 4a \cdot 1$

$$\Rightarrow 4a = \frac{9}{4}, \Rightarrow a = \frac{9}{16}$$

$$\therefore \text{Focus of the parabola} = (a, 0) = \left(\frac{9}{16}, 0\right) \text{ (Ans)}$$

$$\text{Length of latus rectum} = 4a = 4 \times \frac{9}{16} = \frac{9}{4} \text{ (Ans)}$$

$$\text{Equation of directrix } x + a = 0$$

$$\Rightarrow x + \frac{9}{16} = 0, \Rightarrow 16x + 9 = 0 \text{ (Ans)}$$

37. If the extremities of a focal chord of the parabola $y^2 = 4ax$ be $(at_1^2, 2at_1)$ and $(at_2^2, 2at_2)$, prove that $t_1 t_2 = -1$.

Solution: Given equation of parabola $y^2 = 4ax$

$\therefore$ Its focus is $(a, 0)$

Now, the equation of given focal chord be

$$\frac{x - at_1^2}{at_1^2 - at_2^2} = \frac{y - 2at_1}{2at_1 - 2at_2}$$

$$\Rightarrow \frac{a - at_1^2}{at_1^2 - at_2^2} = \frac{0 - 2at_1}{2at_1 - 2at_2} \quad [\because \text{Focus i.e. } (a, 0) \text{ lies on it}]$$

$$\Rightarrow \frac{a(1 - t_1^2)}{a(t_1^2 - t_2^2)} = \frac{2at_1}{2a(t_1 - t_2)}$$

$$\Rightarrow \frac{(1 - t_1^2)}{(t_1^2 - t_2^2)} = \frac{-t_1}{(t_1 - t_2)}$$

$$\Rightarrow \frac{1 - t_1^2}{(t_1 + t_2)(t_1 - t_2)} = \frac{-t_1}{(t_1 - t_2)}$$

$$\Rightarrow -t_1^2 - t_1 t_2 = 1 - t_1^2$$

$$\Rightarrow t_1 t_2 = -1 \text{ (Proved)}$$

38. Find the length of major axis of the ellipse $9x^2 + 16y^2 = 144$.

Solution: Given equation $9x^2 + 16y^2 = 144$

$$\Rightarrow \frac{9x^2}{144} + \frac{16y^2}{144} = 1$$

$$\Rightarrow \frac{x^2}{16} + \frac{y^2}{9} = 1, \Rightarrow \frac{x^2}{4^2} + \frac{y^2}{3^2} = 1$$

$$\therefore \text{Length of major axis} = 2a = 2 \times 4 = 8 \text{ (Ans)}$$

39. Given the ellipse with equation $9x^2 + 25y^2 = 225$, find the major and minor axes, eccentricity, foci and vertices.

Solution:

$$\text{Given } 9x^2 + 25y^2 = 225$$

$$\Rightarrow \frac{9x^2}{225} + \frac{25y^2}{225} = 1$$

$$\Rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1, \Rightarrow \frac{x^2}{5^2} + \frac{y^2}{3^2} = 1$$

$$\text{Comparing with standard form } \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

we get, $a = 5$ and $b = 3$

$$\text{Eccentricity } e = \sqrt{1 - \frac{b^2}{a^2}} = \sqrt{1 - \frac{9}{25}} = \frac{4}{5} \text{ (Ans)}$$

Since the denominator of x^2 is larger, the major axis is along x-axis, minor axis along y-axis. (Ans)

$$\begin{aligned} \text{Foci of the ellipse} &= (\pm ae, 0) = \left(\pm 5 \times \frac{4}{5}, 0\right) \\ &= (\pm 4, 0) \text{ (Ans)} \end{aligned}$$

$$\text{And vertices} = (\pm a, 0) = (\pm 5, 0) \text{ (Ans)}$$

40. Find the equation of the ellipse whose eccentricity is $\frac{4}{5}$ and axes are along the coordinate axes and with foci at $(0, \pm 4)$.

Solution: We are given eccentricity $e = \frac{4}{5}$

and foci $(0, \pm ae) = (0, \pm 4)$

$$\Rightarrow ae = 4, \Rightarrow a \times \frac{4}{5} = 4, \Rightarrow a = 5$$

$$\text{We know, eccentricity } e = \sqrt{1 - \frac{b^2}{a^2}}$$

$$\Rightarrow \frac{4}{5} = \sqrt{1 - \frac{b^2}{a^2}}$$

$$\Rightarrow \frac{16}{25} = 1 - \frac{b^2}{25}$$

$$\Rightarrow \frac{b^2}{25} = 1 - \frac{16}{25} = \frac{9}{25}$$

$$\Rightarrow b^2 = 9, \Rightarrow b = \sqrt{9} = 3$$

$\therefore$ The equation of the ellipse is

$$\frac{x^2}{5^2} + \frac{y^2}{3^2} = 1, \Rightarrow \frac{x^2}{25} + \frac{y^2}{9} = 1 \text{ (Ans)}$$

41. Find the length of the semi-latus rectum of the hyperbola $\frac{x^2}{25} - \frac{y^2}{16} = 1$.

Solution:

$$\text{Given equation } \frac{x^2}{25} - \frac{y^2}{16} = 1$$

$$\text{Comparing with standard form } \frac{x^2}{a^2} - \frac{y^2}{b^2} = 1$$

we get, $a = 5$ and $b = 4$

$$\text{We know length of latus rectum} = \frac{2b^2}{a}$$

$$\begin{aligned} \therefore \text{Length of semi-latus rectum} &= \frac{b^2}{a} \\ &= \frac{16}{5} = \frac{16}{5} \text{ (Ans)} \end{aligned}$$

42. The equation $x = a \cos \theta, y = b \sin \theta$ represents the parametric equation of -

- (a) ellipse
- (b) parabola
- (c) hyperbola
- (d) circle

Solution: Given $x = a \cos \theta, \Rightarrow \frac{x}{a} = \cos \theta$

and $y = b \sin \theta, \Rightarrow \frac{y}{b} = \sin \theta$

$$\therefore \left(\frac{x}{a}\right)^2 + \left(\frac{y}{b}\right)^2 = \cos^2 \theta + \sin^2 \theta$$

$$\Rightarrow \frac{x^2}{a^2} + \frac{y^2}{b^2} = 1 \text{ [Equation of ellipse]}$$

Answer: (a)

43. $x^2 - y^2 = 2022$ represents a ____.

Solution: General equation of hyperbola is

$$x^2 - y^2 = c$$

Where c is constant. In the given question above

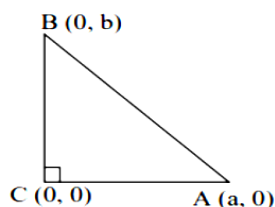
$c = 2022$.

Answer: hyperbola

44. The area of the triangle having vertices $(0, 0)$, $(a, 0)$ and $(0, b)$ is-

- (a) ab
- (b) $\frac{1}{4}ab$
- (c) $\frac{1}{2}ab$
- (d) $\frac{1}{8}ab$

Solution:



Let, the given triangle is $\triangle ABC$.

$\therefore$ Area of $\triangle ABC$

$$= \frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$$

$$= \frac{1}{2} |a(b - 0) + 0(0 - 0) + 0(0 - b)|$$

$$= \frac{1}{2} |ab|$$

Answer: (c)

45. The value of a for which the points $(1, 2)$, $(a, 4)$ and $(3, 6)$ are collinear is-

- (a) 1
- (b) 2
- (c) 3
- (d) 6

Solution:

Given points be collinear if

$$x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2) = 0$$

$$\Rightarrow 1(4 - 6) + a(6 - 2) + 3(2 - 4) = 0$$

$$\Rightarrow -2 + 4a - 6 = 0$$

$$\Rightarrow 4a = 8$$

$$\Rightarrow a = 2$$

Answer: (b)

46. A point divides internally the line segment joining the points $(8, 9)$ and $(-7, 4)$ in the ratio $2 : 3$. The coordinate of the point is-

- (a) $(-2, 7)$
- (b) $(2, -7)$
- (c) $(2, 7)$
- (d) none of these

Solution: Given that $m : n = 2 : 3$

and points are $(8, 9)$ and $(-7, 4)$

$\therefore$ The required coordinate is

$$\left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right) = \left(\frac{2 \times (-7) + 3 \times 8}{2+3}, \frac{2 \times 4 + 3 \times 9}{2+3}\right)$$

$$= (2, 7)$$

Answer: (c)

47. The line joining points $(3, 4)$ and $(-2, -2)$ is divided by X axis in the ratio

- (a) $1 : 2$
- (b) $2 : 3$
- (c) $3 : 2$
- (d) $2 : 1$

Solution: Let the points be divided by X axis at $(a, 0)$ in the ratio $m : n$.

$$\therefore \left(\frac{mx_2 + nx_1}{m+n}, \frac{my_2 + ny_1}{m+n}\right) = (a, 0)$$

$$\Rightarrow \left(\frac{m \cdot (-2) + n \cdot 3}{m+n}, \frac{m \cdot (-2) + n \cdot 4}{m+n}\right) = (a, 0)$$

$$\Rightarrow \frac{-2m + 4n}{m+n} = 0$$

$$\Rightarrow -2m + 4n = 0$$

$$\Rightarrow 2m = 4n$$

$$\Rightarrow \frac{m}{n} = \frac{4}{2} = \frac{2}{1}, \Rightarrow m : n = 2 : 1 \text{ (Ans)}$$

1. If $\frac{d}{dx}[\phi(x)] = f(x)$, then $\int f(x) dx$ is _____.

Solution:

$$\frac{d}{dx}[\phi(x)] = f(x)$$

$$\Rightarrow \int \frac{d}{dx}[\phi(x)] dx = \int f(x) dx$$

Since integration is the reverse of differentiation, we get:

$$\int f(x) dx = \phi(x) + c \quad (\text{Ans})$$

Where c is the constant of integration.

2. Integrate: $\int \cot^2 x dx$.

Solution:

$$\begin{aligned} \int \cot^2 x dx \\ = \int (\operatorname{cosec}^2 x - 1) dx = -\cot x - x + c \quad (\text{Ans}) \end{aligned}$$

3. Integrate: $\int \sin x^\circ dx$

Solution:

$$\begin{aligned} \int \sin x^\circ dx \\ = \int \sin \frac{\pi}{180} x dx \\ = -\frac{\cos \frac{\pi}{180} x}{\frac{\pi}{180}} + c = -\frac{180}{\pi} \cos x^\circ + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

4. $\int \frac{\sin x dx}{\cos x}$ is equal to- (a) $\log \sin x + c$ (b) $\log x + c$ (c) $\sin x + c$ (d) None of these.

Solution:

$$\int \frac{\sin x dx}{\cos x} = \int \tan x dx = \log(\sec x) + c$$

Answer: (d)

5. $\int \frac{\cos x}{\sqrt{1+\cos 2x}} dx =$

(a) $x + c$ (b) $\frac{x}{2} + c$ (c) $\frac{x}{\sqrt{2}} + c$ (d) none of these

Solution:

$$\begin{aligned} \int \frac{\cos x}{\sqrt{1+\cos 2x}} dx \\ = \int \frac{\cos x}{\sqrt{2 \cos^2 x}} dx = \int \frac{\cos x}{\sqrt{2} \cos x} dx = \frac{x}{\sqrt{2}} + c \end{aligned}$$

Answer: (c)

6. Integrate: $\int \sin^2 x dx$

Solution:

$$\begin{aligned} \int \sin^2 x dx \\ = \frac{1}{2} \int 2 \sin^2 x dx \\ = \frac{1}{2} \int (1 - \cos 2x) dx \\ = \frac{1}{2} \left(x - \frac{\sin 2x}{2} \right) + c \\ = \frac{1}{4} (2x - \sin 2x) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

7. Integrate: $\int \frac{dx}{1+\sin x}$

Solution:

$$\begin{aligned} \int \frac{dx}{1+\sin x} \\ = \int \frac{(1-\sin x) dx}{(1+\sin x)(1-\sin x)} \\ = \int \frac{(1-\sin x) dx}{1-\sin^2 x} \\ = \int \frac{(1-\sin x) dx}{\cos^2 x} \\ = \int \sec^2 x dx - \int \sec x \tan x dx \\ = \tan x - \sec x + c \end{aligned}$$

Where c is an arbitrary constant.

8. Evaluate: $\int \frac{dx}{1+\cos x}$

Solution:

$$\begin{aligned} \int \frac{dx}{1+\cos x} \\ = \int \frac{dx}{2 \cos^2 \frac{x}{2}} \\ = \frac{1}{2} \int \sec^2 \frac{x}{2} \\ = \frac{1}{2} \tan \frac{x}{2} + c = \tan \frac{x}{2} + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

9. Integrate: $\int \frac{dx}{1-\cos x}$.

Solution:

$$\begin{aligned} \int \frac{dx}{1-\cos x} &= \int \frac{dx}{2 \sin^2 \frac{x}{2}} \\ &= \frac{1}{2} \int \operatorname{cosec}^2 \frac{x}{2} dx \\ &= \frac{1}{2} \times \frac{-\cot \frac{x}{2}}{\frac{1}{2}} + c = -\cot \frac{x}{2} + c \quad (\text{Ans}) \end{aligned}$$

10. Integrate: $\int \sin^2 x \cos^2 x dx$

Solution:

$$\begin{aligned} \int \sin^2 x \cos^2 x dx \\ = \frac{1}{4} \int (2 \sin x \cos x)^2 dx \\ = \frac{1}{4} \int \sin^2 2x dx \\ = \frac{1}{8} \int 2 \sin^2 2x dx \\ = \frac{1}{8} \int (1 - \cos 4x) dx \\ = \frac{1}{8} \left[x - \frac{\sin 4x}{4} \right] + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

11. Integrate: $\int \frac{\sin 6x - \sin 2x}{\cos 6x - \cos 2x} dx$

Solution:

$$\begin{aligned} & \int \frac{\sin 6x - \sin 2x}{\cos 6x - \cos 2x} dx \\ &= \int \frac{2 \cos \left(\frac{6x+2x}{2}\right) \sin \left(\frac{6x-2x}{2}\right)}{2 \sin \left(\frac{6x+2x}{2}\right) \sin \left(\frac{2x-6x}{2}\right)} dx \\ &= - \int \frac{2 \cos 4x \sin 2x}{2 \sin 4x \sin 2x} dx \\ &= - \int \cot 4x dx \\ &= -\frac{1}{4} \log |\sin 4x| + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

12. Integrate: $\int \frac{\cos x}{(\sin x - a)(\sin x - b)} dx$.

Solution:

$$\begin{aligned} & \int \frac{\cos x}{(\sin x - a)(\sin x - b)} dx \quad \text{Let, } z = \sin x \\ & \Rightarrow \frac{dz}{dx} = \cos x \\ & \Rightarrow dz = \cos x dx \\ &= \int \frac{dz}{(z-a)(z-b)} \\ &= \frac{1}{b-a} \int \frac{(z-a)-(z-b)}{(z-a)(z-b)} dz \\ &= \frac{1}{b-a} \int \left(\frac{1}{z-b} - \frac{1}{z-a} \right) dz \\ &= \frac{1}{b-a} [\log(z-b) - \log(z-a)] + C \\ &= \frac{1}{b-a} \log \left| \frac{z-b}{z-a} \right| + C \\ &= \frac{1}{b-a} \log \left| \frac{\sin x - b}{\sin x - a} \right| + C \quad (\text{Ans}) \end{aligned}$$

C = arbitrary integration constant.

13. Integrate: $\int \frac{dx}{\sin(x-a) \sin(x-b)}$.

Solution:

$$\begin{aligned} & \int \frac{dx}{\sin(x-a) \sin(x-b)} \\ &= \frac{1}{\sin(b-a)} \int \frac{\sin\{(x-a)-(x-b)\}}{\sin(x-a) \sin(x-b)} dx \\ &= \frac{1}{\sin(b-a)} \int \frac{\sin(x-a) \cos(x-b) - \cos(x-a) \sin(x-b)}{\sin(x-a) \sin(x-b)} dx \\ &= \frac{1}{\sin(b-a)} \left[\int \frac{\cos(x-b)}{\sin(x-b)} dx - \int \frac{\cos(x-a)}{\sin(x-a)} dx \right] \\ &= \frac{1}{\sin(b-a)} [\int \cot(x-b) dx - \int \cot(x-a) dx] \\ &= \frac{1}{\sin(b-a)} [\log \sin(x-b) - \log \sin(x-a)] \\ &= \frac{1}{\sin(b-a)} \cdot \log \frac{\sin(x-b)}{\sin(x-a)} + c \quad [\text{Ans.}] \end{aligned}$$

Where c is an arbitrary constant.

14. Integrate: $\int \frac{2^{6x}-1}{2^{2x}-1} dx$

Solution:

$$\begin{aligned} & \int \frac{2^{6x}-1}{2^{2x}-1} dx \\ &= \int \frac{(2^{2x})^3 - 1^3}{2^{2x}-1} dx \\ &= \int \frac{(2^{2x}-1)(2^{4x}+2^{2x}+1)}{(2^{2x}-1)} dx \\ &= \int (2^{4x} + 2^{2x} + 1) dx \\ &= \frac{2^{4x}}{4 \log 2} + \frac{2^{2x}}{2 \log 2} + x + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

15. Integrate: $\int \frac{\sin \sqrt{x}}{\sqrt{x}} dx$

Solution:

$$\begin{aligned} & \text{Let, } \sqrt{x} = z \\ & \text{Or, } x = z^2 \\ & \text{Or, } dx = 2z dz \\ & \therefore \int \frac{\sin \sqrt{x}}{\sqrt{x}} dx \\ &= \int \frac{\sin z}{z} 2z dz \\ &= -2 \cos z + c \\ &= -2 \cos \sqrt{x} + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

16. Integrate: $\int \frac{x}{x^2+3} dx$

Solution:

$$\begin{aligned} & \text{Let, } z = x^2 + 3 \\ & \Rightarrow dz = 2x dx \\ & \Rightarrow \frac{dz}{2} = x dx \\ & \therefore \int \frac{x}{x^2+3} dx \\ &= \int \frac{1}{z} \cdot \frac{dz}{2} \\ &= \frac{1}{2} \int \frac{dz}{z} \\ &= \frac{1}{2} \log z + c \\ &= \frac{1}{2} \log |x^2 + 3| + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

17. Integrate: $\int \frac{dx}{x \log x}$

Solution:

$$\begin{aligned} & \text{Let, } z = \log x \Rightarrow dz = \frac{dx}{x} \\ & \therefore \int \frac{dx}{x \log x} \\ &= \int \frac{dz}{z} = \log z + c = \log(\log x) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

18. Integrate: $\int \frac{dx}{x\{(\log x)^2+9\}}$

Solution:

Let, $z = \log x \Rightarrow dz = \frac{dx}{x}$

$$\begin{aligned} \therefore \int \frac{dx}{x\{(\log x)^2+9\}} &= \int \frac{dz}{z^2+3^2} \\ &= \frac{1}{3} \tan^{-1} \frac{z}{3} + c \\ &= \frac{1}{3} \tan^{-1} \left(\frac{\log x}{3} \right) + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

19. Integrate: $\int \frac{\tan^{-1} x}{1+x^2} dx$

Solution:

Let, $z = \tan^{-1} x \Rightarrow dz = \frac{dx}{1+x^2}$

$$\begin{aligned} \therefore \int \frac{\tan^{-1} x}{1+x^2} dx &= \int z dz = \frac{z^2}{2} + c = \frac{(\tan^{-1} x)^2}{2} + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

20. Integrate: $\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution:

$$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$$

[Let, $z = \sin^{-1} x \therefore \frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow dz = \frac{dx}{\sqrt{1-x^2}}$]

$$\begin{aligned} &= \int z dz \\ &= \frac{z^2}{2} + c = \frac{(\sin^{-1} x)^2}{2} + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

21. Evaluate: $\int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution:

$$\int \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx$$

[Let, $z = \sin^{-1} x \therefore \frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}} \Rightarrow dz = \frac{dx}{\sqrt{1-x^2}}$]

$$\begin{aligned} &= \int z dz \\ &= \frac{z^2}{2} + c = \frac{(\sin^{-1} x)^2}{2} + c \end{aligned}$$

$$\begin{aligned} \therefore \int_0^1 \frac{\sin^{-1} x}{\sqrt{1-x^2}} dx &= \left[\frac{(\sin^{-1} x)^2}{2} \right]_0^1 \\ &= \frac{(\sin^{-1} 1)^2}{2} - \frac{(\sin^{-1} 0)^2}{2} = \frac{\left(\frac{\pi}{2}\right)^2}{2} = \frac{\pi^2}{8} \text{ (Ans)} \end{aligned}$$

22. Integrate: $\int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx$

Solution:

Let, $z = \sin^{-1} x$

$\Rightarrow x = \sin z$

$\therefore \frac{dz}{dx} = \frac{1}{\sqrt{1-x^2}}$

$\Rightarrow dz = \frac{dx}{\sqrt{1-x^2}}$

$$\begin{aligned} \int \frac{x \sin^{-1} x}{\sqrt{1-x^2}} dx &= \int z \sin z dz \\ &= z \int \sin z - \int \left[\frac{d}{dz} (z) \cdot \int \sin z dz \right] dz \\ &= -z \cos z + \int \cos z dz \\ &= \sin z - z \cos z + c \\ &= x - \sin^{-1} x \cdot \cos(\sin^{-1} x) + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

23. Integrate: $\int \frac{e^x + e^{3x}}{e^x + e^{-x}} dx$

Solution:

$$\begin{aligned} \int \frac{e^x + e^{3x}}{e^x + e^{-x}} dx &= \int \frac{e^{2x}(e^x + e^{-x})}{e^x + e^{-x}} dx \\ &= \int e^{2x} dx = \frac{e^{2x}}{2} + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

24. Evaluate: $\int \frac{e^{4 \log x} - e^{3 \log x}}{e^{\log x} - 1} dx$

Solution:

$$\begin{aligned} \int \frac{e^{4 \log x} - e^{3 \log x}}{e^{\log x} - 1} dx &= \int \frac{e^{\log x^4} - e^{\log x^3}}{x - 1} dx \\ &= \int \frac{x^4 - x^3}{x - 1} dx \\ &= \int \frac{x^3(x-1)}{x-1} dx = \int x^3 dx = \frac{x^4}{4} + c \text{ (Ans)} \end{aligned}$$

Where c is arbitrary integration constant.

25. $\int e^{1-\log x} dx$ is equal to- (a) $ex + c$ (b) $e \log|x| + c$ (c) $x \log e + c$ (d) none of these

Solution:

$$\begin{aligned} \int e^{1-\log x} dx &= \int e^{\log e - \log x} dx \\ &= \int e^{\log \left(\frac{e}{x} \right)} dx \\ &= \int \frac{e}{x} dx = e \log|x| + c \end{aligned}$$

Answer: (b)

26. Evaluate: $\int_1^e \log_e x \, dx$

Solution:

$$\begin{aligned} & \int_1^e \log_e x \, dx \\ &= \log_e x \int_1^e 1 \, dx - \int_1^e \left[\frac{d}{dx} (\log_e x) \int 1 \, dx \right] dx \\ &= \log_e x \int_1^e 1 \, dx - \int_1^e \left[\frac{1}{x} \cdot x \right] dx \\ &= [x \log_e x - x]_1^e \\ &= (e \log_e e - e) - (1 \log_e 1 - 1) \\ &= 1 \quad (\text{Ans}) \end{aligned}$$

27. Integrate: $\int x e^x \, dx$

Solution:

$$\begin{aligned} & \int x e^x \, dx \\ &= x \int e^x \, dx - \int \left[\frac{d}{dx} (x) \cdot \int e^x \, dx \right] dx \\ &= x e^x - \int e^x \, dx \\ &= x e^x - e^x + c \\ &= e^x (x - 1) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

28. Evaluate: $\int e^x \cos x \, dx$

Solution:

$$\begin{aligned} \text{Let, } I &= \int e^x \cos x \, dx \\ &= \cos x \int e^x \, dx - \int \left[\frac{d}{dx} (\cos x) \int e^x \, dx \right] dx \\ &= e^x \cos x + \int e^x \sin x \, dx \\ &= e^x \cos x + \sin x \int e^x \, dx - \int e^x \cos x \, dx \\ \Rightarrow I &= e^x \cos x + e^x \sin x - I \\ \Rightarrow 2I &= e^x (\cos x + \sin x) \\ \Rightarrow I &= \frac{e^x}{2} (\cos x + \sin x) \quad (\text{Ans}) \end{aligned}$$

29. Integrate: $\int x e^{x^2} \, dx$

$$\begin{aligned} & \int x e^{x^2} \, dx \\ &= \frac{1}{2} \int 2x e^{x^2} \, dx \\ &= \frac{1}{2} \int e^z \, dz \\ &= \frac{1}{2} e^z + c = \frac{1}{2} e^{x^2} + c \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \text{Let, } z &= x^2 \\ \Rightarrow \frac{dz}{dx} &= 2x \\ \Rightarrow dz &= 2x \, dx \end{aligned}$$

Where c is an arbitrary constant.

30. Integrate: $\int \frac{dx}{1+e^{-x}}$

Solution:

$$\begin{aligned} & \int \frac{dx}{1+e^{-x}} \\ &= \int \frac{e^x dx}{e^x + 1} \\ &= \int \frac{dz}{z} \\ &= \log z + c = \log(e^x + 1) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

31. Integrate: $\int \frac{dx}{e^x + e^{-x}}$

Solution:

$$\begin{aligned} & \int \frac{dx}{e^x + e^{-x}} \\ &= \int \frac{e^x dx}{e^{2x} + 1} \\ &= \int \frac{dz}{z^2 + 1} \\ &= \tan^{-1} z + c \\ &= \tan^{-1} e^x + c \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \text{Let, } z &= e^x \\ \Rightarrow \frac{dz}{dx} &= e^x \\ \Rightarrow dz &= e^x dx \end{aligned}$$

Where c is an arbitrary constant.

32. Integrate: $\int \frac{e^x(1+x)}{\cos^2(xe^x)} \, dx$

Solution:

$$\begin{aligned} \therefore \int \frac{e^x(1+x)}{\cos^2(xe^x)} \, dx & \quad \text{Let,} \\ & \quad z = xe^x \\ &= \int \frac{dz}{\cos^2 z} \\ &= \int \sec^2 z \, dz \\ &= \tan z + c \\ &= \tan xe^x + c \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \Rightarrow \frac{dz}{dx} &= xe^x + e^x \\ \Rightarrow \frac{dz}{dx} &= e^x(1+x) \\ \Rightarrow dz &= e^x(1+x) \, dx \end{aligned}$$

Where c is an arbitrary constant.

33. Integrate: $\int \frac{xe^x}{(1+x)^2} \, dx$

Solution:

$$\begin{aligned} & \int \frac{xe^x}{(1+x)^2} \, dx \\ &= \int \frac{e^x(1+x-1)}{(1+x)^2} \, dx \\ &= \int e^x \left(\frac{1}{1+x} - \frac{1}{(1+x)^2} \right) \, dx \\ &= \int dz = z + c = \frac{e^x}{1+x} + c \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \text{Let, } z &= e^x \cdot \frac{1}{1+x} \\ \Rightarrow \frac{dz}{dx} &= \frac{1}{1+x} \cdot e^x - e^x \cdot \frac{1}{(1+x)^2} \\ \Rightarrow dz &= e^x \left(\frac{1}{1+x} - \frac{1}{(1+x)^2} \right) \, dx \end{aligned}$$

Where c is an arbitrary constant.

34. Integrate: $\int \frac{e^x(x-1)}{x^2} \, dx$

Solution:

$$\begin{aligned} & \int \frac{e^x(x-1)}{x^2} \, dx \\ &= \int e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) \, dx \\ &= \int dz = z + c = \frac{e^x}{x} + c \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} \text{Let, } z &= e^x \cdot \frac{1}{x} \\ \Rightarrow \frac{dz}{dx} &= \frac{1}{x} \cdot e^x - e^x \cdot \frac{1}{x^2} \\ \Rightarrow dz &= e^x \left(\frac{1}{x} - \frac{1}{x^2} \right) \, dx \end{aligned}$$

Where c is an arbitrary constant.

35. Evaluate: $\int e^x \frac{1+x^2}{(1+x)^2} dx$

Solution:

$$\begin{aligned} \int e^x \frac{1+x^2}{(1+x)^2} dx & \quad \text{Let, } z = e^x \cdot \frac{1}{1+x} \\ & \Rightarrow \frac{dz}{dx} = \frac{1}{1+x} \cdot e^x - e^x \cdot \frac{1}{(1+x)^2} \\ & \Rightarrow dz = e^x \left(\frac{1}{1+x} - \frac{1}{(1+x)^2} \right) dx \\ \int e^x \frac{1+2x+x^2-2x}{(1+x)^2} dx & \\ \int e^x \frac{(1+x)^2-2x}{(1+x)^2} dx & \\ \int e^x \left(1 - \frac{2x}{(1+x)^2} \right) dx & \\ \int e^x dx - 2 \int e^x \frac{1+x-1}{(1+x)^2} dx & \\ = e^x - 2 \int e^x \left(\frac{1}{1+x} - \frac{1}{(1+x)^2} \right) dx & \\ = e^x - 2 \int dz & \\ = e^x - 2z + c & \\ = e^x - \frac{2e^x}{1+x} + c \quad (\text{Ans}) & \end{aligned}$$

Where c is an arbitrary constant.

36. Integrate: $\int e^x \frac{x-1}{(x+1)^3} dx$

Solution:

$$\begin{aligned} \int e^x \frac{x-1}{(x+1)^3} dx & \quad \text{Let, } z = e^x \cdot \frac{1}{(x+1)^2} \\ & \Rightarrow \frac{dz}{dx} = \frac{1}{(x+1)^2} \cdot e^x - e^x \cdot \frac{2}{(x+1)^3} \\ & \Rightarrow dz = e^x \left(\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3} \right) dx \\ \int e^x \left[\frac{1}{(x+1)^2} - \frac{2}{(x+1)^3} \right] dx & \\ = \int dz = z + c = \frac{e^x}{(x+1)^2} + c \quad (\text{Ans}) & \end{aligned}$$

Where c is an arbitrary constant.

37. Integrate: $\int \frac{dx}{x\sqrt{x^4-1}}$

Solution:

$$\begin{aligned} \text{Let, } \sqrt{x^4-1} &= z \\ \text{Or, } x^4-1 &= z^2 \\ \text{Or, } x^4 &= 1+z^2 \\ \therefore 4x^3 dx &= 2z dz, \text{ Or, } 2x^3 dx = z dz \\ \text{Now, } \int \frac{dx}{x\sqrt{x^4-1}} & \\ = \frac{1}{2} \int \frac{2x^3 dx}{x^4\sqrt{x^4-1}} & \\ = \frac{1}{2} \int \frac{z dz}{(1+z^2)z} & \\ = \frac{1}{2} \int \frac{dz}{1+z^2} & \\ = \frac{1}{2} \tan^{-1} z + c = \frac{1}{2} \tan^{-1} \sqrt{x^4-1} + c \quad [\text{Ans.}] & \end{aligned}$$

Where c is an arbitrary constant.

38. Integrate: $\int_0^1 \frac{1-x}{1+x} dx$

Solution:

$$\begin{aligned} \int_0^1 \frac{1-x}{1+x} dx & \\ = \int_0^1 \frac{1-(1+x)+2}{1+x} dx & \\ = - \int_0^1 dx + 2 \int_0^1 \frac{1}{1+x} dx & \\ = -[x]_0^1 + [\log(x+1)]_0^1 & \\ = -1 + 2 \log 2 & \\ = 2 \log 2 - 1 \quad (\text{Ans}) & \end{aligned}$$

39. Integrate: $\int \frac{dx}{(1-x)\sqrt{1-x^2}}$

Solution:

$$\begin{aligned} \int \frac{dx}{(1-x)\sqrt{1-x^2}} dx & \quad \text{Let, } x = \cos z \\ & \Rightarrow z = \cos^{-1} x \\ \therefore \frac{dx}{dz} &= -\sin z \\ \Rightarrow dx &= -\sin z dz \\ \int \frac{dx}{(1-x)\sqrt{1-x^2}} dx & \\ = - \int \frac{\sin z dz}{(1-\cos z)\sqrt{1-\cos^2 z}} & \\ = - \int \frac{\sin z dz}{2 \sin^2 \frac{z}{2} \cdot \sin z} & \\ = - \frac{1}{2} \int \operatorname{cosec}^2 \frac{z}{2} dz & \\ = - \frac{1}{2} \left(-\frac{\cot \frac{z}{2}}{\frac{1}{2}} \right) + c & \\ = \cot \frac{z}{2} + c & \\ = \cot \left(\frac{\cos^{-1} x}{2} \right) + c & \\ = \cot \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right) + c = \sqrt{\frac{1+x}{1-x}} + c \quad (\text{Ans}) & \end{aligned}$$

Where c is an arbitrary constant.

40. Evaluate: $\int_0^{\frac{1}{2}} \frac{dx}{(1-2x^2)\sqrt{1-x^2}}$

Solution:

$$\begin{aligned} \int_0^{\frac{1}{2}} \frac{dx}{(1-2x^2)\sqrt{1-x^2}} & \quad \text{Let, } x = \sin \theta \\ & \Rightarrow \frac{dx}{d\theta} = \cos \theta \\ \Rightarrow dx &= \cos \theta d\theta \\ & \quad x = 0, \Rightarrow \theta = 0 \\ & \quad x = \frac{1}{2}, \Rightarrow \theta = \frac{\pi}{6} \\ \int_0^{\frac{1}{2}} \frac{dx}{(1-2x^2)\sqrt{1-x^2}} & \\ = \int_0^{\frac{\pi}{6}} \frac{\cos \theta d\theta}{(1-2\sin^2 \theta)\sqrt{1-\sin^2 \theta}} & \\ = \int_0^{\frac{\pi}{6}} \frac{\cos \theta d\theta}{\cos 2\theta \cos \theta} & \\ = \int_0^{\frac{\pi}{6}} \sec 2\theta d\theta & \\ = \frac{1}{2} [\log(\sec 2\theta + \tan 2\theta)]_0^{\frac{\pi}{6}} & \\ = \frac{1}{2} \left[\log \left(\sec \frac{\pi}{3} + \tan \frac{\pi}{3} \right) - \log(\sec 0 + \tan 0) \right] & \\ = \frac{1}{2} [\log(2 + \sqrt{3}) - \log 1] & \\ = \frac{1}{2} \log(2 + \sqrt{3}) \quad (\text{Ans}) & \end{aligned}$$

41. Integrate: $\int \frac{x-1}{(x-3)(x+2)} dx$

Solution:

$$\begin{aligned} & \int \frac{x-1}{(x-3)(x+2)} dx \\ &= \frac{1}{2} \int \frac{2x-2}{(x-3)(x+2)} dx \\ &= \frac{1}{2} \int \frac{(x-3)+(x+2)-1}{(x-3)(x+2)} dx \\ &= \frac{1}{2} \left[\int \frac{1}{x+2} dx + \int \frac{1}{x-3} dx - \int \frac{1}{(x-3)(x+2)} dx \right] \\ &= \frac{1}{2} [\ln(x+2) + \ln(x-3)] - \frac{1}{2} \int \left[\frac{1}{x-3} - \frac{1}{x+2} \right] dx \\ &= \frac{1}{2} \ln(x+2)(x-3) - \frac{1}{2} [\ln(x-3) - \ln(x+2)] \\ &= \frac{1}{2} [\ln(x+2)(x-3)] - \frac{1}{2} \left[\ln \frac{x-3}{x+2} \right] \\ &= \frac{1}{2} \ln \frac{(x+2)(x-3)}{\frac{x-3}{x+2}} \\ &= \frac{1}{2} \ln(x+2)^2 = \ln(x+2) + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

42. Evaluate: $\int \frac{x dx}{(x-1)(2x+1)}$

Solution:

$$\begin{aligned} & \text{Let, } \frac{A}{x-1} + \frac{B}{2x+1} = \frac{x}{(x-1)(2x+1)} \\ & \Rightarrow A(2x+1) + B(x-1) = x \\ & \text{If } x = 1, \Rightarrow A = \frac{1}{3} \text{ and If } x = -\frac{1}{2}, \Rightarrow B = \frac{1}{3} \\ & \therefore \frac{x}{(x-1)(2x+1)} = \frac{1}{3} \left(\frac{1}{x-1} + \frac{1}{2x+1} \right) \\ & \therefore \int \frac{x dx}{(x-1)(2x+1)} \\ &= \frac{1}{3} \int \left(\frac{1}{x-1} + \frac{1}{2x+1} \right) dx \\ &= \frac{1}{3} \left[\log(x-1) + \frac{1}{2} \log(2x+1) \right] + C \\ &= \frac{1}{3} [\log(x-1) + \log \sqrt{2x+1}] + C \\ &= \frac{1}{3} \log(x-1) \sqrt{2x+1} + C \text{ (Ans)} \\ & C = \text{arbitrary constant.} \end{aligned}$$

43. Integrate: $\int \frac{dx}{(x-1)^2(x-2)}$

Solution: Now, $\frac{1}{(x-1)^2(x-2)}$

$$\begin{aligned} &= \frac{(x-1)-(x-2)}{(x-1)^2(x-2)} \\ &= \frac{1}{(x-1)(x-2)} - \frac{1}{(x-1)^2} \\ &= \frac{(x-1)-(x-2)}{(x-1)(x-2)} - \frac{1}{(x-1)^2} \\ &= \frac{1}{(x-2)} - \frac{1}{(x-1)} - \frac{1}{(x-1)^2} \\ &\therefore \int \frac{dx}{(x-1)^2(x-2)} \\ &= \int \left[\frac{1}{(x-2)} - \frac{1}{(x-1)} - \frac{1}{(x-1)^2} \right] dx \\ &= \log(x-2) - \log(x-1) + \frac{1}{x-1} + c \\ &= \log \frac{(x-2)}{(x-1)} + \frac{1}{x-1} + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

44. Integrate: $\int \frac{dx}{2-x-x^2}$

Solution:

$$\begin{aligned} & \int \frac{dx}{2-x-x^2} = \int \frac{dx}{2-(x+x^2)} \\ &= \int \frac{dx}{2-\left\{x^2+2x \cdot \frac{1}{2} + \left(\frac{1}{2}\right)^2 - \frac{1}{4}\right\}} \\ &= \int \frac{dx}{2-\left(x+\frac{1}{2}\right)^2 + \frac{1}{4}} = \int \frac{dx}{\left(\frac{3}{2}\right)^2 - \left(x+\frac{1}{2}\right)^2} \\ &= \frac{1}{2 \times \frac{3}{2}} \log \left| \frac{\frac{3}{2} + x + \frac{1}{2}}{\frac{3}{2} - x - \frac{1}{2}} \right| + c = \frac{1}{3} \log \left| \frac{2+x}{1-x} \right| + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

45. Integrate: $\int \frac{2^x \log 2 + 10x^9}{2^x + x^{10}} dx$

Solution:

$$\begin{aligned} & \text{Let, } z = 2^x + x^{10} \\ & \Rightarrow \frac{dz}{dx} = 2^x \log 2 + 10x^9 \\ & \Rightarrow dz = (2^x \log 2 + 10x^9) dx \end{aligned}$$

$$\begin{aligned} & \therefore \int \frac{2^x \log 2 + 10x^9}{2^x + x^{10}} dx \\ &= \int \frac{dz}{z} \\ &= \log z + C \\ &= \log(2^x + x^{10}) + C \text{ (Ans)} \end{aligned}$$

46. Integrate: $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx$

Solution:

$$\begin{aligned} & \text{Let, } z^2 = \tan x \\ & \Rightarrow 2z dz = \sec^2 x dx \Rightarrow \frac{2z dz}{\sec^2 x} = dx \\ & \therefore \int \frac{\sqrt{\tan x}}{\sin x \cos x} dx \\ &= \int \frac{z}{\sin x \cos x} \cdot \frac{2z dz}{\sec^2 x} \\ &= 2 \int \frac{z^2}{\tan x} dz \\ &= 2 \int \frac{z^2}{z^2} dz \\ &= 2 \int dz \\ &= 2z + c = 2\sqrt{\tan x} + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

47. Integrate: $\int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx$

Solution:

$$\begin{aligned} & \int \frac{\sin x + \cos x}{\sqrt{1 + \sin 2x}} dx \\ &= \int \frac{\sin x + \cos x}{\sqrt{\sin^2 x + \cos^2 x + 2 \sin x \cos x}} dx \\ &= \int \frac{\sin x + \cos x}{\sqrt{(\sin x + \cos x)^2}} dx \\ &= \int \frac{\sin x + \cos x}{\sin x + \cos x} dx = \int dx = x + c \text{ (Ans)} \end{aligned}$$

Where c is an arbitrary constant.

48. Integrate: $\int \sqrt{1 + \sin x} \, dx$

Solution:

$$\begin{aligned} & \int \sqrt{1 + \sin x} \, dx \\ &= \int \sqrt{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \sin \frac{x}{2} \cos \frac{x}{2}} \, dx \\ &= \int \sqrt{\left(\sin \frac{x}{2} + \cos \frac{x}{2}\right)^2} \, dx \\ &= \int \left(\sin \frac{x}{2} + \cos \frac{x}{2}\right) \, dx \\ &= -\frac{\cos \frac{x}{2}}{\frac{1}{2}} + \frac{\sin \frac{x}{2}}{\frac{1}{2}} + c \\ &= 2 \left(\sin \frac{x}{2} - \cos \frac{x}{2}\right) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

49. Integrate: $\int \frac{\cos x - \sin x}{\cos x + \sin x} (1 + \sin 2x) \, dx$

Solution:

$$\begin{aligned} & \int \frac{\cos x - \sin x}{\cos x + \sin x} (1 + \sin 2x) \, dx \\ &= \int \frac{\cos x - \sin x}{\cos x + \sin x} (\sin^2 x + \cos^2 x + 2 \sin x \cos x) \, dx \\ &= \int \frac{\cos x - \sin x}{\cos x + \sin x} (\cos x + \sin x)^2 \, dx \\ &= \int (\cos x - \sin x)(\cos x + \sin x) \, dx \\ &= \int (\cos^2 x - \sin^2 x) \, dx \\ &= \int \cos 2x \, dx = \frac{\sin 2x}{2} + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

50. Evaluate: $\int \frac{dx}{\sqrt{(x-a)(b-x)}}$

Solution:

$$\text{Let, } x - a = z^2 \therefore dx = 2z \, dz$$

$$\text{Or, } x = z^2 + a$$

$$\begin{aligned} & \therefore \int \frac{dx}{\sqrt{(x-a)(b-x)}} \\ &= \int \frac{2z \, dz}{\sqrt{z^2 \{b - (z^2 + a)\}}} \\ &= \int \frac{2z \, dz}{z \sqrt{\{(\sqrt{b-a})^2 - z^2\}}} \\ &= 2 \sin^{-1} \frac{z}{\sqrt{b-a}} + c \\ &= 2 \sin^{-1} \frac{\sqrt{x-a}}{\sqrt{b-a}} + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

51. Evaluate: $\int \frac{dx}{\sqrt{(x-2)(3-x)}}$

Solution:

$$\begin{aligned} & \int \frac{dx}{\sqrt{(x-2)(3-x)}} = \int \frac{dx}{\sqrt{3x - x^2 - 6 + 2x}} \\ &= \int \frac{dx}{\sqrt{5x - x^2 - 6}} = \int \frac{dx}{\sqrt{-6 - (x^2 - 5x + \frac{25}{4}) + \frac{25}{4}}} \\ &= \int \frac{dx}{\sqrt{-6 - \left\{x^2 - 2 \cdot x \cdot \frac{5}{2} + \left(\frac{5}{2}\right)^2 - \frac{25}{4}\right\}}} \\ &= \int \frac{dx}{\sqrt{-6 - \left(x - \frac{5}{2}\right)^2 + \frac{25}{4}}} = \int \frac{dx}{\sqrt{\left(\frac{1}{2}\right)^2 - \left(x - \frac{5}{2}\right)^2}} \\ &= \sin^{-1} \left[\frac{\left(x - \frac{5}{2}\right)}{\left(\frac{1}{2}\right)} \right] + c = \sin^{-1}(2x - 5) + c \quad (\text{Ans}) \end{aligned}$$

Where c is an arbitrary constant.

52. Integrate: $\int \frac{dx}{\sin x + \cos x}$

Solution:

$$\begin{aligned} & \int \frac{dx}{\sin x + \cos x} \\ &= \frac{1}{\sqrt{2}} \cdot \int \frac{1}{\frac{1}{\sqrt{2}} \sin x + \frac{1}{\sqrt{2}} \cos x} \, dx \\ &= \frac{1}{\sqrt{2}} \cdot \int \frac{1}{\sin x \cdot \cos \frac{\pi}{4} + \sin \frac{\pi}{4} \cos x} \, dx \\ &= \frac{1}{\sqrt{2}} \cdot \int \frac{1}{\sin \left(x + \frac{\pi}{4}\right)} \, dx \\ &= \frac{1}{\sqrt{2}} \cdot \int \operatorname{cosec} \left(x + \frac{\pi}{4}\right) \, dx \\ &= \frac{1}{\sqrt{2}} \cdot \log \left| \operatorname{cosec} \left(x + \frac{\pi}{4}\right) - \cot \left(x + \frac{\pi}{4}\right) \right| + c \end{aligned}$$

Where c is an arbitrary constant.

53. Evaluate: $\int \frac{dx}{5 + 4 \cos x}$

Solution:

$$\begin{aligned} & \int \frac{dx}{5 + 4 \cos x} \\ &= \int \frac{dx}{5 \left(\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2}\right) + 4 \left(\cos^2 \frac{x}{2} - \sin^2 \frac{x}{2}\right)} \\ &= \int \frac{dx}{\sin^2 \frac{x}{2} + 9 \cos^2 \frac{x}{2}} \\ &= \int \frac{\sec^2 \frac{x}{2} \, dx}{9 + \tan^2 \frac{x}{2}} \quad \left[\begin{array}{l} \text{[Let, } \tan \frac{x}{2} = z \\ \therefore \frac{1}{2} \sec^2 \frac{x}{2} \, dx = dz \\ \text{Or, } \sec^2 \frac{x}{2} \, dx = 2dz] \end{array} \right. \\ &= 2 \int \frac{dz}{3^2 + z^2} \\ &= 2 \cdot \frac{1}{3} \tan^{-1} \frac{z}{3} + c \\ &= \frac{2}{3} \tan^{-1} \frac{1}{3} \tan \frac{x}{2} + c \\ &= \frac{2}{3} \tan^{-1} \tan \frac{x}{6} + c \\ &= \frac{2}{3} \times \frac{x}{6} + c = \frac{x}{9} + c \quad [\text{Ans.}] \end{aligned}$$

Where c is an arbitrary constant.

54. Integrate: $\int \frac{dx}{\sqrt{x(a^2-x)}}$

Solution:

Let, $x = a^2 \sin^2 \theta$

Or, $\sin \theta = \frac{\sqrt{x}}{a}$ Or, $\theta = \sin^{-1} \frac{\sqrt{x}}{a}$

and, $dx = a^2 2 \sin \theta \cos \theta d\theta$

$\therefore \int \frac{dx}{\sqrt{x(a^2-x)}} = \int \frac{2a^2 \sin \theta \cos \theta d\theta}{\sqrt{a^2 \sin^2 \theta (a^2 - a^2 \sin^2 \theta)}}$

$= \int \frac{2a^2 \sin \theta \cos \theta d\theta}{a \sin \theta \sqrt{a^2(1-\sin^2 \theta)}}$

$= \int \frac{2a^2 \sin \theta \cos \theta d\theta}{a^2 \sin \theta \sqrt{\cos^2 \theta}} = \int \frac{2a^2 \sin \theta \cos \theta d\theta}{a^2 \sin \theta \cos \theta}$

$= 2 \int d\theta = 2\theta + c = 2 \sin^{-1} \frac{\sqrt{x}}{a} + c$ (Ans)

Where c is an arbitrary constant.

55. Integrate: $\int \frac{\cos x dx}{\sqrt{(1+\sin x)(2+\sin x)}}$

Solution:

$\int \frac{\cos x dx}{\sqrt{(1+\sin x)(2+\sin x)}}$

$= \int \frac{dz}{\sqrt{(1+z-2)z}}$

$= \int \frac{dz}{\sqrt{(z-1)z}}$

$= \int \frac{dz}{\sqrt{(1+z-2)z}}$

$= \int \frac{dz}{\sqrt{z^2 - 2z + \frac{1}{2} - \left(\frac{1}{2}\right)^2}}$

$= \int \frac{dz}{\sqrt{\left(z - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2}}$

$= \log \left| \left(z - \frac{1}{2}\right) + \sqrt{\left(z - \frac{1}{2}\right)^2 - \left(\frac{1}{2}\right)^2} \right| + c$

$= \log \left| \left(\sin x + \frac{3}{2}\right) + \sqrt{(1+\sin x)(2+\sin x)} \right| + c$

Where c is an arbitrary constant.

56. Evaluate: $\int_0^{\frac{\pi}{2}} \frac{\cos x dx}{(1+\sin x)(2+\sin x)}$

Solution:

$\int_0^{\frac{\pi}{2}} \frac{\cos x dx}{(1+\sin x)(2+\sin x)}$

$= \int_0^1 \frac{dz}{(1+z)(2+z)}$

$= \int_0^1 \frac{(2+z)-(1+z)}{(1+z)(2+z)} dz$

$= \int_0^1 \left\{ \frac{1}{(1+z)} - \frac{1}{(2+z)} \right\} dz$

$= [\log(1+z) - \log(2+z)]_0^1$

$= \left[\log \left(\frac{1+z}{2+z} \right) \right]_0^1$

$= \log \left(\frac{1+1}{2+1} \right) - \log \left(\frac{1+0}{2+0} \right)$

$= \log \left(\frac{2}{3} \right) - \log \left(\frac{1}{2} \right) = \log \left(\frac{\frac{2}{3}}{\frac{1}{2}} \right) = \log \left(\frac{4}{3} \right)$ (Ans)

Let, $z = \sin x$

$\therefore \frac{dz}{dx} = \cos x$

$\Rightarrow dz = \cos x dx$

x	0	$\frac{\pi}{2}$
z	0	1

57. Evaluate: $\int \frac{\cos x}{(2 \sin x - 3)(\sin x + 2)} dx$

Solution:

$\int \frac{\cos x}{(2 \sin x - 3)(\sin x + 2)} dx$

$= \int \frac{1}{(2z-3)(z+2)} dz$

$= \int \frac{1}{2z^2 + z - 6} dz$

$= \frac{1}{2} \int \frac{dz}{z^2 + \frac{z}{2} - 3}$

$= \frac{1}{2} \int \frac{dz}{z^2 + 2z \cdot \frac{1}{4} + \left(\frac{1}{4}\right)^2 - \frac{1}{16} - 3}$

$= \frac{1}{2} \int \frac{dz}{\left(z + \frac{1}{4}\right)^2 - \frac{49}{16}} = \frac{1}{2} \int \frac{dz}{\left(z + \frac{1}{4}\right)^2 - \left(\frac{7}{4}\right)^2}$

$= \frac{1}{2 \times \frac{7}{4}} \log \left| \frac{z + \frac{1}{4} + \frac{7}{4}}{z + \frac{1}{4} - \frac{7}{4}} \right| + c = \frac{2}{7} \log \left| \frac{\sin x - \frac{3}{2}}{\sin x + 2} \right| + c$ (Ans)

Where c is an arbitrary constant.

58. Integrate: $\int \frac{\cos x}{\cos(x+a)} dx$

Solution:

$\int \frac{\cos x}{\cos(x+a)} dx$

$= \int \frac{\cos(x+a-a)}{\cos(x+a)} dx$

$= \int \frac{\cos(x+a) \cos a + \sin(x+a) \sin a}{\cos(x+a)} dx$

$= \cos a \int dx + \sin a \int \tan(x+a) dx$

$= x \cos a + \sin a \log |\sec(x+a)| + c$ (Ans)

Where c is an arbitrary constant.

59. The value of $\int_0^{\pi/2} \sin x dx$ is- (a) 1 (b) 0 (c) -1 (d)

None of these.

Solution:

$\int_0^{\pi/2} \sin x dx = [-\cos x]_0^{\pi/2}$

$= -\left[\cos \frac{\pi}{2} - \cos 0\right] = -(0 - 1) = 1$

Answer: (a)

60. Integrate: $\int_0^{\pi/2} \sin 2x dx$

Solution:

$\int_0^{\pi/2} \sin 2x dx = -\left[\frac{\cos 2x}{2}\right]_0^{\pi/2}$

$= -\left[\frac{-1}{2} - \frac{1}{2}\right] = 1$ (Ans)

61. Integrate: $\int_0^{\pi/4} \tan^2 x dx$

Solution:

$\int_0^{\pi/4} \tan^2 x dx$

$= \int_0^{\pi/4} (\sec^2 x - 1) dx$

$= [\tan x - x]_0^{\pi/4}$

$= \left(\tan \frac{\pi}{4} - \frac{\pi}{4}\right) - 0 = 1 - \frac{\pi}{4}$ (Ans)

62. Integrate: $\int_1^2 \left(\frac{x^2-1}{x^2}\right) e^{x+\frac{1}{x}} dx$

Solution:

Let, $z = x + \frac{1}{x}$

$\therefore \frac{dz}{dx} = 1 - \frac{1}{x^2} \Rightarrow dz = \left(1 - \frac{1}{x^2}\right) dx$

$\therefore \int \left(\frac{x^2-1}{x^2}\right) e^{x+\frac{1}{x}} dx$

$= \int \left(1 - \frac{1}{x^2}\right) e^{x+\frac{1}{x}} dx$

$= \int e^z dz = e^z + c = e^{x+\frac{1}{x}} + c$

$\therefore \int_1^2 \left(\frac{x^2-1}{x^2}\right) e^{x+\frac{1}{x}} dx$

$= \left[e^{x+\frac{1}{x}}\right]_1^2 = e^{2+\frac{1}{2}} - e^{1+\frac{1}{1}}$

$= e^2\sqrt{e} - e^2 = e^2(\sqrt{e} - 1) \text{ (Ans)}$

63. Integrate: $\int_0^{\pi/2} \frac{\cos x dx}{\sin x + \cos x}$

Solution:

Let, $I = \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx \dots\dots\dots(i)$

$\Rightarrow I = \int_0^{\pi/2} \frac{\cos(\frac{\pi}{2}-x)}{\sin(\frac{\pi}{2}-x) + \cos(\frac{\pi}{2}-x)} dx$

$\Rightarrow I = \int_0^{\pi/2} \frac{\sin x}{\cos x + \sin x} dx \dots\dots\dots(ii)$

Doing eqn. (i)+(ii)

$I + I = \int_0^{\pi/2} \frac{\cos x}{\sin x + \cos x} dx + \int_0^{\pi/2} \frac{\sin x}{\cos x + \sin x} dx$

$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx$

$\Rightarrow 2I = \int_0^{\pi/2} 1 dx, \Rightarrow 2I = [x]_0^{\pi/2}$

$\Rightarrow 2I = \frac{\pi}{2} - 0, \Rightarrow I = \frac{\pi}{4} \text{ (Ans)}$

64. Show that $\int_0^{\pi/2} \frac{\sin x dx}{\sin x + \cos x} = \frac{\pi}{4}$

Solution:

Let, $I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx \dots\dots\dots(i)$

$\Rightarrow I = \int_0^{\pi/2} \frac{\sin(\frac{\pi}{2}-x)}{\sin(\frac{\pi}{2}-x) + \cos(\frac{\pi}{2}-x)} dx$

$\Rightarrow I = \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx \dots\dots\dots(ii)$

Doing eqn. (i)+(ii)

$I + I = \int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x} dx + \int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x} dx$

$\Rightarrow 2I = \int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x} dx$

$\Rightarrow 2I = \int_0^{\pi/2} 1 dx, \Rightarrow 2I = [x]_0^{\pi/2}$

$\Rightarrow 2I = \frac{\pi}{2} - 0, \Rightarrow I = \frac{\pi}{4} \text{ (Proved)}$

65. Integrate: $\int_0^{\pi} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

Solution:

Let, $I = \int_0^{\pi} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx \dots\dots\dots(i)$

$\Rightarrow I = \int_0^{\pi} \frac{\sqrt{\cos(\frac{\pi}{2}-x)}}{\sqrt{\sin(\frac{\pi}{2}-x)} + \sqrt{\cos(\frac{\pi}{2}-x)}} dx$

$\Rightarrow I = \int_0^{\pi} \frac{\sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx \dots\dots\dots(ii)$

Doing eqn. (i)+(ii)

$I + I = \int_0^{\pi} \frac{\sqrt{\cos x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx + \int_0^{\pi} \frac{\sqrt{\sin x}}{\sqrt{\cos x} + \sqrt{\sin x}} dx$

$\Rightarrow 2I = \int_0^{\pi} \frac{\sqrt{\cos x} + \sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx$

$\Rightarrow 2I = \int_0^{\pi} 1 dx, \Rightarrow 2I = [x]_0^{\pi}$

$\Rightarrow 2I = \pi - 0, \Rightarrow I = \frac{\pi}{2} \text{ (Ans)}$

66. Evaluate: $\int_0^{\pi} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx$

Solution:

Let, $I = \int_0^{\pi} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx \dots\dots\dots(i)$

$\Rightarrow I = \int_0^{\pi} \frac{\sqrt{\tan(\frac{\pi}{2}-x)}}{\sqrt{\tan(\frac{\pi}{2}-x)} + \sqrt{\cot(\frac{\pi}{2}-x)}} dx$

$\Rightarrow I = \int_0^{\pi} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}} dx \dots\dots\dots(ii)$

Doing eqn. (i)+(ii)

$I + I = \int_0^{\pi} \frac{\sqrt{\tan x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx + \int_0^{\pi} \frac{\sqrt{\cot x}}{\sqrt{\cot x} + \sqrt{\tan x}} dx$

$\Rightarrow 2I = \int_0^{\pi} \frac{\sqrt{\tan x} + \sqrt{\cot x}}{\sqrt{\tan x} + \sqrt{\cot x}} dx$

$\Rightarrow 2I = \int_0^{\pi} 1 dx, \Rightarrow 2I = [x]_0^{\pi}$

$\Rightarrow 2I = \pi - 0, \Rightarrow I = \frac{\pi}{2} \text{ (Ans)}$

67. $\int_0^1 |x| dx = \underline{\hspace{2cm}}$

Solution:

$\int_0^1 |x| dx = \int_0^1 x dx \quad [\because |x| = x \text{ for } x \geq 0]$

$= \left[\frac{x^2}{2}\right]_0^1 = \frac{1^2}{2} - \frac{0^2}{2} = \frac{1}{2} \text{ (Ans)}$

68. Integrate: $\int_{-1}^1 |x| dx$

Solution:

$\int_{-1}^1 |x| dx = \int_{-1}^0 |x| dx + \int_0^1 |x| dx$

$= -\int_{-1}^0 x dx + \int_0^1 x dx$

$= -\left[\frac{x^2}{2}\right]_{-1}^0 + \left[\frac{x^2}{2}\right]_0^1$

$= -\left[0 - \frac{1}{2}\right] + \left[\frac{1}{2} - 0\right]$

$= \frac{1}{2} + \frac{1}{2} = 1 \text{ (Ans)}$

69. Integrate: $\int_0^2 |x-1| dx$

Solution:

$$\begin{aligned} & \int_0^2 |x-1| dx \\ &= \int_0^1 |x-1| dx + \int_1^2 |x-1| dx \\ &= -\int_0^1 (x-1) dx + \int_1^2 (x-1) dx \\ &= -\left[\frac{x^2}{2} - x\right]_0^1 + \left[\frac{x^2}{2} - x\right]_1^2 \\ &= -\left[\frac{1}{2} - 1\right] + \left[\left(\frac{2^2}{2} - 2\right) - \left(\frac{1^2}{2} - 1\right)\right] \\ &= \frac{1}{2} + \frac{1}{2} = 1 \quad (\text{Ans}) \end{aligned}$$

70. $\int_0^1 \frac{1-x}{1+x} dx = \underline{\hspace{2cm}}$

Solution:

$$\begin{aligned} & \text{Let, } 1+x = z, \Rightarrow dx = dz \text{ \& } x = z-1 \\ & \text{and } x=0, \Rightarrow z=1 \text{ \& } x=1, \Rightarrow z=2 \\ & \therefore \int_0^1 \frac{1-x}{1+x} dx = \int_1^2 \frac{1-(z-1)}{z} dz = \int_1^2 \frac{2-z}{z} dz \\ &= \int_1^2 \left(\frac{2}{z} - 1\right) dz \\ &= [2 \log z - z]_1^2 \\ &= (2 \log 2 - 2) - (1 \log 1 - 1) \\ &= \log 2^2 - 2 - 0 + 1 \\ &= \log(4) - 1 \quad (\text{Ans}) \end{aligned}$$

71. Evaluate: $\int_1^2 \log x dx$.

Solution:

$$\begin{aligned} & \text{Now, } \int \log x dx \\ &= \log x \int dx - \int \left[\frac{d}{dx}(\log x) \int dx\right] dx \\ &= x \log x - \int \left(\frac{1}{x} \cdot x\right) dx \\ &= x \log x - x \\ &\therefore \int_1^2 \log x dx = [x \log x - x]_1^2 \\ &= (2 \log 2 - 2) - (1 \log 1 - 1) \\ &= \log(2^2) - 2 - 0 + 1 \\ &= \log(4) - 1 \quad (\text{Ans}) \end{aligned}$$

72. The value of $\int_{-1}^1 e^{|x|} dx$ is-

- (a) $e-1$
(b) $2(e-1)$
(c) $e+1$
(d) none of these

Solution:

$$\begin{aligned} & \int_{-1}^1 e^{|x|} dx = \int_{-1}^0 e^{|x|} dx + \int_0^1 e^{|x|} dx \\ &= \int_{-1}^0 e^{-x} dx + \int_0^1 e^x dx \quad \begin{array}{l} |x| = x \text{ when } x \geq 0 \\ |x| = -x \text{ when } x < 0 \end{array} \\ &= [-e^{-x}]_{-1}^0 + [e^x]_0^1 \\ &= -[e^0 - e] + [e - 1] \\ &= -1 + e + e - 1 \\ &= 2e - 2 = 2(e-1) \end{aligned}$$

Answer: (b)

73. Evaluate: $\int_0^1 e^{2x} dx$

Solution:

$$\begin{aligned} & \int_0^1 e^{2x} dx \\ &= \left[\frac{e^{2x}}{2}\right]_0^1 = \frac{1}{2}(e^2 - e^0) = \frac{1}{2}(e^2 - 1) \quad (\text{Ans}) \end{aligned}$$

74. Integrate: $\int_0^{\pi/2} \frac{dx}{2+\cos x}$

Solution: $\int_0^{\pi/2} \frac{dx}{2+\cos x}$

$$\begin{aligned} &= \int_0^{\pi/2} \frac{dx}{1+1+\cos x} \\ &= \int_0^{\pi/2} \frac{dx}{\sin^2 \frac{x}{2} + \cos^2 \frac{x}{2} + 2 \cos^2 \frac{x}{2}} \\ &= \int_0^{\pi/2} \frac{dx}{\sin^2 \frac{x}{2} + 3 \cos^2 \frac{x}{2}} \\ &= \int_0^{\pi/2} \frac{\sec^2 \frac{x}{2} dx}{\tan^2 \frac{x}{2} + 3} \\ &= \int_0^1 \frac{2dz}{z^2 + (\sqrt{3})^2} \\ &= 2 \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{z}{\sqrt{3}} \right]_0^1 \\ &= 2 \left[\frac{1}{\sqrt{3}} \tan^{-1} \frac{1}{\sqrt{3}} - 0 \right] \\ &= 2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{\pi}{6} = \frac{\pi}{3\sqrt{3}} \quad (\text{Ans}) \end{aligned}$$

$$\begin{aligned} & \text{Let, } z = \tan \frac{x}{2} \\ & \Rightarrow 2dz = \sec^2 \frac{x}{2} dx \\ & \text{And } x \rightarrow \frac{\pi}{2} \Rightarrow z \rightarrow 1 \\ & x \rightarrow 0 \Rightarrow z \rightarrow 0 \end{aligned}$$

75. Integrate: $\int_0^{\pi} \frac{x \sin x dx}{1+\cos^2 x}$

Solution:

$$\begin{aligned} & \text{Let, } I = \int_0^{\pi} \frac{x \sin x dx}{1+\cos^2 x} \\ &= \int_0^{\pi} \frac{(\pi-x) \sin(\pi-x) dx}{1+\cos^2(\pi-x)} \\ &= \pi \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx - \int_0^{\pi} \frac{x \sin x}{1+\cos^2 x} dx \\ &\therefore I = \pi \int_0^{\pi} \frac{\sin x}{1+\cos^2 x} dx - I \\ &\text{or, } 2I = \pi \int_0^{\pi} \frac{\sin x dx}{1+\cos^2 x} \quad \begin{array}{l} \text{Let, } -\cos x = z \\ x \rightarrow 0 \Rightarrow z \rightarrow -1 \\ x \rightarrow \pi \Rightarrow z \rightarrow 1 \\ \therefore \sin x dx = dz \end{array} \\ &\text{or, } 2I = \pi \int_{-1}^1 \frac{dz}{1+z^2} \\ &\text{or, } 2I = \pi [\tan^{-1} z]_{-1}^1 \\ &\text{or, } 2I = \pi \left[\frac{\pi}{4} - \left(-\frac{\pi}{4}\right) \right] \\ &\text{or, } 2I = \pi \left[\frac{\pi}{4} + \frac{\pi}{4} \right] \\ &\text{or, } 2I = \frac{2\pi^2}{4} \text{ or, } I = \frac{\pi^2}{4} \quad (\text{Ans}) \end{aligned}$$

76. Integrate: $\int_0^1 \frac{dx}{\sqrt{1-x^2}}$

Solution:

$$\begin{aligned} & \int_0^1 \frac{dx}{\sqrt{1-x^2}} = [\sin^{-1} x]_0^1 \\ &= \sin^{-1} 1 - \sin^{-1} 0 \\ &= \frac{\pi}{2} - 0 = \frac{\pi}{2} \quad (\text{Ans}) \end{aligned}$$

77. The value of the integral $\int_0^1 \frac{dx}{1+x^2}$ is- (a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$ (c) 0

(d) None of these

Solution:

$$\begin{aligned} \int_0^1 \frac{dx}{1+x^2} &= [\tan^{-1} x]_0^1 \\ &= \tan^{-1}(1) - \tan^{-1} 0 = \frac{\pi}{4} - 0 = \frac{\pi}{4} \end{aligned}$$

Answer: (b)

78. Integrate: $\int_0^\pi \cos^3 x \, dx$

Solution:

$$\begin{aligned} \text{Let, } I &= \int_0^\pi \cos^3 x \, dx \\ \Rightarrow I &= \int_0^\pi \cos^3(\pi - x) \, dx \\ \Rightarrow I &= -\int_0^\pi \cos^3 x \, dx \\ \Rightarrow I &= -I, \Rightarrow 2I = 0, \Rightarrow I = 0 \text{ (Ans)} \end{aligned}$$

79. What is the value of $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \, dx$?

Solution:

$$\begin{aligned} \text{Let, } f(x) &= \sin^7 x \\ \Rightarrow f(-x) &= \sin^7(-x) \\ \Rightarrow f(-x) &= -\sin^7 x \\ \Rightarrow f(-x) &= -f(x) \text{ [Odd function]} \\ \text{As } \sin^7 x &\text{ is an odd function} \\ \therefore \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^7 x \, dx &= 0 \text{ (Ans)} \end{aligned}$$

80. Evaluate: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x \cos^5 x \, dx$.

Solution:

$$\begin{aligned} \text{Let, } f(x) &= \sin^9 x \cos^5 x \\ \Rightarrow f(-x) &= \sin^9(-x) \cos^5(-x) \\ \Rightarrow f(-x) &= -\sin^9 x \cos^5 x = -f(x) \\ \therefore f(x) &\text{ is an odd function.} \\ \text{So, } \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sin^9 x \cos^5 x \, dx &= 0 \text{ (Ans)} \end{aligned}$$

81. Evaluate: $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \sin x^3 \, dx$

Solution:

$$\begin{aligned} \text{Let, } f(x) &= x^2 \sin x^3 \\ \Rightarrow f(-x) &= (-x)^2 \sin(-x)^3 \\ \Rightarrow f(-x) &= -x^2 \sin x^3 = -f(x) \\ \therefore f(x) &\text{ is an odd function.} \\ \text{Since } f(x) &\text{ is odd, and we are integrating over a} \\ &\text{symmetric interval } \left[-\frac{\pi}{2}, \frac{\pi}{2} \right] \\ \text{So, } \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} x^2 \sin x^3 \, dx &= 0 \text{ (Ans)} \end{aligned}$$

82. What is the value of $\int_{-\pi}^{\pi} \frac{x}{1+\cos^2 x} \, dx$?

Solution:

$$\begin{aligned} \text{Let, } f(x) &= \frac{x}{1+\cos^2 x} \\ \Rightarrow f(-x) &= \frac{-x}{1+\cos^2(-x)} \\ \Rightarrow f(-x) &= -\frac{x}{1+\cos^2 x} \\ \Rightarrow f(-x) &= -f(x) \text{ [Odd function]} \\ \text{As } \frac{x}{1+\cos^2 x} &\text{ is an odd function} \\ \therefore \int_{-\pi}^{\pi} \frac{x}{1+\cos^2 x} \, dx &= 0 \text{ (Ans)} \end{aligned}$$

83. Integrate: $\int_3^5 \frac{x^2}{x^2-4} \, dx$

$$\begin{aligned} \text{Solution: } \int_3^5 \frac{x^2}{x^2-4} \, dx &= \int_3^5 \frac{x^2-4+4}{x^2-4} \, dx \\ &= \int_3^5 1 \, dx + \int_3^5 \frac{4}{x^2-2^2} \, dx \\ &= \left[x + 4 \cdot \frac{1}{2 \cdot 2} \log \left| \frac{x-2}{x+2} \right| \right]_3^5 \\ &= \left[\left(5 + 4 \cdot \frac{1}{2 \cdot 2} \log \left| \frac{5-2}{5+2} \right| \right) - \left(3 + 4 \cdot \frac{1}{2 \cdot 2} \log \left| \frac{3-2}{3+2} \right| \right) \right] \\ &= 2 + \log \left(\frac{3}{5} \right) - \log \left(\frac{1}{5} \right) \\ &= 2 + \log \left(\frac{3 \times 5}{7 \times 1} \right) = 2 + \log \left(\frac{15}{7} \right) \text{ (Ans)} \end{aligned}$$

84. Evaluate: $\int_0^{\frac{\pi}{4}} \log(1 + \tan x) \, dx$

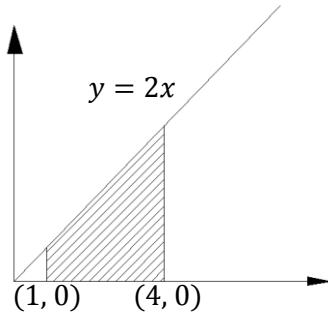
Solution:

$$\begin{aligned} \text{Let, } I &= \int_0^{\frac{\pi}{4}} \log(1 + \tan x) \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \tan \left(\frac{\pi}{4} - x \right) \right] \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \frac{\tan \frac{\pi}{4} - \tan x}{1 + \tan \frac{\pi}{4} \tan x} \right] \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[1 + \frac{1 - \tan x}{1 + \tan x} \right] \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[\frac{1 + \tan x + 1 - \tan x}{1 + \tan x} \right] \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log \left[\frac{2}{1 + \tan x} \right] \, dx \\ \Rightarrow I &= \int_0^{\frac{\pi}{4}} \log 2 \, dx - \int_0^{\frac{\pi}{4}} \log(1 + \tan x) \, dx \\ \Rightarrow I &= \log 2 \int_0^{\frac{\pi}{4}} dx - I \\ \Rightarrow 2I &= \log 2 [x]_0^{\frac{\pi}{4}} = \frac{\pi}{4} \log 2 \\ \Rightarrow I &= \frac{\pi}{8} \log 2 \text{ (Ans)} \end{aligned}$$

85. The area bounded by straight line $y = 2x$, above x axis from $x = 1$ to $x = 4$ is:

- (a) 16 sq. unit
(b) 15 sq. unit
(c) 7.5 sq. unit
(d) None of these

Solution:



Given, $y = 2x$

∴ Area bounded given equations

$$= \int_1^4 y \, dx$$

$$= \int_1^4 2x \, dx$$

$$= \left[\frac{2x^2}{2} \right]_1^4 = 16 - 1 = 15 \text{ sq. unit}$$

Answer: (b)

86. The area of the triangle enclosed by the line $y = -x$, the x axis and $x = 4$ is

- (a) 4 (b) 6 (c) 10 (d) 8

Solution: Required area

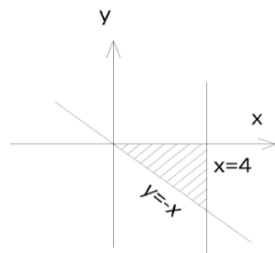
$$= \int_0^4 y \, dx$$

$$= - \int_0^4 x \, dx$$

$$= - \left[\frac{x^2}{2} \right]_0^4 = \frac{16}{2} = -8$$

∴ Area cannot be (-ve)

Answer: (d)



87. The area bounded by parabola $y^2 = x$, the line $x = 0$ and $y = 3$ is

- (a) 4 (b) 5 (c) 9 (d) 6

Solution:

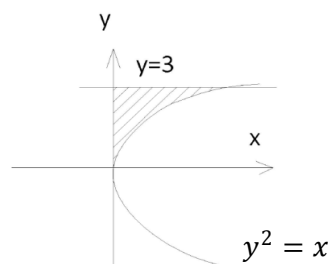
Required area

$$= \int_0^3 x \, dy$$

$$= \int_0^3 y^2 \, dy$$

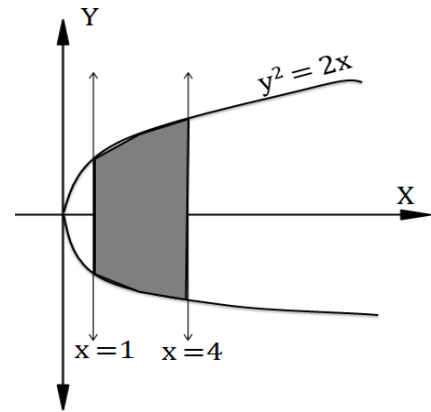
$$= \left[\frac{y^3}{3} \right]_0^3 = \frac{27}{3} = 9$$

Answer: (c)



88. Mark the area bounded by $y^2 = 2x$, $x = 1$ and $x = 4$ and find the enclosed area.

Solution:



Given equation of parabola $y^2 = 2x$

It can be written as $y = \pm\sqrt{2x}$

Computing the area in 1st quadrant ($y \geq 0$) and double the result the area

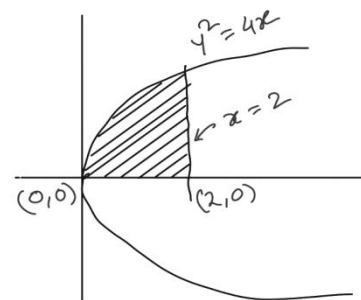
$$A = 2 \int_1^4 y \, dx = 2 \int_1^4 \sqrt{2x} \, dx = 2\sqrt{2} \int_1^4 x^{\frac{1}{2}} \, dx$$

$$= 2\sqrt{2} \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_1^4 = 2\sqrt{2} \times \frac{2}{3} \left(4^{\frac{3}{2}} - 1^{\frac{3}{2}} \right)$$

$$= \frac{4\sqrt{2}}{3} (8 - 1) = \frac{28\sqrt{2}}{3} \text{ Sq. unit (Ans)}$$

89. Find the area bounded by the parabola $y^2 = 4x$ above X -axis and ordinate at $x = 2$.

Solution: Given, $y^2 = 4x$, $\Rightarrow y = 2\sqrt{x}$



∴ Area bounded by the given equations is-

$$A = \int_0^2 y \, dx$$

$$= \int_0^2 2\sqrt{x} \, dx \quad [\because y = 2\sqrt{x}]$$

$$= 2 \int_0^2 x^{\frac{1}{2}} \, dx = 2 \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^2 = 2 \times \frac{2}{3} \times 2^{\frac{3}{2}}$$

$$= \frac{4}{3} \times 2\sqrt{2} = \frac{8\sqrt{2}}{3} \text{ Sq. unit (Ans)}$$

90. Show that the area bounded by the curves $y = x^2$ and $y = x$ is $\frac{1}{6}$.

Solution:

Given equations

$$y = x^2 \dots\dots\dots(1)$$

$$y = x \dots\dots\dots(2)$$

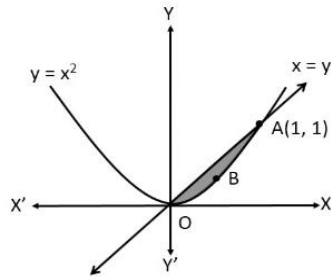
By solving (1) and (2)

$$x = 1, y = 1$$

∴ Area bounded

$$= \int_0^1 x \, dx - \int_0^1 x^2 \, dx$$

$$= \left[\frac{x^2}{2} - \frac{x^3}{3} \right]_0^1 = \frac{1}{2} - \frac{1}{3} = \frac{1}{6} \text{ (Proved)}$$



91. Shade the area enclosed by the parabola $y^2 = 4x$ and $x^2 = 4y$. Use method of integration to find this area.

Solution:

Given equations

$$y^2 = 4x \dots\dots\dots(1)$$

$$x^2 = 4y \dots\dots\dots(2)$$

By solving (1) and (2)

$$x = 4, y = 4$$

∴ Area enclosed

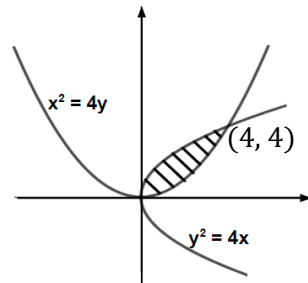
$$= \int_0^4 \sqrt{4x} \, dx - \int_0^4 \frac{x^2}{4} \, dx$$

$$= 2 \int_0^4 x^{\frac{1}{2}} \, dx - \frac{1}{4} \int_0^4 x^2 \, dx$$

$$= 2 \left[\frac{x^{\frac{3}{2}}}{\frac{3}{2}} \right]_0^4 - \frac{1}{4} \left[\frac{x^3}{3} \right]_0^4$$

$$= 2 \times \frac{2}{3} \times 4^{\frac{3}{2}} - \frac{1}{4} \times \frac{4^3}{3}$$

$$= \frac{32}{3} - \frac{16}{3} = \frac{16}{3} \text{ (Ans)}$$



92. Find the area bounded by the circle $x^2 + y^2 = 4$ and the straight line $y = x$, in the first quadrant.

Solution:

$$\text{Now, } x^2 + y^2 = 4$$

$$\text{Or, } x^2 + y^2 = 2^2$$

$$\text{Or, } y = \sqrt{4 - x^2}$$

So, radius of circle = 2

∴ Required area

$$= \frac{1}{2} \int_0^2 y \, dx$$

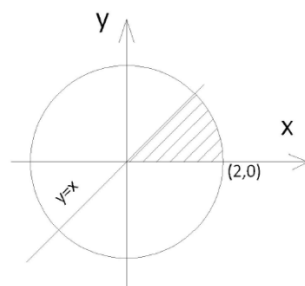
$$= \frac{1}{2} \int_0^2 \sqrt{4 - x^2} \, dx$$

$$= \frac{1}{2} \left[\frac{x\sqrt{4-x^2}}{2} + \frac{4}{2} \sin^{-1} \frac{x}{2} \right]_0^2$$

$$= \frac{1}{2} \left[\frac{2\sqrt{4-2^2}}{2} + \frac{4}{2} \sin^{-1} \frac{2}{2} \right]$$

$$= \frac{1}{2} \left[\frac{2\sqrt{4-2^2}}{2} + \frac{4}{2} \sin^{-1} \frac{2}{2} \right]$$

$$= \frac{1}{2} \times 2 \times \frac{\pi}{2} = \frac{\pi}{2} \text{ (Ans)}$$



93. What is the volume of revolution of the area enclosed by $y = \sqrt{\cos x}$, x axis and the lines $x = 0$, $x = \frac{\pi}{4}$ about x axis.

Solution:

Volume of revolution

$$= \int_0^{\frac{\pi}{4}} \pi y^2 \, dx$$

$$= \pi \int_0^{\frac{\pi}{4}} \cos x \, dx$$

$$= \pi [\sin x]_0^{\frac{\pi}{4}}$$

$$= \pi \left[\sin \frac{\pi}{4} - \sin 0 \right]$$

$$= \frac{\pi}{\sqrt{2}} \text{ (Ans)}$$

94. Find the volume of the solid generated by revolving the area bounded by the curve $y = \sin x$, $x = 0$, $x = \frac{\pi}{2}$ about the X axis.

Solution:

Required volume,

$$V = \pi \int_0^{\frac{\pi}{2}} y^2 \, dx$$

$$= \pi \int_0^{\frac{\pi}{2}} \sin^2 x \, dx$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} 2 \sin^2 x \, dx$$

$$= \frac{\pi}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) \, dx$$

$$= \frac{\pi}{2} \left[x - \frac{\sin 2x}{2} \right]_0^{\frac{\pi}{2}}$$

$$= \frac{\pi}{2} \left[\left(\frac{\pi}{2} - \frac{\sin 2 \cdot \frac{\pi}{2}}{2} \right) - 0 \right] = \frac{\pi^2}{4} \text{ (Ans)}$$

95. The line segment $x + y = 1$, $0 \leq y \leq 1$ is revolved about Y axis to generate a cone. Find the volume of the cone.

Solution: The given line equation is $x + y = 1$

$$\Rightarrow x = 1 - y$$

Since the region is revolved about the Y -axis, the radius of the disk at height y is:

$$x = 1 - y$$

∴ The volume of the cone is-

$$V = \pi \int_0^1 (1 - y)^2 \, dy$$

$$= \pi \int_0^1 (1 - 2y + y^2) \, dy$$

$$= \pi \left[y - y^2 + \frac{y^3}{3} \right]_0^1$$

$$= \pi \left[1 - 1^2 + \frac{1^3}{3} \right] = \frac{\pi}{3} \text{ (Ans)}$$

1. The order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^2 = x^2 \left(\frac{dy}{dx}\right)^3 \text{ is-}$$

- (a) 2 and 2
(b) 2 and 1
(c) 2 and 3
(d) none of these

Explanation: The order of a differential equation is defined to be that of the highest order derivative it contains. The degree of a differential equation is defined as the power to which the highest order derivative is raised. Therefore order and degree of given equation is 2 and 2 respectively.

Answer: (a)

2. The order and degree of the differential equation

$$\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^5 + y = 3 \sin x \text{ are- (a) 3 and 2 (b) 1 and 5 (c) 2 and 3 (d) None of these}$$

Solution: Given equation

$$\left(\frac{d^3y}{dx^3}\right)^2 + \left(\frac{dy}{dx}\right)^5 + y = 3 \sin x$$

Order = Highest order derivative = 3

Degree = Power of highest order derivative = 2

Answer: (a)

3. The order and degree of the differential equation

$$\left(\frac{d^2y}{dx^2}\right)^5 - x^3 \left(\frac{dy}{dx}\right)^3 = 0 \text{ are-}$$

- (a) 1 and 2 (b) 2 and 5 (c) 2 and 3 (d) none

$$\text{Solution: Given, } \left(\frac{d^2y}{dx^2}\right)^5 - x^3 \left(\frac{dy}{dx}\right)^3 = 0$$

∴ Order of the equation

= Highest order of derivative = 2

And degree of the equation

= Power of highest order derivative = 5

Answer: (b)

4. The order and degree of the differential equation

$$\left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}} = c \frac{d^2y}{dx^2} \text{ is-}$$

- (a) 2 and 1 (b) 2 and 2 (c) 1 and 2 (d) 2 and 3

$$\text{Solution: Given, } \left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^{\frac{3}{2}} = c \frac{d^2y}{dx^2}$$

$$\Rightarrow \left\{1 + \left(\frac{dy}{dx}\right)^2\right\}^3 = \left(c \frac{d^2y}{dx^2}\right)^2$$

∴ Order of the equation

= Highest order of derivative = 2

And degree of the equation

= Power of highest order derivative = 2

Answer: (b)

5. The differential equation whose general solution is $y = A \sin x + B \cos x$ is-

$$(a) \frac{d^2y}{dx^2} = y$$

$$(b) \frac{d^2y}{dx^2} = -y$$

$$(c) \frac{d^2y}{dx^2} = 0$$

- (d) none of these

Solution: Given, $y = A \sin x + B \cos x$

$$\Rightarrow \frac{dy}{dx} = A \cos x - B \sin x$$

$$\Rightarrow \frac{d}{dx} \left(\frac{dy}{dx}\right) = -A \sin x - B \cos x$$

$$\Rightarrow \frac{d^2y}{dx^2} = -(A \sin x + B \cos x)$$

$$\Rightarrow \frac{d^2y}{dx^2} = -y$$

Answer: (b)

6. General solution of the differential equation $\frac{dy}{dx} = 1$ is- (a) $y = c$ (b) $x = 0$ (c) $y = x$ (d) none of these

$$\text{Solution: Given, } \frac{dy}{dx} = 1, \Rightarrow dy = dx$$

Integrating both sides we get,

$$\int dy = \int dx$$

$$\Rightarrow y = x + c$$

Answer: (d)

7. Solution of the equation $xy \, dx = dy$ is

$$(a) xy = c$$

$$(b) x^2 + \log y = c$$

$$(c) \frac{1}{2} x^2 = \log y + c$$

- (d) none of these

Explanation: Given equation $xy \, dx = dy$

$$\Rightarrow x dx = \frac{dy}{y}, \Rightarrow \int x dx = \int \frac{dy}{y}$$

$$\Rightarrow \frac{x^2}{2} = \log y + c$$

Answer: (c)

8. General solution of the differential equation

$$\frac{dy}{dx} = \frac{1}{x} \text{ is-}$$

$$(a) xy = c$$

$$(b) y = x + c$$

$$(c) y = \log x + c$$

- (d) none of these

$$\text{Explanation: Given, } \frac{dy}{dx} = \frac{1}{x}, \Rightarrow dy = \frac{dx}{x}$$

$$\Rightarrow \int dy = \int \frac{dx}{x}$$

$$\Rightarrow y = \log x + c$$

Answer: (c)

9. Solution of the D.E. $\frac{dy}{dx} = \frac{y}{a}$ is-

- (a) $\frac{1}{2}y^2 = x + c$
 (b) $x = a \log y + c$
 (c) $x^2 = a \log y$
 (d) none of these

Solution: Given, $\frac{dy}{dx} = \frac{y}{a}$

$$\Rightarrow \frac{dy}{y} = \frac{1}{a} dx$$

$$\Rightarrow \int \frac{dy}{y} = \frac{1}{a} \int dx$$

$$\Rightarrow \log y = \frac{1}{a} x$$

$$\Rightarrow x = a \log y + c$$

Answer: (b)

10. Solve: $\frac{d^2y}{dx^2} = 0$

- (a) $y = x + c$
 (b) $y = ax + b$
 (c) $x = ab$
 (d) none of these

Solution: Given, $\frac{d^2y}{dx^2} = 0$

$$\Rightarrow \frac{dy_1}{dx} = 0 \quad \left[\because y_1 = \frac{dy}{dx} \right]$$

$$\Rightarrow \int dy_1 = \int 0 dx + a$$

$$\Rightarrow y_1 = a$$

$$\Rightarrow \frac{dy}{dx} = a$$

$$\Rightarrow \int dy = a \int dx + b$$

$$\Rightarrow y = ax + b$$

Answer: (b)

11. The differential equation of all straight lines having gradient 2 is -

- (a) $\frac{dy}{dx} = 0$
 (b) $\frac{dy}{dx} + 2 = 0$
 (c) $\frac{dy}{dx} = 2$
 (d) none of these

Explanation: General equation of straight line is

$$y = mx + c \quad [\text{Where } m = \text{gradient}]$$

$$\therefore \frac{dy}{dx} = m$$

$$\text{Given gradient } m = 2$$

$$\therefore \frac{dy}{dx} = 2$$

Answer: (c)

12. The differential equation of the family of circles with centre at the origin is-

- (a) $x dx + y dy = 0$
 (b) $y dx + x dy = 0$
 (c) $x^2 dx + y^2 dy = 0$
 (d) $x dy - y dx = 0$

Solution: The general equation of circle with centre at origin is $x^2 + y^2 = a^2$

Differentiate both side with respect to x we get,

$$2x + 2y \frac{dy}{dx} = 0$$

$$\Rightarrow x + y \frac{dy}{dx} = 0$$

$$\Rightarrow x dx + y dy = 0$$

Answer: (a)

13. The number of arbitrary constant in the general solution of $3 \frac{d^2y}{dx^2} + 5 \frac{dy}{dx} + x = 0$ is-

- (a) 1 (b) 2 (c) 3 (d) 4

Explanation: The Order of the given differential equation is 2, because $\frac{d^2y}{dx^2}$ is the highest differential appearing in the equation. So the number of arbitrary constants is 2.

Answer: (b)

14. If the differential equation $(2x^2 + 4y)dx + k(4x + y - 1) = 0$ is exact, then the value of k is-

- (a) 1 (b) 2 (c) -1 (d) -2

Solution: Given equation is

$$(2x^2 + 4y)dx + k(4x + y - 1) = 0$$

$$\text{Here, } M = 2x^2 + 4y$$

$$\therefore \frac{\partial M}{\partial y} = 0 + 4 = 4$$

$$\text{And, } N = k(4x + y - 1)$$

$$\therefore \frac{\partial N}{\partial x} = k(4 + 0) = 4k$$

As the given equation is exact

$$\therefore \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}, \Rightarrow 4 = 4k, \Rightarrow k = 1$$

Answer: (a)

15. The integrating factor of $\frac{dy}{dx} + 3x^2y = 6x$ is-

- (a) e^{3x} (b) e^{x^3} (c) x^3 (d) none of these

Solution: Given, $\frac{dy}{dx} + 3x^2y = 6x$

$$\Rightarrow \frac{dy}{dx} + Py = Q \quad [P = 3x^2, Q = 6x]$$

$$\therefore \text{Integrating Factor (IF)} = e^{\int P dx}$$

$$\Rightarrow \text{I.F.} = e^{\int 3x^2 dx}$$

$$= e^{3 \times \frac{x^3}{3}} = e^{x^3}$$

Answer: (b)

16. The Integrating factor (IF) of the differential equation $\frac{dy}{dx} + \frac{1}{x}y = \sin x$ is- (a) x (b) $\frac{1}{x}$ (c) $\log_e x$ (d) none of these

Solution: Given, $\frac{dy}{dx} + \frac{1}{x} \cdot y = \sin x$

$$\Rightarrow \frac{dy}{dx} + Py = Q \text{ [Where } P = \frac{1}{x} \text{ and } Q = \sin x]$$

As, it is a linear differential equation

$\therefore$ Integrating Factor

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

Answer: (a)

17. I.F. of equation $\frac{dy}{dx} - \frac{1}{x} \cdot y = x^2$

(a) x

(b) $\frac{1}{x}$

(c) $-x$

(d) none of these

Explanation: Given, $\frac{dy}{dx} - \frac{1}{x} \cdot y = x^2$

$$\Rightarrow \frac{dy}{dx} + Py = Q \text{ [Where } P = -\frac{1}{x} \text{ and } Q = x^2]$$

As, it is a linear differential equation

$\therefore$ Integrating Factor

$$\begin{aligned} \text{I.F.} &= e^{\int P dx} = e^{\int -\frac{1}{x} dx} = e^{-\log x} \\ &= e^{\log x^{-1}} \\ &= x^{-1} = \frac{1}{x} \end{aligned}$$

Answer: (b)

18. Integration factor of equation $\frac{dy}{dx} + \frac{1}{x} \cdot y = e^x$ is-

(a) x

(b) $\frac{1}{x}$

(c) $\log x$

(d) e^x

Solution: Given, $\frac{dy}{dx} + \frac{1}{x} \cdot y = e^x$

$$\Rightarrow \frac{dy}{dx} + Py = Q \text{ [Where } P = \frac{1}{x} \text{ and } Q = e^x]$$

As, it is a linear differential equation

$\therefore$ Integrating Factor

$$\text{I.F.} = e^{\int P dx} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

Answer: (a)

19. The I.F. of the equation $\cos x \frac{dy}{dx} + y \sin x = 1$ is-

(a) $\cos x$

(b) $-\sec^2 x$

(c) $\sec x$

(d) $\log(\sec x)$

Solution: Given equation $\cos x \frac{dy}{dx} + y \sin x = 1$

$$\Rightarrow \frac{dy}{dx} + \frac{\sin x}{\cos x} y = \frac{1}{\cos x}$$

$$\Rightarrow \frac{dy}{dx} + \tan x \times y = \sec x$$

$$\Rightarrow \frac{dy}{dx} + Py = Q \text{ [Where } P = \tan x \text{ and } Q = \sec x]$$

As, it is a linear differential equation

$\therefore$ Integrating Factor

$$\text{I.F.} = e^{\int P dx} = e^{\int \tan x dx} = e^{\log(\sec x)} = \sec x$$

Answer: (c)

20. C.F. of differential equation $(D^2 - 2D + 1)y = e^x$ is-

(a) $(c_1 + c_2)e^x$

(b) $(c_1 + xc_2)e^x$

(c) $(c_1 + c_2)e^{-x}$

(d) none of these

Solution: Given equation $(D^2 - 2D + 1)y = e^x$

$$\therefore \text{A.E. is } m^2 - 2m + 1 = 0$$

$$\Rightarrow (m - 1)^2 = 0$$

$$\Rightarrow m - 1 = 0$$

$$\Rightarrow m = 1$$

$$\therefore \text{C.F.} = (c_1 + xc_2)e^{mx} = (c_1 + xc_2)e^x$$

Answer: (b)

21. The complementary function (CF) of

$$\frac{d^2y}{dx^2} - 16 = xe^{3x} \text{ is } \underline{\hspace{2cm}}.$$

Solution: Homogeneous part of given equation-

$$\frac{d^2y}{dx^2} - 16 = 0$$

The auxiliary equation is:

$$m^2 - 16 = 0$$

$$\Rightarrow m^2 = 4^2, \Rightarrow m = +1, -1$$

The complementary function is:

$$\text{CF} = c_1 e^{4x} + c_2 e^{-4x} \text{ (Ans)}$$

Where c_1 and c_2 are arbitrary constants.

22. C.F. of the equation $\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^x$ is

- (a) $c_1e^{-2x} + c_2e^{-2x}$
 (b) $c_1e^{2x} + c_2e^{2x}$
 (c) $(c_1 + xc_2)e^{-2x}$
 (d) none of these

Solution: Given equation,

$$\frac{d^2y}{dx^2} + 4\frac{dy}{dx} + 4y = e^x$$

$$\Rightarrow (D^2 + 4D + 4)y = e^x$$

$$\text{A.E. is } m^2 + 4m + 4 = 0$$

$$\Rightarrow m^2 + 2 \cdot m \cdot 2 + 2^2 = 0$$

$$\Rightarrow (m + 2)^2 = 0$$

$$\Rightarrow m + 2 = 0, \Rightarrow m = -2$$

$$\therefore \text{C.F.} = (c_1 + xc_2)e^{-2x}$$

Answer: (c)

23. Complementary function of the differential equation $\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x$ is-

- (a) $(c_1 + c_2)e^x$ (b) $(c_1 + c_2x)e^x$ (c) $(c_1 + c_2)e^{-x}$
 (d) $(c_1 + c_2x)e^{-x}$

Solution: Given equation,

$$\frac{d^2y}{dx^2} - 2\frac{dy}{dx} + y = e^x$$

$$\Rightarrow (D^2 - 2D + 1)y = e^x$$

$$\text{A.E. is } m^2 - 2m + 1 = 0$$

$$\Rightarrow (m - 1)^2 = 0$$

$$\Rightarrow m = 1$$

This gives repeated root

$$\therefore \text{C.F.} = (c_1 + c_2x)e^x$$

Answer: (b)

24. C.F. of the equation $\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x$ is -

- (a) $c_1e^{2x} + c_2e^x$
 (b) $(c_1 + xc_2)e^{2x}$
 (c) $c_1e^{-2x} + c_2e^{-x}$
 (d) none of these

Solution: Given equation,

$$\frac{d^2y}{dx^2} - 3\frac{dy}{dx} + 2y = x$$

$$\Rightarrow (D^2 - 3D + 2)y = x$$

$$\text{A.E. is } m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m - 2) - 1(m - 2) = 0$$

$$\Rightarrow (m - 2)(m - 1) = 0$$

$$\Rightarrow m = 1, 2$$

$$\therefore \text{C.F.} = c_1e^{2x} + c_2e^x$$

Answer: (a)

25. If the roots of Auxiliary Equation (AE) of a linear differential equation of 2nd order with constant coefficient be $(2 \pm 3i)$, find the complementary function(CF).

Solution: If the roots of AE are complex conjugate of form: $r = \alpha \pm \beta i$

Then CF = $e^{\alpha x} (c_1 \cos \beta x + c_2 \sin \beta x)$

According to given condition $\alpha = 2, \beta = 3$

$$\therefore \text{CF} = e^{2x} (c_1 \cos 3x + c_2 \sin 3x) \quad (\text{Ans})$$

26. $y = c_1 + c_2e^x$ is the solution of the differential equation _____.

Solution: Given equation $y = c_1 + c_2e^x$

$$\therefore \frac{dy}{dx} = c_2e^x \text{ and } \frac{d^2y}{dx^2} = c_2e^x$$

$$\Rightarrow \frac{d^2y}{dx^2} = \frac{dy}{dx}$$

$$\Rightarrow \frac{d^2y}{dx^2} - \frac{dy}{dx} = 0 \quad (\text{Ans})$$

27. P.I. of $(D^2 - 3D + 2)y = e^{3x}$ is-

- (a) $\frac{e^{2x}}{3}$
 (b) $\frac{e^{3x}}{2}$
 (c) $\frac{e^{3x}}{3}$

(d) none of these

Explanation: Given, $(D^2 - 3D + 2)y = e^{3x}$

$$\begin{aligned} \text{P.I.} &= \frac{Q}{D^2 - 3D + 2} \\ &= \frac{e^{3x}}{3^2 - 3 \times 3 + 2} = \frac{e^{3x}}{2} \end{aligned}$$

Answer: (b)

28. P.I. of $(D^2 - 4)y = \sin 3x$ is-

- (a) $\frac{\sin 3x}{13}$
 (b) $-\frac{\sin 3x}{13}$
 (c) $-\frac{3 \sin 3x}{13}$

(d) none of these

Solution: Given, $(D^2 - 4)y = \sin 3x$

Here $a = 3$

$$\begin{aligned} \therefore \text{P.I.} &= \frac{\sin 3x}{D^2 - 4} \\ &= \frac{\sin 3x}{-3^2 - 4} \quad [\text{Replacing } D^2 \text{ by } -a^2] \\ &= \frac{\sin 3x}{-13} \end{aligned}$$

Answer: (b)

29. Find $\frac{1}{D^2+5}(\sin 2x)$.

Solution:

$$\text{Formula: } \frac{1}{D^2+a^2} \sin bx = \frac{\sin bx}{a^2+b^2}$$

Here in this question, $a^2 = 5$ and $b = 2$

$$\therefore \frac{1}{D^2+5}(\sin 2x) = \frac{\sin 2x}{5+2^2} = \frac{\sin 2x}{9} \quad (\text{Ans})$$

30. Particular Integral (PI) of the differential equation

$$\frac{d^2y}{dx^2} + 4y = 5e^{2x} \text{ is } \underline{\hspace{2cm}}.$$

Solution: Given, $\frac{d^2y}{dx^2} + 4y = 5e^{2x}$

$$\Rightarrow (D^2 + 4)y = 5e^{2x}$$

$$\therefore \text{PI} = \frac{5e^{2x}}{D^2+4} = \frac{5e^{2x}}{2^2+4} = \frac{5e^{2x}}{8} \quad (\text{Ans})$$

31. P.I. of the equation $\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x}$

(a) $\frac{e^{4x}}{4}$

(b) $\frac{e^{4x}}{2}$

(c) $\frac{e^{-4x}}{4}$

(d) none of these

Solution: Given equation,

$$\frac{d^2y}{dx^2} - 5\frac{dy}{dx} + 6y = e^{4x}$$

$$\Rightarrow (D^2 - 5D + 6)y = e^{4x}$$

$$\therefore \text{P.I.} = \frac{e^{4x}}{D^2-5D+6} = \frac{e^{4x}}{4^2-5 \times 4+6} = \frac{e^{4x}}{2}$$

Answer: (b)

32. Find the differential equation of all straight lines passing through origin.

Solution: Let $y = mx$ be the required equation.

Differentiating both side with respect to x we get,

$$\frac{dy}{dx} = m$$

$$\therefore \text{The required equation is } y = \frac{dy}{dx}x \quad (\text{Ans})$$

33. The degree of the homogenous function

$u(x, y) = \frac{xy}{x-y}$ is-

(a) 1

(b) 2

(c) $\frac{1}{2}$

(d) 0

Explanation: In a homogeneous function the degree of all the terms are equal. Here it can be clearly seen that degree of x and y are same which is equal to 1. Hence, the function is of degree 1.

Answer: (a)

34. Solve the equation $ydx - xdy = 0$.

Solution: $ydx - xdy = 0$

$$\Rightarrow ydx = xdy, \Rightarrow \frac{dx}{x} = \frac{dy}{y}$$

$$\Rightarrow \int \frac{dx}{x} = \int \frac{dy}{y} + \log c$$

$$\Rightarrow \log x = \log y + \log c$$

$$\Rightarrow \log x - \log y = \log c$$

$$\Rightarrow \log\left(\frac{x}{y}\right) = \log c$$

$$\Rightarrow \frac{x}{y} = c \quad (\text{Ans})$$

c is the integrating constant.

35. Solve: $(x + 2y)\frac{dy}{dx} = y$

Solution: Let $v = \frac{y}{x}, \Rightarrow y = vx$

$$\therefore \frac{dy}{dx} = v + x \frac{dv}{dx}$$

We are given $(x + 2y)\frac{dy}{dx} = y$

$$\Rightarrow \frac{dy}{dx} = \frac{y}{x+2y}$$

$$\Rightarrow v + x \frac{dv}{dx} = \frac{vx}{x+2vx} = \frac{vx}{x(1+2v)}$$

$$\Rightarrow x \frac{dv}{dx} = \frac{v}{1+2v} - v = \frac{v-v-2v^2}{1+2v} = \frac{-2v^2}{1+2v}$$

$$\Rightarrow \frac{1+2v}{2v^2} dv = \frac{dx}{x}$$

$$\Rightarrow \left(\frac{1}{2v^2} + \frac{1}{v}\right) dv = \frac{dx}{x}$$

Integrating both sides

$$\int \left(\frac{1}{2v^2} + \frac{1}{v}\right) dv = \int \frac{dx}{x}$$

$$\Rightarrow -\frac{1}{2v} + \log v = \log x + c$$

$$\Rightarrow -\frac{x}{2y} + \log\left(\frac{y}{x}\right) = \log x + c \quad (\text{Ans})$$

c is the integrating constant.

36. Solve: $\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$

Solution:

$$\frac{dy}{dx} + \frac{y}{x} = \frac{y^2}{x^2}$$

$$\Rightarrow v + x \frac{dv}{dx} + v = v^2$$

$$\Rightarrow x \frac{dv}{dx} = v^2 - 2v$$

$$\Rightarrow \frac{dv}{v^2-2v} = \frac{dx}{x}$$

$$\Rightarrow \frac{1}{2} \int \frac{v-(v-2)}{v(v-2)} dv = \int \frac{dx}{x} + \log c$$

$$\Rightarrow \frac{1}{2} \int \left\{ \frac{1}{(v-2)} - \frac{1}{v} \right\} dv = \log x + \log c$$

$$\Rightarrow \frac{1}{2} [\log(v-2) - \log v] = \log cx$$

$$\Rightarrow \log \frac{(v-2)}{v} = 2 \log cx$$

$$\Rightarrow \log \frac{(v-2)}{v} = \log c^2 x^2, \Rightarrow \frac{v-2}{v} = c^2 x^2$$

$$\Rightarrow \frac{y}{x} - 2 = c^2 x^2 \times \frac{y}{x}$$

$$\Rightarrow y - 2x = c^2 x^2 y \quad (\text{Ans})$$

Let, $\frac{y}{x} = v$

$$\Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

37. Solve: $\frac{ydy+xdx}{x^2+y^2} = 0$

Solution: Given, $\frac{ydy+xdx}{x^2+y^2} = 0$

$$\Rightarrow ydy + xdx = 0$$

$$\Rightarrow \int y dy + \int x dx = c$$

$$\Rightarrow \frac{y^2}{2} + \frac{x^2}{2} = c$$

$$\Rightarrow x^2 + y^2 = 2c \quad (\text{Ans})$$

c is an arbitrary integration constant.

38. Solve: $\frac{dy}{dx} = \frac{y}{x} (\log \frac{y}{x} + 1)$

Solution:

$$\frac{dy}{dx} = \frac{y}{x} (\log \frac{y}{x} + 1)$$

$$\Rightarrow v + x \frac{dv}{dx} = v(\log v + 1)$$

$$\Rightarrow v + x \frac{dv}{dx} = v \log v + v$$

$$\Rightarrow x \frac{dv}{dx} = v \log v$$

$$\Rightarrow \frac{dv}{v \log v} = \frac{dx}{x}$$

$$\Rightarrow \int \frac{dv}{v \log v} = \int \frac{dx}{x} + \log c \quad [\log c \text{ taken as constant}]$$

$$\Rightarrow \log(\log v) = \log x + \log c$$

$$\Rightarrow \log(\log v) = \log cx$$

$$\Rightarrow \log v = cx$$

$$\Rightarrow \log \left(\frac{y}{x} \right) = cx \quad (\text{Ans})$$

39. Solve: $\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$

Solution:

$$\frac{dy}{dx} = \frac{y}{x} + \tan \frac{y}{x}$$

$$\Rightarrow v + x \frac{dv}{dx} = v + \tan v$$

$$\Rightarrow x \frac{dv}{dx} = \tan v$$

$$\Rightarrow \frac{dv}{\tan v} = \frac{dx}{x}$$

$$\Rightarrow \int \cot v dv = \int \frac{dx}{x} + \log c$$

$[\log c \text{ taken as constant}]$

$$\Rightarrow \log(\sin v) = \log x + \log c$$

$$\Rightarrow \log(\sin v) = \log cx$$

$$\Rightarrow \sin v = cx$$

$$\Rightarrow \sin \left(\frac{y}{x} \right) = cx$$

$$\Rightarrow \frac{y}{x} = \sin^{-1}(cx)$$

$$\Rightarrow y = x \sin^{-1}(cx) \quad (\text{Ans})$$

$$\text{Let, } \frac{y}{x} = v$$

$$\Rightarrow y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

40. Solve: $xdy - ydx = (x^2 + y^2)dx$

Solution:

$$xdy - ydx = (x^2 + y^2)dx$$

$$\Rightarrow x \frac{dy}{dx} - y = x^2 + y^2$$

$$\Rightarrow x^2 \frac{dv}{dx} = x^2 + v^2 x^2$$

$$\Rightarrow x^2 \frac{dv}{dx} = x^2(1 + v^2)$$

$$\Rightarrow \frac{dv}{1+v^2} = dx$$

$$\Rightarrow \int \frac{dv}{1+v^2} = \int dx + c$$

$$\Rightarrow \tan^{-1} v = x + c$$

$$\Rightarrow v = \tan(x + c)$$

$$\Rightarrow \frac{y}{x} = \tan(x + c)$$

$$\Rightarrow y = x \tan(x + c) \quad (\text{Ans})$$

c is arbitrary constant

$$\text{Let, } y = vx$$

$$\Rightarrow \frac{dy}{dx} = v + x \frac{dv}{dx}$$

$$\Rightarrow x \frac{dy}{dx} = vx + x^2 \frac{dv}{dx}$$

$$\Rightarrow x \frac{dy}{dx} - y = x^2 \frac{dv}{dx}$$

41. Solve: $(xdx + ydy) + \frac{xdy-ydx}{x^2+y^2} = 0$

Solution:

$$(xdx + ydy) + \frac{xdy-ydx}{x^2+y^2} = 0$$

$$\Rightarrow \frac{1}{2}(2xdx + 2ydy) + \frac{\frac{xdy-ydx}{x^2}}{\frac{x^2+y^2}{x^2}} = 0$$

$$\Rightarrow \frac{1}{2}d(x^2 + y^2) + \frac{d\left(\frac{y}{x}\right)}{1+\left(\frac{y}{x}\right)^2} = 0$$

$$\Rightarrow \frac{1}{2} \int d(x^2 + y^2) + \int \frac{d\left(\frac{y}{x}\right)}{1+\left(\frac{y}{x}\right)^2} = c$$

$$\Rightarrow \frac{1}{2}(x^2 + y^2) + \tan^{-1} \left(\frac{y}{x} \right) = c \quad (\text{Ans})$$

Where c is an arbitrary constant

42. Solve: $\frac{dy}{dx} + y \cot x = 2 \cos x$

Solution:

$$\frac{dy}{dx} + y \cot x = 2 \cos x$$

$$\Rightarrow \frac{dy}{dx} + Py = Q \quad [P = \cot x, Q = 2 \cos x]$$

$$\therefore \text{I. F.} = e^{\int P dx} = e^{\int \cot x dx} = e^{\log |\sin x|} = \sin x$$

$\therefore$ The required solution is

$$\frac{d}{dx}(y \times \text{I. F.}) = Q \times \text{I. F.}$$

$$\Rightarrow \frac{d}{dx}(y \sin x) = 2 \cos x \sin x$$

$$\Rightarrow \int d(y \sin x) = \int 2 \cos x \sin x dx + c$$

$$\Rightarrow y \sin x = \int \sin 2x dx + c$$

$$\Rightarrow y \sin x = -\frac{\cos 2x}{2} + c$$

$$\Rightarrow y \sin x + \frac{1}{2} \cos 2x = c \quad (\text{Ans})$$

43. Solve: $x \frac{dy}{dx} + y = xe^x$, given $y = 1$ when $x = 0$.

Solution:

$$\begin{aligned} x \frac{dy}{dx} + y &= xe^x \\ \Rightarrow \frac{dy}{dx} + \frac{1}{x} \cdot y &= e^x \dots (i) \\ \Rightarrow \frac{dy}{dx} + Py &= Q \end{aligned}$$

So, it is a linear equation in y .

$$\therefore \text{I.F.} = e^{\int \frac{1}{x} dx} = e^{\log x} = x$$

Multiplying equation (i) by x we get,

$$\begin{aligned} x \frac{dy}{dx} + y &= xe^x \\ \Rightarrow \frac{d}{dx}(xy) &= xe^x \\ \Rightarrow d(xy) &= xe^x dx \\ \Rightarrow \int d(xy) &= \int xe^x dx \\ \Rightarrow xy &= xe^x - e^x + c \\ \Rightarrow 0 &= 0 - 1 + c \quad [\text{Given, } y = 1, \text{ when } x = 0] \\ \Rightarrow c &= 1 \end{aligned}$$

$\therefore$ The required solution is

$$xy = xe^x - e^x + 1 \quad (\text{Ans})$$

44. Solve: $\frac{dy}{dx} + y = x$.

Solution: $\frac{dy}{dx} + y = x$

$$\begin{aligned} \Rightarrow \frac{dy}{dx} &= x - y \\ \Rightarrow 1 - \frac{dz}{dx} &= z & \text{Let, } z = x - y \\ \Rightarrow 1 - z &= \frac{dz}{dx} & \Rightarrow \frac{dz}{dx} = 1 - \frac{dy}{dx} \\ \Rightarrow dx &= \frac{dz}{1-z} & \Rightarrow \frac{dy}{dx} = 1 - \frac{dz}{dx} \\ \Rightarrow \int dx &= -\int \frac{dz}{z-1} + c \\ \Rightarrow x &= -\log(z-1) + c \\ \Rightarrow x + \log(x-y-1) &= c \quad (\text{Ans}) \\ c &\text{ is an arbitrary constant.} \end{aligned}$$

45. Solve: $\frac{d^2y}{dx^2} - 4 = 0$

Solution: $\frac{d^2y}{dx^2} - 4 = 0$

$$\begin{aligned} \Rightarrow \frac{d^2y}{dx^2} &= 4 \\ \Rightarrow \frac{d}{dx} \left(\frac{dy}{dx} \right) &= 4 \\ \Rightarrow d \left(\frac{dy}{dx} \right) &= 4dx \\ \Rightarrow \int d \left(\frac{dy}{dx} \right) &= 4 \int dx + c_1 \\ \Rightarrow \frac{dy}{dx} &= 4x + c_1 \\ \Rightarrow \int dy &= \int (4x + c_1) dx + c_2 \\ \Rightarrow y &= 4 \times \frac{x^2}{2} + c_1x + c_2 \\ \Rightarrow y &= 2x^2 + c_1x + c_2 \quad (\text{Ans}) \\ c_1 \text{ and } c_2 &\text{ are integrating constant.} \end{aligned}$$

46. The equation $(x^2 + y^2)dx - 2xydy = 0$ is a homogeneous equation true or false justify.

Solution:

$$\begin{aligned} (x^2 + y^2)dx - 2xydy &= 0 \\ \Rightarrow 2xydy &= (x^2 + y^2)dx \\ \Rightarrow \frac{dy}{dx} &= \frac{(x^2 + y^2)}{2xy} \\ &= \frac{1}{2} \left(\frac{x^2}{xy} + \frac{y^2}{xy} \right) \\ &= \frac{1}{2} \left(\frac{x}{y} + \frac{y}{x} \right) = \frac{1}{2} \left\{ \left(\frac{y}{x} \right)^{-1} + \left(\frac{y}{x} \right) \right\} \\ \therefore \frac{dy}{dx} &= f \left(\frac{y}{x} \right) \end{aligned}$$

So, the given equation is homogeneous. (True)

47. Check the exactness: $(xy - 2x^2) + xy \frac{dy}{dx} = 0$

Solution: Given equation,

$$\begin{aligned} (xy - 2x^2) + xy \frac{dy}{dx} &= 0 \\ \Rightarrow (xy - 2x^2)dx + xydy &= 0 \\ \text{Comparing with } Mdx + Ndy &= 0 \text{ we get,} \\ M &= xy - 2x^2 \\ \therefore \frac{\partial M}{\partial y} &= x \end{aligned}$$

Again, $N = xy$

$$\therefore \frac{\partial N}{\partial x} = y$$

$$\text{So, } \frac{\partial M}{\partial y} \neq \frac{\partial N}{\partial x}$$

Hence, the given equation is not exact type.

48. Test the equation

$$(2ax + by)ydx + (ax + 2by)xdy = 0$$

for exact type.

Solution: Given equation,

$$\begin{aligned} (2ax + by)ydx + (ax + 2by)xdy &= 0 \\ \text{Comparing with } Mdx + Ndy &= 0 \text{ we get,} \\ M &= (2ax + by)y = 2axy + by^2 \\ \therefore \frac{\partial M}{\partial y} &= 2ax + 2by = 2(ax + by) \end{aligned}$$

Again, $N = (ax + 2by)x = ax^2 + 2bxy$

$$\therefore \frac{\partial N}{\partial x} = 2ax + 2by = 2(ax + by)$$

$$\text{So, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

Hence, the given equation is exact type.

49. Find the differential equation whose general solutions are $y = Ae^x + Be^{-x}$.

Solution: $y = Ae^x + Be^{-x}$

$$\therefore \frac{dy}{dx} = Ae^x - Be^{-x}$$

Again differentiating we get,

$$\begin{aligned} \frac{d^2y}{dx^2} &= Ae^x + Be^{-x} \\ \Rightarrow \frac{d^2y}{dx^2} &= y, \Rightarrow \frac{d^2y}{dx^2} - y = 0 \quad (\text{Ans}) \end{aligned}$$

50. If the differential equation

$(2x^2 + 4y)dx + k(4x + y - 1)dy = 0$ is exact then value of k is ____.

Solution: Given equation,

$$(2x^2 + 4y)dx + k(4x + y - 1)dy = 0$$

Comparing with $Mdx + Ndy = 0$ we get,

$$M = 2x^2 + 4y \quad \therefore \frac{\partial M}{\partial y} = 4$$

$$\text{And, } N = k(4x + y - 1)$$

$$\therefore \frac{\partial N}{\partial x} = k(4) = 4k$$

As the given equation is exact type

$$\text{So, } \frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$$

$$\Rightarrow 4 = 4k, \Rightarrow k = 1 \quad (\text{Ans})$$

51. Solve: $2xy + 3x^2 = \frac{dy}{dx}(2y - x^2 + 1)$

Solution:

$$2xy + 3x^2 = \frac{dy}{dx}(2y - x^2 + 1)$$

$$\Rightarrow 2xydx + 3x^2dx = 2ydy - x^2dy + dy$$

$$\Rightarrow 2xydx + x^2dy + 3x^2dx - 2ydy - dy = 0$$

$$\Rightarrow d(x^2y) + 3x^2dx - 2ydy - dy = 0$$

$$\Rightarrow \int d(x^2y) + 3 \int x^2dx - 2 \int ydy - \int dy = c$$

$$\Rightarrow x^2y + x^3 - y^2 - y = c \quad (\text{Ans})$$

Where c = arbitrary constant.

52. Solve: $\frac{dy}{dx} = \frac{y}{x} + x^2$. Given $y = 1$ when $x = 1$.

Solution:

$$\frac{dy}{dx} = \frac{y}{x} + x^2$$

$$\Rightarrow \frac{dy}{dx} - \frac{y}{x} = x^2$$

$$\Rightarrow \frac{dy}{dx} + Py = x^2 \quad [P = -\frac{1}{x}, Q = x^2]$$

$$\therefore \text{I. F.} = e^{-\int \frac{1}{x} dx} = e^{-\log x} = e^{\log x^{-1}} = x^{-1} = \frac{1}{x}$$

$\therefore$ The required solution is

$$\frac{d}{dx}(y \times \text{I. F.}) = Q \times \text{I. F.}$$

$$\Rightarrow \frac{d}{dx}\left(y \times \frac{1}{x}\right) = x^2 \times \frac{1}{x}$$

$$\Rightarrow d\left(\frac{y}{x}\right) = x dx$$

$$\Rightarrow \int d\left(\frac{y}{x}\right) = \int x dx + c$$

$$\Rightarrow \frac{y}{x} = \frac{x^2}{2} + c$$

$$\Rightarrow \frac{1}{1} = \frac{1^2}{2} + c, \Rightarrow c = \frac{1}{2}$$

$$\therefore \frac{y}{x} = \frac{x^2}{2} + \frac{1}{2}$$

$$\Rightarrow 2y = x^3 + x \quad (\text{Ans})$$

53. Solve: $\frac{dy}{dx} + xy = xy^3$

Solution:

$$\frac{dy}{dx} + xy = xy^3$$

$$\Rightarrow y^{-3} \frac{dy}{dx} + xy^{-2} = x$$

$$\Rightarrow -\frac{1}{2} \frac{du}{dx} + xu = x$$

$$\Rightarrow \frac{du}{dx} - 2xu = -2x$$

Now, it is a linear equation in u .

$$\therefore \text{I.F.} = e^{\int (-2x) dx} = e^{-x^2}$$

From above equation we get,

$$\Rightarrow e^{-x^2} \frac{du}{dx} - 2xue^{-x^2} = -2xe^{-x^2}$$

$$\Rightarrow e^{-x^2} du - 2xue^{-x^2} dx = -2xe^{-x^2} dx$$

$$\Rightarrow d(ue^{-x^2}) = e^{-x^2} d(-x^2)$$

$$\Rightarrow \int d(ue^{-x^2}) = \int e^{-x^2} d(-x^2)$$

$$\Rightarrow ue^{-x^2} = e^{-x^2} + c$$

$$\Rightarrow y^{-2}e^{-x^2} = e^{-x^2} + c$$

$$\Rightarrow y^{-2} = 1 + ce^{x^2} \quad (\text{Ans})$$

54. Solve: $ydy = (y^2 - 1)(x + 1)dx$

Solution:

$$ydy = (y^2 - 1)(x + 1)dx$$

$$\Rightarrow \frac{ydy}{y^2 - 1} = (x + 1)dx$$

$$\Rightarrow \int \frac{2ydy}{y^2 - 1} = 2 \int (x + 1) dx + c$$

$$\Rightarrow \log(y^2 - 1) = 2\left(\frac{x^2}{2} + x\right) + c$$

$$\Rightarrow \log(y^2 - 1) = x^2 + 2x + c$$

$$\Rightarrow y^2 - 1 = e^{(x^2 + 2x + c)}$$

$$\Rightarrow y^2 = 1 + e^{(x^2 + 2x)} \cdot e^c$$

$$\Rightarrow y^2 = 1 + e^{(x^2 + 2x)} \cdot C \quad (\text{Ans})$$

Where C is an arbitrary constant.

55. Solve: $(D^2 - 3D + 2)y = e^{3x}$

Solution:

$$(D^2 - 3D + 2)y = e^{3x}$$

Let $y = Ae^{mx}$ be a trial solution of given equation.

$$\therefore \text{Auxiliary equation is } m^2 - 3m + 2 = 0$$

$$\Rightarrow m^2 - 2m - m + 2 = 0$$

$$\Rightarrow m(m - 2) - 1(m - 2) = 0$$

$$\Rightarrow (m - 2)(m - 1) = 0$$

$$\Rightarrow m = 2, 1$$

$$\therefore \text{C.F.} = c_1 e^{2x} + c_2 e^x$$

$$\text{P.I.} = \frac{1}{D^2 - 3D + 2} e^{3x}$$

$$= \frac{1}{3^2 - 3 \times 3 + 2} e^{3x} = \frac{e^{3x}}{2}$$

So, the general solution is $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow y = c_1 e^{2x} + c_2 e^x + \frac{e^{3x}}{2} \quad (\text{Ans})$$

56. Solve: $(D^2 - 4D + 4)y = e^{-2x}$

Solution:

$$(D^2 - 4D + 4)y = e^{-2x}$$

Let $y = Ae^{mx}$ be a trial solution of given equation.

$$\therefore \text{Auxiliary equation is } m^2 - 4m + 4 = 0$$

$$\Rightarrow m^2 - 2 \cdot m \cdot 2 + 2^2 = 0$$

$$\Rightarrow (m - 2)^2 = 0$$

$$\Rightarrow m = 2, 2$$

$$\therefore \text{C.F.} = (c_1 + c_2x)e^{2x}$$

$$\text{P.I.} = \frac{1}{(D-2)^2} e^{-2x} = \frac{1}{(-2-2)^2} e^{-2x} = \frac{e^{-2x}}{16}$$

So, the general solution is $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow y = (c_1 + c_2x)e^{2x} + \frac{e^{-2x}}{16} \quad (\text{Ans})$$

c_1, c_2 are arbitrary independent constants independent to each other.

57. Solve: $\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x^3$

Solution:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = x^3$$

$$\Rightarrow (D^2 + 2D + 1)y = x^3$$

$$\text{Auxiliary Equation is } m^2 + 2m + 1 = 0$$

$$\Rightarrow (m + 1)^2 = 0$$

$$\Rightarrow m = -1, -1$$

$$\therefore \text{C.F.} = (c_1 + c_2x)e^{-x}$$

$$\text{P.I.} = \frac{1}{D^2 + 2D + 1} \times x^3$$

$$= \frac{1}{(D+1)^2} \times x^3$$

$$= (D + 1)^{-2} x^3$$

$$= (1 - 2D + 3D^2 - 4D^3 + \dots) x^3$$

$$= x^3 - 2\frac{d}{dx}(x^3) + 3\frac{d}{dx}\left\{\frac{d}{dx}(x^3)\right\} - \dots$$

$$= x^3 - 6x^2 + 18x - 24$$

$\therefore$ The required general solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = (c_1 + c_2x)e^{-x} + x^3 - 6x^2 + 18x - 24 \quad (\text{Ans})$$

c_1, c_2 are arbitrary independent constants independent to each other.

58. Solve: $\frac{d^2y}{dx^2} - 4a\frac{dy}{dx} + 4a^2y = x^2e^{2ax}$

Solution:

$$\frac{d^2y}{dx^2} - 4a\frac{dy}{dx} + 4a^2y = x^2e^{2ax}$$

$$\Rightarrow (D^2 - 4aD + 4a^2)y = x^2e^{2ax}$$

$$\therefore \text{A.E. is } m^2 - 4am + 4a^2 = 0$$

$$\Rightarrow m^2 - 2 \cdot m \cdot 2a + (2a)^2 = 0$$

$$\Rightarrow (m - 2a)^2 = 0$$

$$\Rightarrow m = 2a, 2a$$

$$\therefore \text{C.F.} = (c_1 + c_2x)e^{2ax}$$

$$\text{and P.I.} = \frac{x^2e^{2ax}}{D^2 - 4aD + 4a^2}$$

$$= \frac{x^2e^{2ax}}{(D-2a)^2}$$

$$= \frac{x^2e^{2ax}}{(D+2a-2a)^2}$$

$$= e^{2ax} \times \frac{x^2}{D^2}$$

$$= e^{2ax} \times \frac{1}{D} \cdot \frac{x^3}{3}$$

$$= e^{2ax} \times \frac{x^4}{12}$$

$\therefore$ The complete solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = (c_1 + c_2x)e^{2ax} + e^{2ax} \times \frac{x^4}{12} \quad (\text{Ans})$$

c_1, c_2 are arbitrary independent constants independent to each other.

59. Solve: $\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = x^2e^x$

Solution:

$$\frac{d^2y}{dx^2} + 5\frac{dy}{dx} + 6y = x^2e^x$$

$$\Rightarrow (D^2 + 5D + 6)y = x^2e^x$$

$$\therefore \text{A.E. is } m^2 + 5m + 6 = 0$$

$$\Rightarrow m^2 + 2m + 3m + 6 = 0$$

$$\Rightarrow m(m + 2) + 3(m + 2) = 0$$

$$\Rightarrow (m + 2)(m + 3) = 0$$

$$\therefore m = -2, -3$$

$$\therefore \text{C.F.} = c_1e^{-2x} + c_2e^{-3x}$$

$$\text{and P.I.} = \frac{x^2e^x}{D^2 + 5D + 6}$$

$$= \frac{x^2e^x}{1^2 + 5 \cdot 1 + 6}$$

$$= \frac{x^2e^x}{12}$$

$\therefore$ The complete solution is

$$y = \text{C.F.} + \text{P.I.}$$

$$\Rightarrow y = c_1e^{-2x} + c_2e^{-3x} + \frac{x^2e^x}{12} \quad (\text{Ans})$$

c_1, c_2 are arbitrary independent constants independent to each other.

60. Solve: $2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 3y = 0$, Given $y = 9$, $\frac{dy}{dx} = 1$ when $x = 0$.

Solution:

$$2\frac{d^2y}{dx^2} + 5\frac{dy}{dx} - 3y = 0$$

$$\Rightarrow (2D^2 + 5D - 3)y = 0$$

Let, $y = Ae^{mx}$ be a trial solution

So, auxiliary equation is

$$2m^2 + 5m - 3 = 0$$

$$\Rightarrow 2m^2 + 6m - m - 3 = 0$$

$$\Rightarrow 2m(m + 3) - 1(m + 3) = 0$$

$$\Rightarrow (m + 3)(2m - 1) = 0$$

$$\Rightarrow m = \frac{1}{2}, -3$$

So, general solution is

$$y = c_1 e^{\frac{x}{2}} + c_2 e^{-3x}$$

$$\Rightarrow 9 = c_1 + c_2 \text{ [Given } y = 9 \text{ when } x = 0]$$

$$\text{Now } \frac{dy}{dx} = \frac{1}{2}c_1 e^{\frac{x}{2}} - 3c_2 e^{-3x}$$

$$\Rightarrow 1 = \frac{c_1}{2} - 3c_2 \text{ [Given } \frac{dy}{dx} = 1 \text{ when } x = 0]$$

$$\Rightarrow 2 = c_1 - 6c_2$$

$$\Rightarrow 2 = 9 - c_2 - 6c_2 \text{ [}\because 9 = c_1 + c_2]$$

$$\Rightarrow 7c_2 = 7,$$

$$\Rightarrow c_2 = 1, \therefore c_1 = 8$$

$\therefore$ The required solution is

$$y = 8e^{\frac{x}{2}} + e^{-3x} \text{ (Ans)}$$

61. Solve: $\frac{dy}{dx} = -\frac{x+y \cos x}{1+\sin x}$

Solution:

$$\frac{dy}{dx} = -\frac{x+y \cos x}{1+\sin x}$$

$$\Rightarrow (1 + \sin x)dy = -(x + y \cos x)dx$$

$$\Rightarrow (1 + \sin x)dy + (x + y \cos x)dx = 0$$

Here, $M = 1 + \sin x$, $N = x + y \cos x$

$$\therefore \frac{\partial M}{\partial x} = \cos x \text{ and } \frac{\partial N}{\partial y} = \cos x$$

So, the given equation is exact type

$\therefore$ The solution is

$$\int (x + y \cos x) dx + \int dy = c$$

[keeping y as constant]

$$\Rightarrow \frac{x^2}{2} + y \sin x + y = c \text{ (Ans)}$$

Where c is an arbitrary constant.

62. Solve: $\frac{dy}{dx} = (x + y)^2$, when $x = 0, y = 1$

Solution:

$$\frac{dy}{dx} = (x + y)^2$$

$$\Rightarrow \frac{dz}{dx} - 1 = z^2$$

$$\Rightarrow \frac{dz}{dx} = z^2 + 1$$

$$\Rightarrow \frac{dz}{z^2+1} = dx$$

$$\Rightarrow \int \frac{dz}{z^2+1} = \int dx + c$$

$$\Rightarrow \tan^{-1} z = x + c$$

$$\Rightarrow \tan^{-1}(x + y) = x + c$$

$$\Rightarrow \tan^{-1}(1) = c \text{ [}\because x = 0, y = 1]$$

$$\Rightarrow c = \frac{\pi}{4}$$

$\therefore$ The required solution is

$$\tan^{-1}(x + y) = x + \frac{\pi}{4}$$

$$\Rightarrow x + y = \tan\left(x + \frac{\pi}{4}\right) \text{ (Ans)}$$

63. Solve: $(1 + x^3)dy - x^2y dx = 0$, if $y = 2$, when $x = 1$

Solution:

$$(1 + x^3)dy - x^2y dx = 0$$

$$\Rightarrow (1 + x^3)dy = x^2y dx$$

$$\Rightarrow \frac{dy}{y} = \frac{x^2 dx}{1+x^3}$$

$$\Rightarrow \int \frac{dy}{y} = \frac{1}{3} \int \frac{3x^2 dx}{1+x^3} + \log c \text{ [log } c = \text{constant]}$$

$$\Rightarrow \log y = \frac{1}{3} \log(1 + x^3) + \log c$$

$$\Rightarrow 3 \log y = \log(1 + x^3) + 3 \log c$$

$$\Rightarrow \log y^3 = \log(1 + x^3) + \log c^3$$

$$\Rightarrow \log y^3 = \log(1 + x^3)c^3$$

$$\Rightarrow y^3 = (1 + x^3)c^3$$

$$\Rightarrow 2^3 = (1 + 1^3)c^3$$

$$\Rightarrow 8 = 2c^3, \Rightarrow c^3 = 4$$

$\therefore$ The required solution is $y^3 = 4(1 + x^3)$ (Ans)

64. Solve the differential equation:

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$$

Solution: Let, $y = e^{mx}$ be the particular solution of given diff. equation.

$$\Rightarrow \frac{dy}{dx} = me^{mx} \text{ and } \frac{d^2y}{dx^2} = m^2 e^{mx}$$

$$\therefore \frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$$

$$\Rightarrow m^2 e^{mx} - 10me^{mx} + 25e^{mx} = 0$$

$$\Rightarrow e^{mx}(m^2 - 10m + 25) = 0$$

$$\Rightarrow m^2 - 10m + 25 = 0 \text{ [} e^{mx} \neq 0]$$

$$\Rightarrow (m - 5)^2 = 0, \Rightarrow m = 5, 5$$

Since, roots are real and equal. Therefore the general solution is $y = (c_1 + c_2 x)e^{5x}$.

65. Solve the differential equation:

$$\frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = e^{5x}$$

Solution: Let, $y = e^{mx}$ be the particular solution of given diff. equation.

$$\Rightarrow \frac{dy}{dx} = me^{mx} \text{ and } \frac{d^2y}{dx^2} = m^2e^{mx}$$

$$\therefore \frac{d^2y}{dx^2} - 10\frac{dy}{dx} + 25y = 0$$

$$\Rightarrow m^2e^{mx} - 10me^{mx} + 25e^{mx} = 0$$

$$\Rightarrow e^{mx}(m^2 - 10m + 25) = 0$$

$$\Rightarrow m^2 - 10m + 25 = 0 \quad [e^{mx} \neq 0]$$

$$\Rightarrow (m - 5)^2 = 0$$

$$\Rightarrow m = 5, 5$$

$$\therefore \text{C.F.} = (c_1 + c_2x)e^{5x}$$

$$\text{P.I.} = \frac{1}{(D-2)^2}e^{5x} = \frac{1}{(5-2)^2}e^{5x} = \frac{e^{5x}}{9}$$

So, the general solution is $y = \text{C.F.} + \text{P.I.}$

$$\Rightarrow y = (c_1 + c_2x)e^{5x} + \frac{e^{5x}}{9} \quad (\text{Ans})$$

c_1, c_2 are arbitrary independent constants independent to each other.

66. Find the Integrating Factor (I. F.) of $\frac{dy}{dx} + 2xy = 9x$

Solution: Given, $\frac{dy}{dx} + 2xy = 9x$

$$\Rightarrow \frac{dy}{dx} + Py = Q \quad [P = 2x, Q = 9x]$$

$$\therefore \text{Integrating Factor (IF)} = e^{\int P dx}$$

$$\Rightarrow \text{I.F.} = e^{\int 2x dx}$$

$$= e^{2 \times \frac{x^2}{2}} = e^{x^2} \quad (\text{Ans})$$

67. Solve: $\cos y dx + (1 + 2e^{-x}) \sin y dy = 0$ when $x = 0, y = \frac{\pi}{4}$

Solution:

$$\cos y dx + (1 + 2e^{-x}) \sin y dy = 0$$

$$\Rightarrow \cos y dx = -(1 + 2e^{-x}) \sin y dy$$

$$\Rightarrow \frac{dx}{1+2e^{-x}} = -\frac{\sin y}{\cos y} dy$$

$$\Rightarrow \int \frac{e^x dx}{e^x + 2} = -\int \tan y dy + \log c \quad [\log c = \text{constant}]$$

$$\Rightarrow \log(e^x + 2) + \log(\sec y) = \log c$$

$$\Rightarrow \log(e^x + 2)(\sec y) = \log c$$

$$\Rightarrow (e^x + 2)(\sec y) = c$$

$$\Rightarrow (e^0 + 2)\left(\sec \frac{\pi}{4}\right) = c$$

$$\Rightarrow c = 3\sqrt{2}$$

$\therefore$ The required solution is

$$(e^x + 2)(\sec y) = 3\sqrt{2} \quad (\text{Ans})$$

68. Solve: $\frac{dy}{dx} = 1 + y + x^2 + yx^2$

Solution:

$$\frac{dy}{dx} = 1 + y + x^2 + yx^2$$

$$\Rightarrow \frac{dy}{dx} = (1 + y) + x^2(1 + y)$$

$$\Rightarrow \frac{dy}{dx} = (1 + y)(1 + x^2)$$

$$\Rightarrow \frac{dy}{(1+y)} = (1 + x^2)dx$$

$$\Rightarrow \int \frac{dy}{(1+y)} = \int (1 + x^2) dx + c$$

$$\Rightarrow \log(1 + y) = x + \frac{x^3}{3} + c \quad (\text{Ans})$$

69. Find the Particular Integral (P.I.) of the differential equation $\frac{d^2y}{dx^2} - y = 1$.

Solution:

$$\text{Given, } \frac{d^2y}{dx^2} - y = 1$$

$$\Rightarrow (D^2 - 1)y = 1$$

$$\therefore \text{P.I.} = \frac{1}{D^2 - 1} \times 1$$

$$= -\frac{1}{1 - D^2} \times 1$$

$$= -(1 - D^2)^{-1} \times 1$$

$$= -(1 + D^2 + \dots) \times 1$$

$$= -1 - 0 = -1 \quad (\text{Ans})$$

{Prepared by [NatiTute](#) YouTube Channel}

1. If $u = e^{xy}$, then the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y}$ is ____.

Solution: $u = e^{xy}$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{\partial}{\partial x}(e^{xy}) + \frac{\partial}{\partial y}(e^{xy})$$

$$= ye^{xy} + xe^{xy} = (x + y)e^{xy} \quad (\text{Ans})$$

2. If $u = \tan^{-1} \frac{y}{x}$ find $\frac{\partial u}{\partial x}$

Solution:

Given, $u = \tan^{-1} \frac{y}{x}$

Differentiating partially with respect to x we get,

$$\frac{\partial u}{\partial x} = \frac{\partial}{\partial x} \left(\tan^{-1} \frac{y}{x} \right)$$

$$= \frac{1}{1 + \frac{y^2}{x^2}} \cdot \frac{\partial}{\partial x} \left(\frac{y}{x} \right)$$

$$= \frac{x^2}{x^2 + y^2} \left(-\frac{y}{x^2} \right) = -\frac{y}{x^2 + y^2} \quad (\text{Ans})$$

3. If $z = 3x^2y$ then $\frac{\partial^2 z}{\partial x \partial y} = ?$

- (a) $3y$
(b) $6x$
(c) $6y$
(d) none of these

Explanation:

Given, $z = 3x^2y \therefore \frac{\partial z}{\partial y} = 3x^2$

$$\Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) = \frac{\partial}{\partial x} (3x^2)$$

$$\Rightarrow \frac{\partial^2 z}{\partial x \partial y} = 6x$$

Answer: (b)

4. If $u = 7xy^2$, find $\frac{\partial^2 u}{\partial x \partial y}$.

Solution:

Given, $u = 7xy^2$

$$\therefore \frac{\partial u}{\partial y} = 7x \cdot 2y = 14xy$$

$$\Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} (14xy)$$

$$\Rightarrow \frac{\partial^2 u}{\partial x \partial y} = 14y$$

5. If $u = \log(x^2 + y^2)$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2$

Solution:

Given, $u = \log(x^2 + y^2)$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{x^2 + y^2} \times 2x, \Rightarrow x \frac{\partial u}{\partial x} = \frac{2x^2}{x^2 + y^2}$$

$$\frac{\partial u}{\partial y} = \frac{1}{x^2 + y^2} \times 2y, \Rightarrow y \frac{\partial u}{\partial y} = \frac{2y^2}{x^2 + y^2}$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{2x^2}{x^2 + y^2} + \frac{2y^2}{x^2 + y^2}$$

$$= \frac{2}{x^2 + y^2} \times (x^2 + y^2)$$

$$= 2 \quad (\text{Proved})$$

6. If $u = xy + yz + zx$ then the value of $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z}$

at point (1, 1, 1) is-

- (a) 2
(b) 6
(c) 3
(d) 0

Explanation:

Given, $u = xy + yz + zx$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = y + z + x + z + y + x$$

$$= 2(x + y + z)$$

$$\therefore \text{At point } (1, 1, 1)$$

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = 2(1 + 1 + 1) = 6$$

Answer: (b)

7. If $u = \sin^{-1} \frac{y}{x}$, then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

- (a) 1
(b) 0
(c) 2
(d) none of these

Explanation:

Given, $u = \sin^{-1} \frac{y}{x}$

$$\Rightarrow \sin u = \frac{y}{x} = x^0 u \left(\frac{y}{x} \right)$$

So, $\sin u$ is a homogeneous function of degree 0.

According to Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad [\text{Here } n = 0]$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \times u = 0$$

Answer: (b)

8. If $f(x, y) = \frac{x}{y}$, then find the value of $x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y}$ is equal to- (a) f (b) $-f$ (c) 1 (d) 0

Solution: Given $f(x, y) = \frac{x}{y} = x^0 f \left(\frac{x}{y} \right)$

So, f is a homogeneous function of degree 0.

According to Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \quad [\text{Here } n = 0]$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \times u = 0$$

Answer: (d)

9. State Euler's theorem on homogeneous function.

Solution:

Euler's Theorem on Homogeneous Function: Euler's theorem states that if u is a homogeneous function of degree n of the variables x, y, z then-

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

$$\text{or } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = nu$$

10. Euler's theorem can be applied to function which is-

- (a) linear
- (b) homogeneous
- (c) differentiable
- (d) none of these

Explanation: Euler's theorem states that if u is a homogeneous function of degree n of the variables x, y, z then-

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu$$

Answer: (b)

11. Verify Euler's theorem for the function

$$u = 3x^2 + 4y^2$$

Solution:

$$\text{Given, } u = 3x^2 + 4y^2$$

$$\Rightarrow u = x^2 \left(3 + 4 \frac{y^2}{x^2} \right) = x^2 \left\{ 3 + 4 \left(\frac{y}{x} \right)^2 \right\}$$

$\therefore u$ is homogeneous function in x, y of degree 2

$$\text{By Euler's theorem } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

$$\text{Now, } \frac{\partial u}{\partial x} = 6x, \Rightarrow x \frac{\partial u}{\partial x} = 6x^2$$

$$\text{and } \frac{\partial u}{\partial y} = 8y, \Rightarrow y \frac{\partial u}{\partial y} = 8y^2$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$$

$$= 6x^2 + 8y^2 = 2(3x^2 + 4y^2) = 2u$$

$\therefore$ Given function satisfy Euler's theorem.

12. If $u = e^{\frac{x}{y}}$; find the value of $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$

Solution:

$$\text{Given, } u = e^{\frac{x}{y}}$$

$$\Rightarrow u = e^{\frac{x}{y}} = e^{\frac{1}{\frac{y}{x}}} = x^0 u \left(\frac{y}{x} \right)$$

$\therefore u$ is a homogeneous function of degree 0.

By, Euler's theorem

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = 0 \times u = 0 \quad (\text{Ans})$$

13. If $u = x^2 + xy + y^2$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

- (a) u
- (b) $2u$
- (c) $3u$
- (d) none of these

Solution:

$$\text{Given, } u = x^2 + xy + y^2$$

$$\Rightarrow u = x^2 \left(1 + \frac{y}{x} + \frac{y^2}{x^2} \right)$$

$$\Rightarrow u = x^2 f \left(\frac{y}{x} \right)$$

$\therefore u(x, y)$ is a homogeneous function of degree 2.

$\therefore$ By Euler's theorem we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

Answer: (b)

14. If $u = 5x^2 + 7xy + 3y^2$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

- (a) u
- (b) $2u$
- (c) $4u$
- (d) none of these

Solution:

$$\text{Given, } u = 5x^2 + 7xy + 3y^2$$

$$\Rightarrow u = x^2 \left(5 + 7 \frac{y}{x} + 3 \frac{y^2}{x^2} \right)$$

$$\Rightarrow u = x^2 f \left(\frac{y}{x} \right)$$

$\therefore u(x, y)$ is a homogeneous function of degree 2.

$\therefore$ By Euler's theorem we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 2u$$

Answer: (b)

15. If $u = e^{xy}$ then $\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = ?$

Solution:

$$\text{Given, } u = e^{xy}$$

$$\Rightarrow \frac{\partial u}{\partial x} = ye^{xy} \quad \text{and} \quad \frac{\partial u}{\partial y} = xe^{xy}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = ye^{xy} + xe^{xy} = (x + y)e^{xy} \quad (\text{Ans})$$

16. If $u = 7xy^2$ find $\frac{\partial^2 u}{\partial x \partial y}$.

Solution:

$$\text{Given, } u = 7xy^2$$

$$\therefore \frac{\partial u}{\partial y} = 14xy$$

$$\therefore \frac{\partial^2 u}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial u}{\partial y} \right) = \frac{\partial}{\partial x} (14xy) = 14y \quad (\text{Ans})$$

17. If $u(x, y) = \sqrt{\frac{x^2}{x+y}}$ then $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = ?$

Solution:

$$\text{Given, } u(x, y) = \sqrt{\frac{x^2}{x+y}}$$

$$\Rightarrow u(x, y) = \frac{x}{\sqrt{x+y}}$$

$$= \frac{x}{\sqrt{x} \sqrt{1+\frac{y}{x}}}$$

$$= x^{\frac{1}{2}} \times \frac{1}{\sqrt{1+\frac{y}{x}}} = x^{\frac{1}{2}} \phi \left(\frac{y}{x} \right)$$

$\therefore u(x, y)$ is a homogeneous function of degree $\frac{1}{2}$

$\therefore$ By Euler's theorem we get

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} u \quad (\text{Ans})$$

18. If $z = \tan^{-1} \left(\frac{x-y}{\sqrt{x}+\sqrt{y}} \right)$, show that

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = \frac{1}{4} \sin 2z$$

Solution:

$$\text{Given, } z = \tan^{-1} \left(\frac{x-y}{\sqrt{x}+\sqrt{y}} \right)$$

$$\Rightarrow \tan z = \frac{x-y}{\sqrt{x}+\sqrt{y}}$$

$$\Rightarrow \tan z = \frac{x \left(1 - \frac{y}{x} \right)}{\sqrt{x} \left(1 + \sqrt{\frac{y}{x}} \right)}$$

$$\Rightarrow \tan z = x^{\frac{1}{2}} f \left(\frac{y}{x} \right)$$

$\therefore \tan z$ is a homogeneous function in x, y of degree $\frac{1}{2}$

$\therefore$ By Euler's theorem we get,

$$x \frac{\partial(\tan z)}{\partial x} + y \frac{\partial(\tan z)}{\partial y} = \frac{1}{2} \tan z$$

$$\Rightarrow x \sec^2 z \frac{\partial z}{\partial x} + y \sec^2 z \frac{\partial z}{\partial y} = \frac{1}{2} \tan z$$

$$\begin{aligned} \Rightarrow x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} &= \frac{1}{2} \tan z \times \frac{1}{\sec^2 z} \\ &= \frac{1}{2} \times \frac{\sin z}{\cos z} \times \cos^2 z \\ &= \frac{1}{2} \times 2 \sin z \cos z \\ &= \frac{1}{4} \sin 2z \text{ (Proved)} \end{aligned}$$

19. If $u = \cos^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$, show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0$$

Solution:

$$\text{Given, } u = \cos^{-1} \left(\frac{x+y}{\sqrt{x}+\sqrt{y}} \right)$$

$$\Rightarrow \cos u = \frac{x+y}{\sqrt{x}+\sqrt{y}}$$

$$\Rightarrow \cos u = \frac{x \left(1 + \frac{y}{x} \right)}{\sqrt{x} \left(1 + \sqrt{\frac{y}{x}} \right)}$$

$$\Rightarrow \cos u = x^{\frac{1}{2}} f \left(\frac{y}{x} \right)$$

$\therefore \cos u$ is a homogeneous function in x, y of degree $\frac{1}{2}$

$\therefore$ By Euler's theorem we get,

$$x \frac{\partial(\cos u)}{\partial x} + y \frac{\partial(\cos u)}{\partial y} = \frac{1}{2} \cos u$$

$$\Rightarrow x(-\sin u) \frac{\partial u}{\partial x} + y(-\sin u) \frac{\partial u}{\partial y} = \frac{1}{2} \cos u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = -\frac{1}{2} \frac{\cos u}{\sin u}$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + \frac{1}{2} \cot u = 0 \text{ (Proved)}$$

20. If $u = \sin^{-1} \left(\frac{x-y}{\sqrt{x}+\sqrt{y}} \right)$, show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u$$

Solution:

$$\text{Given, } u = \sin^{-1} \left(\frac{x-y}{\sqrt{x}+\sqrt{y}} \right)$$

$$\Rightarrow \sin u = \frac{x-y}{\sqrt{x}+\sqrt{y}}$$

$$\Rightarrow \sin u = \frac{x \left(1 - \frac{y}{x} \right)}{\sqrt{x} \left(1 + \sqrt{\frac{y}{x}} \right)}$$

$$\Rightarrow \sin u = x^{\frac{1}{2}} f \left(\frac{y}{x} \right)$$

$\therefore \sin u$ is a homogeneous function in x, y of degree $\frac{1}{2}$

$\therefore$ By Euler's theorem we get,

$$x \frac{\partial(\sin u)}{\partial x} + y \frac{\partial(\sin u)}{\partial y} = \frac{1}{2} \sin u$$

$$\Rightarrow x(\cos u) \frac{\partial u}{\partial x} + y(\cos u) \frac{\partial u}{\partial y} = \frac{1}{2} \sin u$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \frac{\sin u}{\cos u}$$

$$\Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{1}{2} \tan u \text{ (Proved)}$$

21. If $u = \log(e^x + e^y)$, find $\frac{\partial u}{\partial x}$.

Solution:

$$u = \log(e^x + e^y)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{e^x + e^y} \times \frac{\partial(e^x + e^y)}{\partial x} = \frac{e^x}{e^x + e^y}$$

$$\frac{\partial u}{\partial y} = \frac{1}{e^x + e^y} \times \frac{\partial(e^x + e^y)}{\partial y} = \frac{e^y}{e^x + e^y}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{e^x}{e^x + e^y} + \frac{e^y}{e^x + e^y} = \frac{e^x + e^y}{e^x + e^y} = 1 \text{ (Ans)}$$

22. If $u = \sin^{-1} \frac{x^2+y^2}{x+y}$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \tan u$.

Solution:

$$\text{Given, } u = \sin^{-1} \frac{x^2+y^2}{x+y}$$

$$\Rightarrow \sin u = \frac{x^2+y^2}{x+y}$$

$$\Rightarrow \sin u = \frac{x^2 \left(1 + \frac{y^2}{x^2} \right)}{x \left(1 + \frac{y}{x} \right)} = x^1 \frac{1 + \left(\frac{y}{x} \right)^2}{1 + \frac{y}{x}} = x^1 f \left(\frac{y}{x} \right)$$

$\therefore \sin u$ is a homogeneous function of degree 1

By Euler's theorem,

$$x \frac{\partial(\sin u)}{\partial x} + y \frac{\partial(\sin u)}{\partial y} = 1 \times (\sin u)$$

$$\Rightarrow x \cos u \frac{\partial u}{\partial x} + y \cos u \frac{\partial u}{\partial y} = \sin u$$

$$\Rightarrow \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \cos u = \sin u$$

$$\Rightarrow \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) = \frac{\sin u}{\cos u} = \tan u \text{ (Proved)}$$

23. If $u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$, show that

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \sin 2u$$

Solution:

$$\text{Given, } u = \tan^{-1} \left(\frac{x^3+y^3}{x-y} \right)$$

$$\Rightarrow \tan u = \frac{x^3+y^3}{x-y} = \frac{x^3 \left(1 + \frac{y^3}{x^3} \right)}{x \left(1 - \frac{y}{x} \right)} = x^2 f \left(\frac{y}{x} \right)$$

$\therefore \tan u$ is a homogeneous function of degree 2.

So, by Euler's theorem we get,

$$x \frac{\partial (\tan u)}{\partial x} + y \frac{\partial (\tan u)}{\partial y} = 2 \tan u$$

$$\Rightarrow x \sec^2 u \frac{\partial u}{\partial x} + y \sec^2 u \frac{\partial u}{\partial y} = 2 \tan u$$

$$\Rightarrow \left(x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} \right) \sec^2 u = 2 \tan u$$

$$\begin{aligned} \Rightarrow x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{2 \tan u}{\sec^2 u} \\ &= 2 \frac{\sin u}{\cos u} \times \cos^2 u \\ &= 2 \sin u \cos u \\ &= \sin 2u \quad (\text{Proved}) \end{aligned}$$

24. If $u = \log(x^3 + y^3 - x^2y - xy^2)$ then show that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{2}{x+y}$$

Solution:

$$\begin{aligned} u &= \log(x^3 + y^3 - x^2y - xy^2) \\ &= \log\{(x+y)^3 - 3xy(x+y) - xy(x+y)\} \\ &= \log\{(x+y)^3 - 4xy(x+y)\} \\ &= \log[(x+y)\{(x+y)^2 - 4xy\}] \\ &= \log(x+y)(x-y)^2 \\ &= \log(x+y) + \log(x-y)^2 \\ &= \log(x+y) + 2 \log(x-y) \end{aligned}$$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{x+y} + \frac{2}{x-y} \quad \text{and} \quad \frac{\partial u}{\partial y} = \frac{1}{x+y} - \frac{2}{x-y}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{1}{x+y} + \frac{2}{x-y} + \frac{1}{x+y} - \frac{2}{x-y}$$

$$\Rightarrow \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} = \frac{2}{x+y} \quad (\text{Proved})$$

25. If $u = e^{xyz}$ then show that

$$\frac{\partial^3 u}{\partial x \partial y \partial z} = (1 + 3xyz + x^2y^2z^2)e^{xyz}$$

Solution: Given, $u = e^{xyz}$

$$\therefore \frac{\partial u}{\partial z} = xye^{xyz}$$

$$\Rightarrow \frac{\partial}{\partial y} \left(\frac{\partial u}{\partial z} \right) = xy \cdot zxe^{xyz} + e^{xyz} \cdot x$$

$$\Rightarrow \frac{\partial^2 u}{\partial y \partial z} = (x^2yz + x)e^{xyz}$$

$$\Rightarrow \frac{\partial}{\partial x} \left(\frac{\partial^2 u}{\partial y \partial z} \right) = (x^2yz + x)yze^{xyz} + (2xyz + 1)e^{xyz}$$

$$\begin{aligned} \Rightarrow \frac{\partial^3 u}{\partial x \partial y \partial z} &= (x^2yz + x)yze^{xyz} + (2xyz + 1)e^{xyz} \\ &= (x^2y^2z^2 + xyz + 2xyz + 1)e^{xyz} \\ &= (1 + 3xyz + x^2y^2z^2)e^{xyz} \quad (\text{Proved}) \end{aligned}$$

26. If $u = \log \left(\frac{x^4+y^4}{x+y} \right)$, show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3$

Solution: Given, $u = \log \left(\frac{x^4+y^4}{x+y} \right)$

$$\Rightarrow u = \log(x^4 + y^4) - \log(x + y)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{4x^3}{x^4+y^4} - \frac{1}{x+y}$$

$$\Rightarrow x \frac{\partial u}{\partial x} = \frac{4x^4}{x^4+y^4} - \frac{x}{x+y}$$

$$\text{and } \frac{\partial u}{\partial y} = \frac{4y^3}{x^4+y^4} - \frac{1}{x+y}$$

$$\Rightarrow y \frac{\partial u}{\partial y} = \frac{4y^4}{x^4+y^4} - \frac{y}{x+y}$$

$$\begin{aligned} \therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} &= \frac{4x^4}{x^4+y^4} - \frac{x}{x+y} + \frac{4y^4}{x^4+y^4} - \frac{y}{x+y} \\ &= 4 \left(\frac{x^4+y^4}{x^4+y^4} \right) - \left(\frac{x+y}{x+y} \right) \\ &= 4 - 1 = 3 \quad (\text{Proved}) \end{aligned}$$

27. If $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$ prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} = 0$

Solution: Given, $u = \frac{y}{z} + \frac{z}{x} + \frac{x}{y}$

Differentiating partially with respect to x we get,

$$\frac{\partial u}{\partial x} = -\frac{z}{x^2} + \frac{1}{y}, \Rightarrow x \frac{\partial u}{\partial x} = -\frac{z}{x} + \frac{x}{y}$$

Differentiating partially with respect to y we get,

$$\frac{\partial u}{\partial y} = \frac{1}{z} - \frac{x}{y^2}, \Rightarrow y \frac{\partial u}{\partial y} = \frac{y}{z} - \frac{x}{y}$$

Differentiating partially with respect to z we get,

$$\frac{\partial u}{\partial z} = -\frac{y}{z^2} + \frac{1}{x}, \Rightarrow z \frac{\partial u}{\partial z} = -\frac{y}{z} + \frac{z}{x}$$

$$\begin{aligned} \therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} + z \frac{\partial u}{\partial z} &= -\frac{z}{x} + \frac{x}{y} + \frac{y}{z} - \frac{x}{y} - \frac{y}{z} + \frac{z}{x} = 0 \quad (\text{Proved}) \end{aligned}$$

28. If $u = \frac{x^2+y^2}{x+y}$, then show that

$$\left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right)^2 = 4 \left(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right)$$

Solution:

$$\text{Given, } u = \frac{x^2+y^2}{x+y}$$

$$\therefore \frac{\partial u}{\partial x} = \frac{2x}{x+y} - \frac{x^2+y^2}{(x+y)^2} \quad \text{and} \quad \frac{\partial u}{\partial y} = \frac{2y}{x+y} - \frac{x^2+y^2}{(x+y)^2}$$

$$\begin{aligned} \text{Now, } \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} &= \frac{2x}{x+y} - \frac{x^2+y^2}{(x+y)^2} - \frac{2y}{x+y} + \frac{x^2+y^2}{(x+y)^2} \\ &= \frac{2x}{x+y} - \frac{2y}{x+y} = \frac{2(x-y)}{x+y} \end{aligned}$$

$$\therefore \text{L.H.S} = \left(\frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right)^2 = \left\{ \frac{2(x-y)}{x+y} \right\}^2$$

$$\begin{aligned} \text{R.H.S} &= 4 \left(1 - \frac{\partial u}{\partial x} - \frac{\partial u}{\partial y} \right) \\ &= 4 \left\{ 1 - \frac{2x}{x+y} + \frac{x^2+y^2}{(x+y)^2} - \frac{2y}{x+y} + \frac{x^2+y^2}{(x+y)^2} \right\} \\ &= 4 \left\{ 1 - 2 + \frac{2(x^2+y^2)}{(x+y)^2} \right\} \\ &= 4 \left\{ \frac{(x+y)^2 + (x-y)^2}{(x+y)^2} - 1 \right\} \\ &= 4 \left\{ \frac{(x+y)^2 + (x-y)^2 - (x+y)^2}{(x+y)^2} \right\} \\ &= 4 \left\{ \frac{(x-y)^2}{(x+y)^2} \right\} = \left\{ \frac{2(x-y)}{x+y} \right\}^2 \end{aligned}$$

$$\therefore \text{L. H. S} = \text{R. H. S} \quad (\text{Proved})$$

29. If $u = \tan^{-1}\left(\frac{y}{x}\right)$ then find $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y}$

Solution: Given, $u = \tan^{-1}\left(\frac{y}{x}\right)$

$$\Rightarrow u = x^0 f\left(\frac{y}{x}\right)$$

Here, u is a homogeneous function of the degree 0.
By Euler's theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu \text{ [Here } n = 0]$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0 \text{ (Ans)}$$

30. If $u = \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x}$ show that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 0$.

Solution:

$$\begin{aligned} \text{Given, } u &= \sin^{-1}\frac{x}{y} + \tan^{-1}\frac{y}{x} \\ &= \sin^{-1}\frac{1}{\left(\frac{y}{x}\right)} + \tan^{-1}\left(\frac{y}{x}\right) \\ &= x^0 f\left(\frac{y}{x}\right) \end{aligned}$$

$\therefore u$ is a homogeneous function of degree 0

Here, $n = 0$

So, by Euler's theorem,

$$x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = nu = 0 \times u = 0 \text{ (Proved)}$$

31. Verify Euler's theorem for

$$u = x^3 + 3x^2y + 3xy^2 + y^3$$

Solution:

$$\text{Given, } u = x^3 + 3x^2y + 3xy^2 + y^3$$

$$\Rightarrow u = (x + y)^3 = x^3 \left(1 + \frac{y}{x}\right)^3 = x^3 f\left(\frac{y}{x}\right)$$

$\therefore u$ is a homogeneous function of degree 3

$$\text{By Euler's theorem } x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3u$$

$$\text{Now, } u = (x + y)^3$$

$$\therefore \frac{\partial u}{\partial x} = 3(x + y)^2, \Rightarrow x \frac{\partial u}{\partial x} = 3x(x + y)^2$$

$$\text{and } \frac{\partial u}{\partial y} = 3(x + y)^2, \Rightarrow y \frac{\partial u}{\partial y} = 3y(x + y)^2$$

$$\therefore x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 3x(x + y)^2 + 3y(x + y)^2$$

$$= 3(x + y)^2(x + y)$$

$$= 3(x + y)^3$$

$$= 3u$$

$\therefore$ Euler's theorem is valid for given equation.

32. If $f(x, y) = 2x^2 - xy + 2y^3$ then find $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$ at the point (1, 2).

Solution:

$$\text{Given, } f(x, y) = 2x^2 - xy + 2y^3$$

$$\therefore \frac{\partial f}{\partial x} = 4x - y$$

$$\Rightarrow \left[\frac{\partial f}{\partial x}\right]_{(1,2)} = 4.1 - 2 = 2 \text{ (Ans)}$$

$$\text{and } \frac{\partial f}{\partial y} = -x + 6y^2$$

$$\Rightarrow \left[\frac{\partial f}{\partial y}\right]_{(1,2)} = -1 + 6.2^2 = 23 \text{ (Ans)}$$

33. If $u = \log(x^3 + y^3 + z^3 - 3xyz)$ show that

$$\frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z}$$

Solution:

$$\text{Given, } u = \log(x^3 + y^3 + z^3 - 3xyz)$$

$$= \log\{(x + y + z)(x^2 + y^2 + z^2 - xy - yz - zx)\}$$

$$= \log(x + y + z) + \log(x^2 + y^2 + z^2 - xy - yz - zx)$$

$$\therefore \frac{\partial u}{\partial x} = \frac{1}{x+y+z} + \frac{2x-y-z}{x^2+y^2+z^2-xy-yz-zx}$$

$$\frac{\partial u}{\partial y} = \frac{1}{x+y+z} + \frac{2y-z-x}{x^2+y^2+z^2-xy-yz-zx}$$

$$\frac{\partial u}{\partial z} = \frac{1}{x+y+z} + \frac{2z-y-x}{x^2+y^2+z^2-xy-yz-zx}$$

$$\therefore \frac{\partial u}{\partial x} + \frac{\partial u}{\partial y} + \frac{\partial u}{\partial z} = \frac{3}{x+y+z} + \frac{2x-y-z+2y-z-x+2z-y-x}{x^2+y^2+z^2-xy-yz-zx}$$

$$= \frac{3}{x+y+z} + \frac{0}{x^2+y^2+z^2-xy-yz-zx}$$

$$= \frac{3}{x+y+z} \text{ (Proved)}$$

34. Define homogeneous function.

Solution:

Homogeneous Function:

A function $f(x, y)$ is said to be of degree (or order) n in the variables x and y if it can be expressed in the form

$$f(x, y) = x^n \phi\left(\frac{y}{x}\right) \text{ or } y^n \psi\left(\frac{x}{y}\right)$$

or

$$f(kx, ky) = k^n f(x, y)$$

Similarly,

A function $f(x, y, z)$ is said to be of degree (or order) n in the variables x, y and z if it can be expressed in the form

$$f(x, y, z) = x^n \phi\left(\frac{y}{x}, \frac{z}{x}\right) \text{ or } y^n \psi\left(\frac{x}{y}, \frac{z}{y}\right) \text{ or } z^n g\left(\frac{x}{z}, \frac{y}{z}\right)$$

or

$$f(kx, ky, kz) = k^n f(x, y, z)$$

35. Find the degree of function $x^2 \sin\left(\frac{y^2}{x^2}\right) + y^2$.

$$\text{Solution: Let, } f(x, y) = x^2 \sin\left(\frac{y^2}{x^2}\right) + y^2$$

$$\Rightarrow f(kx, ky) = k^2 x^2 \sin\left(\frac{k^2 y^2}{k^2 x^2}\right) + k^2 y^2$$

$$\Rightarrow f(kx, ky) = k^2 \left\{x^2 \sin\left(\frac{y^2}{x^2}\right) + y^2\right\}$$

$\therefore$ Degree of given function is that power of k i.e. 2

1. If the mean of $x - 2, 10, x + 3, 7$ is 9 then find x .

Solution:

Given values $x - 2, 10, x + 3, 7$

Mean = 9

Number of value = 4

$$\text{Now, mean} = \frac{x-2+10+x+3+7}{4}$$

$$\Rightarrow 9 = \frac{2x+18}{4}$$

$$\Rightarrow 2x + 18 = 36$$

$$\Rightarrow 2x = 18$$

$$\Rightarrow x = 9 \quad (\text{Ans})$$

2. The median of data 6, 4, -1, 0, 4, 8, 9 is-

(a) 0

(b) 4

(c) 6

(d) none of these

Solution: Given data: 6, 4, -1, 0, 4, 8, 9

Arranging the data in an ascending order of magnitude we get, -1, 0, 4, 4, 6, 8, 9

Since number of values in the array is $n = 7$ which is an odd number. Therefore the median is

$$\frac{n+1}{2} = \frac{7+1}{2} = 4^{\text{th}} \text{ value of the series.}$$

Hence, the required median is 4

Answer: (b)

3. Compute the median of the data: 10, 5, 9, 6, 4, 8, 7, 6, 5

Solution: Given data: 10, 5, 9, 6, 4, 8, 7, 6, 5

Arranging the data in an ascending order of magnitude we get, 4, 5, 5, 6, 6, 7, 8, 9, 10

Since number of values in the array is $n = 9$ which is an odd number. Therefore the median is

$$\frac{n+1}{2} = \frac{9+1}{2} = 5^{\text{th}} \text{ value of the series.}$$

Hence, the required median is 6 (Ans)

4. Median of the data 2.5, 2.0, 2.6, 2.2, 2.9 and 2.1 is

(a) 2.2

(b) 2.383

(c) 2.35

(d) none of these

Solution: Given data: 2.5, 2.0, 2.6, 2.2, 2.9, 2.1

Arranging the data in an ascending order

2.0, 2.1, 2.2, 2.5, 2.6, 2.9

$$\therefore \text{Median} = \frac{2.2+2.5}{2} = 2.35$$

Answer: (c)

5. Find the mode of the data: 10, 12, 8, 10, 12, 11, 10, 12, 7, 11, 12.

Solution:

Given data: 10, 12, 8, 10, 12, 11, 10, 12, 7, 11, 12

Now arranging the data in an ascending order we get: 7, 8, 10, 10, 10, 11, 11, 12, 12, 12, 12

$\therefore$ Mode = The value corresponding to highest frequency = 12 (Ans)

6. Find the mode of the data: 7, -1, 2, 3, 5, 0, 2, 3, 2, 3, 0, 4, 2, 5, 1

Solution:

Given data: 7, -1, 2, 3, 5, 0, 2, 3, 2, 3, 0, 4, 2, 5, 1

Now arranging the data in an ascending order we get: -1, 0, 0, 1, 2, 2, 2, 2, 3, 3, 3, 4, 5, 5, 7

$\therefore$ Mode = The value corresponding to highest frequency = 2 (Ans)

7. Find the median of the data:

1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 3, 4, 1, 2, 1.

Solution: Sorting the numbers:

1, 1, 1, 1, 1, 2, 2, 2, 2, 3, 3, 3, 4, 4, 4

There are 15 numbers. So the position of median

$$= \frac{15+1}{2} = \frac{16}{2} = 8$$

Looking at the ordered data the 8th value is 2

$\therefore$ The median of the given data is 2. (Ans)

8. What is the meaning of mean, median and mode?

Answer: For any given data,

i) **Mean** is the average of given data values and this can be calculated by dividing the sum of all data values by number of data values.

ii) **Median** is the middlemost value of the data set when data values are arranged either in ascending or descending order.

iii) **Mode** is the most frequently occurred data value.

9. The mean, median and mode of the data: 0, 5, -1, 5, 2, 1, -1, 4, 3, 0, 2, 3, 3 are-

(a) 2, 3, 3

(b) 2, 2, 3

(c) 3, 3, 3

(d) none of these

Explanation:

$$\text{Mean} = \frac{0+5-1+5+2+1-1+4+3+0+2+3+3}{13} = \frac{26}{13} = 2$$

Now arranging the data in an ascending order

-1, -1, 0, 0, 1, 2, 2, 3, 3, 3, 4, 5, 5

Since number of values in the array is $n = 13$

which is an odd number. Therefore the median is

$$\frac{n+1}{2} = \frac{13+1}{2} = 7^{\text{th}} \text{ value of the series.}$$

Hence, the required median is 2

$\therefore$ Mode = The value corresponding to highest frequency = 3

$\therefore$ Mean = 2, Median = 2, Mode = 3

Answer: (b)

10. The means of the following data is-

Age in year (x)	12	13	14	15
No. of boys (f)	2	3	2	1

- (a) 12.5 years
(b) 13 years
(c) 13.25 years
(d) none of these

Solution:

$$\text{Mean} = \frac{(12 \times 2) + (13 \times 3) + (14 \times 2) + (15 \times 1)}{2 + 3 + 2 + 1}$$

$$= \frac{106}{8} = 13.25$$

Answer: (c)

11. If $y = 2x - 5$ and $\bar{x} = 7$, find the value of $\bar{y}$.

Solution:

Given, $y = 2x - 5$
 $\Rightarrow \bar{y} = 2\bar{x} - 5$
 $\Rightarrow \bar{y} = 2(7) - 5$ [$\because \bar{x} = 7$]
 $\Rightarrow \bar{y} = 14 - 5 = 9$ (Ans)

12. Find the variance of 2, 4, 6, 8, 10

Solution:

Given data: 2, 4, 6, 8, 10
 $\text{Mean } \bar{x} = \frac{2+4+6+8+10}{5} = \frac{30}{5} = 6$
 $\therefore \text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$
 $= \frac{(2-6)^2 + (4-6)^2 + (6-6)^2 + (8-6)^2 + (10-6)^2}{5}$
 $= \frac{16+4+0+4+16}{5} = \frac{40}{5} = 8$ (Ans)

13. Find the variance of: 5, 6, 7, 8, 10

Solution:

Given data: 5, 6, 7, 8, 10
 $\text{Mean } \bar{x} = \frac{5+6+7+8+10}{5} = \frac{36}{5} = 7.2$
 $\therefore \text{Variance} = \frac{\sum (x - \bar{x})^2}{n}$
 $= \frac{(5-7.2)^2 + (6-7.2)^2 + (7-7.2)^2 + (8-7.2)^2 + (10-7.2)^2}{5}$
 $= \frac{4.84+1.44+0.04+0.64+7.84}{5} = \frac{14.8}{5} = 2.96$ (Ans)

14. If $\text{var}(x) = 5$ and $y = 5x + 6$ then $\text{var}(y) = ?$

- (a) 125
(b) 150
(c) 6
(d) none

Explanation:

Given, $y = 5x + 6$
 $\Rightarrow \text{var}(y) = 5^2 \times \text{var}(x)$
 $= 12 \times 5 = 125$ (Ans)

15. What is the relation between mean, median and mode?

Answer:

Mode = 3(Median) - 2(Mean)
 or, Mean - Mode = 3 (Mean - Median)

16. If $\text{var}(x) = 3$ and $y = 2x + 3$ then $\text{var}(y) = ?$

- (a) 6
(b) 12
(c) 15
(d) 9

Solution: Given $\text{var}(x) = 3$

$y = 2x + 3$
 $\Rightarrow \text{var}(y) = 2^2 \times \text{var}(x)$
 $= 4 \times 3 = 12$

Answer: (b)

17. If $\text{var}(x) = 3$, $\text{var}(y) = 27$ and $y = ax + b$ then the value of $a = ?$

- (a) 3
(b) -3
(c) ± 3
(d) none of these

Solution:

Given, $\text{var}(x) = 3$, $\text{var}(y) = 27$
 and $y = ax + b$
 $\Rightarrow \text{var}(y) = a^2 \text{var}(x)$
 $\Rightarrow 27 = a^2 \times 3$
 $\Rightarrow 3a^2 = 27$
 $\Rightarrow a^2 = 9$
 $\Rightarrow a = \pm \sqrt{9} = \pm 3$
 Answer: (c)

18. Find the arithmetic mean of first five prime numbers.

Answer:

The first five prime numbers are 2, 3, 5, 7, 11

$\therefore$ The mean of these numbers
 $= \frac{2+3+5+7+11}{5} = \frac{28}{5} = 5.6$ (Ans)

19. Find the standard deviation of the following numbers: 1, 2, 3, 4, 5, 6, 7, 8, 9.

Solution:

Given data: 1, 2, 3, 4, 5, 6, 7, 8, 9

$\text{Mean } \bar{x} = \frac{1+2+3+\dots+9}{9} = \frac{9(9+1)}{2 \times 9} = 5$

Frequency $N = 9$

x	$d_i = x - \bar{x}$	d_i^2
1	-4	16
2	-3	9
3	-2	4
4	-1	1
5	5	0
6	1	1
7	2	4
8	3	9
9	4	16
		$\sum d_i^2 = 60$

$\therefore$ Standard deviation = $\sqrt{\frac{\sum d_i^2}{N}}$
 $= \sqrt{\frac{60}{9}} = 2.58$ (Ans)

20. Find the median of the distribution given below:

x	1-10	11-20	21-30	31-40	41-50	51-60
$f(x)$	3	10	9	13	7	15

Solution:

Class interval	Class boundary	Frequency	Cumulative frequency
1-10	0.5-10.5	3	3
11-20	10.5-20.5	10	13
21-30	20.5-30.5	9	22
31-40	30.5-40.5	13	35
41-50	40.5-50.5	7	42
51-60	50.5-60.5	15	57
		N=57	

Median = the value corresponding to cumulative frequency $\frac{N}{2} = \frac{57}{2} = 28.5$

From the above table 28.5 lies in 30.5-40.5

Lower limit of median class $L = 30.5$

Cumulative frequency of the class preceding the median class $F = 22$

Simple frequency of median class $f = 13$

Class width of median class $i = 10$

$$\begin{aligned}\therefore \text{Median} &= L + \frac{\frac{N}{2} - F}{f} \times i \\ &= 30.5 + \frac{28.5 - 22}{13} \times 10 \\ &= 35.5 \quad (\text{Ans})\end{aligned}$$

21. Find the median of the distribution given below:

x	15-19	20-24	25-29	30-34	35-39	40-44	45-49
$f(x)$	4	8	18	20	25	16	9

Solution:

Class interval	Class boundary	Frequency	Cumulative frequency
15-19	14.5-19.5	4	4
20-24	19.5-24.5	8	12
25-29	24.5-29.5	18	30
30-34	29.5-34.5	20	50
35-39	34.5-39.5	25	75
40-44	39.5-44.5	16	91
45-49	44.5-49.5	9	100
		N=100	

Median = the value corresponding to cumulative frequency $\frac{N}{2} = \frac{100}{2} = 50$

From the above table 50 lies in 29.5-34.5

Lower limit of median class $L = 29.5$

Cumulative frequency of the class preceding the median class $F = 30$

Simple frequency of median class $f = 20$

Class width of median class $i = 5$

$$\begin{aligned}\therefore \text{Median} &= L + \frac{\frac{N}{2} - F}{f} \times i \\ &= 29.5 + \frac{50 - 30}{20} \times 5 \\ &= 34.5 \quad (\text{Ans})\end{aligned}$$

22. Calculate mean and standard deviation from the following data:

x_i	4	8	11	17	20	24	32
f_i	3	5	9	5	4	3	1

Solution: Let assumed mean $a = 17$

x_i	f_i	d_i	d_i^2	$f_i d_i$	$f_i d_i^2$
4	3	-13	169	-39	507
8	5	-9	81	-45	405
11	9	-6	36	-54	324
17	5	0	0	0	0
20	4	3	9	12	36
24	3	7	49	21	147
32	1	15	225	15	225

Now, $N = \sum f_i = 30$

$\sum f_i d_i = -90$, $\sum f_i d_i^2 = 1644$

The mean $\bar{x} = a + \frac{\sum f_i d_i}{N} = 17 - \frac{90}{30} = 14 \quad (\text{Ans})$

$$\begin{aligned}\text{Standard Deviation} &= \sqrt{\frac{\sum f_i d_i^2}{N} - \left(\frac{\sum f_i d_i}{N}\right)^2} \\ &= \sqrt{\frac{1644}{30} - \left(\frac{-90}{30}\right)^2} \\ &= \sqrt{54.8 - 9} \\ &= \sqrt{45.8} = 6.76 \quad (\text{Ans})\end{aligned}$$

23. Calculate the Mean and S.D. from the following frequency distribution:

Class	20-29	30-39	40-49	50-59	60-69	70-79	80-89	90-99
Frq.	5	12	15	20	18	10	6	4

Solution:

Let assume mean $a = 54.5$, Class width $h = 10$

class	Mid value (x)	f	$d = \frac{(x - a)}{h}$	fd	fd^2
20-29	24.5	5	-3	-15	45
30-39	34.5	12	-2	-24	48
40-49	44.5	15	-1	-15	15
50-59	54.5	20	0	0	0
60-69	64.5	18	1	18	18
70-79	74.5	10	2	20	40
80-89	84.5	6	3	18	54
90-99	94.5	4	4	16	64

Now, $\sum f = 90$, $\sum fd = 18$, $\sum fd^2 = 284$

$$\therefore \bar{d} = \frac{\sum fd}{\sum f} = \frac{18}{90} = 0.2$$

$$\begin{aligned}\text{Mean, } \bar{x} &= a + h\bar{d} \\ &= 54.5 + 10 \times 0.2 = 56.5 \quad (\text{Ans})\end{aligned}$$

$$\begin{aligned}\text{Standard deviation of } x &= h \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2} \\ &= 10 \times \sqrt{\frac{284}{90} - (0.2)^2} \\ &= 10 \times 1.765 \\ &= 17.65 \quad (\text{Ans})\end{aligned}$$

24. A distribution consists of three components with frequencies 200, 250, 300 having means 25, 10 and 15; standard deviation 3, 4 and 5 respectively. Prove that the mean of combined distribution is 16 and standard deviation is 7.2 approximately.

Solution: Given data

Frequencies: $n_1 = 200, n_2 = 250, n_3 = 300$

Means: $\bar{x}_1 = 25, \bar{x}_2 = 10, \bar{x}_3 = 15$

Standard deviations: $s_1 = 3, s_2 = 4, s_3 = 5$

∴ Combined Mean

$$\bar{x} = \frac{n_1\bar{x}_1 + n_2\bar{x}_2 + n_3\bar{x}_3}{n_1 + n_2 + n_3} = \frac{200 \times 25 + 250 \times 10 + 300 \times 15}{200 + 250 + 300} = 16 \text{ (Proved)}$$

Now,

$$d_1 = \bar{x}_1 - \bar{x} = 25 - 16 = 9$$

$$d_2 = \bar{x}_2 - \bar{x} = 10 - 16 = -6$$

$$d_3 = \bar{x}_3 - \bar{x} = 15 - 16 = -1$$

∴ Combined standard deviation

$$s = \sqrt{\frac{n_1(s_1^2 + d_1^2) + n_2(s_2^2 + d_2^2) + n_3(s_3^2 + d_3^2)}{n_1 + n_2 + n_3}} = \sqrt{\frac{200(3^2 + 9^2) + 250(4^2 + (-6)^2) + 300(5^2 + (-1)^2)}{200 + 250 + 300}} = \sqrt{\frac{1800 + 13000 + 7800}{200 + 250 + 300}} = 7.19 \approx 7.2 \text{ (Proved)}$$

25. The table below gives the weights of 50 school boys nearest to kg. Calculate the Mean and S.D. of the distribution:

Wt. in kg (x)	40-42	43-45	46-48	49-51	52-54	55-57
f	3	7	10	14	9	7

Solution:

Let assume mean $a = 47$, Class width $h = 3$

class	Mid value (x)	f	$d = \frac{(x - a)}{h}$	fd	fd^2
40-42	41	3	-2	-6	12
43-45	44	7	-1	-7	7
46-48	47	10	0	0	0
49-51	50	14	1	14	14
52-54	53	9	2	18	36
55-57	56	7	3	21	63

Now, $\sum f = 50, \sum fd = 40, \sum fd^2 = 132$

$$\therefore \bar{d} = \frac{\sum fd}{\sum f} = \frac{40}{50} = 0.8$$

Mean, $\bar{x} = a + h\bar{d}$

$$= 47 + 3 \times 0.8 = 49.4 \text{ (Ans)}$$

$$\begin{aligned} \text{Standard deviation of } x &= h \sqrt{\frac{\sum fd^2}{\sum f} - \left(\frac{\sum fd}{\sum f}\right)^2} \\ &= 3 \times \sqrt{\frac{132}{50} - (0.8)^2} \\ &= 3 \times \sqrt{2} \\ &= 4.24 \text{ (Ans)} \end{aligned}$$

26. Find the mean of the first 10 natural numbers.

Solution:

The first 10 natural numbers are

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

$$\begin{aligned} \text{Mean} &= \frac{1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 10}{10} \\ &= \frac{10(10+1)}{2 \times 10} = \frac{11}{2} = 5.5 \text{ (Ans)} \end{aligned}$$

27. Find the value of y from the following observations if these are already arranged in ascending order.

The Median is 63.

20, 24, 42, y , $y + 2$, 73, 75, 80, 99

Solution:

As the number of observations made is odd, so the median will be the middle term, i.e. $y + 2$.

Therefore, $y + 2 = 63$

$$\Rightarrow y = 63 - 2 = 61 \text{ (Ans)}$$

28. Find the mode of the following items.

0, 5, 5, 1, 6, 4, 3, 0, 2, 5, 5, 6

Solution:

On arranging the items in ascending order, we get:

0, 0, 1, 2, 3, 4, 5, 5, 5, 5, 6, 6

As we can see 5 occurs the maximum number of times.

Therefore, the mode of the given items = 5 (Ans)

29. If the mean of first n natural numbers is 15, then find n .

Solution: We know that the sum of first n natural

$$\text{numbers} = \frac{n(n+1)}{2}$$

Mean of the first n natural numbers

$$= \frac{\text{Sum of first } n \text{ natural numbers}}{n}$$

$$= \frac{\frac{n(n+1)}{2}}{n} = \frac{n+1}{2}$$

$$\text{According to the given, } \frac{n+1}{2} = 15$$

$$\Rightarrow n + 1 = 30$$

$$\Rightarrow n = 29 \text{ (Ans)}$$

30. For a moderately skewed distribution, the mean and median are respectively 26.8 and 27.9. What is the mode of the distribution?

Solution:

Given, Mean = 26.8, Median = 27.9

Using the relationship between mean, median and mode,

$$\begin{aligned} \text{Mode} &= 3 \text{ Median} - 2 \text{ Mean} \\ &= 3 \times 27.9 - 2 \times 26.8 \\ &= 83.7 - 53.6 \\ &= 30.1 \end{aligned}$$

∴ The mode of the distribution is 30.1. (Ans)

31. Calculate the arithmetic mean, range, median and mode of the given data: 2,4,7,4,9,5,7,3,6,7

Solution:

$$\text{Arithmetic mean} = \frac{2+4+7+4+9+5+7+3+6+7}{10} = \frac{54}{10} = 5.4 \quad (\text{Ans})$$

Median:

Arranging the data in ascending order:

2,3,4,4,5,6,7,7,9

Here the middle terms are 5 and 6

The average of 5 and 6

$$= \frac{5+6}{2} = \frac{11}{2} = 5.5 \quad \therefore \text{Median} = 5.5 \quad (\text{Ans})$$

Mode: The most repeating values is 7.

$\therefore$ Mode is 7 (Ans)

Range

= Highest observation – Lowest observation

$$= 9 - 2 = 7 \quad (\text{Ans})$$

32. If mode is 63 and median is 77, calculate arithmetic mean.

Solution:

Given: Mode = 63, Median = 77

As per the empirical relationship between mean,

median and mode: Mode = 3(Median) – 2(Mean)

Substituting the given values in the formula.

$$63 = 3(77) - 2(\text{Mean})$$

$$\text{or, } 2(\text{Mean}) = 231 - 63$$

$$\text{or, Mean} = \frac{168}{2} = 84$$

Hence, mean is 84. (Ans)

33. Calculate mode when arithmetic mean is 146 and median is 130.

Solution: Given, Mean = 146, Median = 130

As per the empirical relationship between mean, median and mode: Mode = 3(Median) – 2(Mean)
Substituting the given values in the formula.

$$\text{Mode} = 3(130) - 2(146)$$

$$\text{or, Mode} = 390 - 292$$

Hence, Mode is 98. (Ans)

34. What does Standard Deviation tell you?

Answer: Standard deviation tells us how far is the mean from each observation in the given data set. In other words, it shows the typical deviation from the mean.

35. What is the standard deviation example?

Answer: Consider the data set: 2, 1, 3, 2, 4. The mean and the sum of squares of deviations of the observations from the mean will be 2.4 and 5.2, respectively. Thus, the standard deviation will be

$$\sqrt{\frac{5.2}{5}} = \sqrt{1.04} = 1.0198$$

{Prepared by [NatiTute](#) YouTube Channel}

1. A die is thrown once, then the probability of obtaining a six is-

(a) $\frac{1}{3}$
 (b) $\frac{1}{2}$
 (c) $\frac{1}{6}$
 (d) none of these

Solution: When a dice is thrown, number of possible outcomes = 1, 2, 3, 4, 5, 6

The total outcome = 6

Here, 6 occurs one time

No. of favorable outcomes = 1

∴ Required probability = $\frac{1}{6}$

Answer: (c)

2. The probability of getting at least any one of the following events, point 'six' or 'one' on the top in rolling of an unbiased dice once is-

(a) $\frac{1}{6}$
 (b) $\frac{1}{9}$
 (c) $\frac{1}{3}$
 (d) none of these

Solution: When a dice is thrown, number of possible outcomes = 1, 2, 3, 4, 5, 6

The total outcome = 6

No. of favorable outcomes {1, 6} = 2

∴ Required probability = $\frac{2}{6} = \frac{1}{3}$

Answer: (c)

3. The probability of getting an even number from the numbers up to 50 (including 50) is ____.

Solution: The numbers from 1 to 50 form our sample space. So, the total number of possible outcomes is: $n(S) = 50$

The even numbers in this range are:

2, 4, 6, 50

So, the number of favorable outcomes is:

$n(E) = 25$

∴ Probability $P(E) = \frac{n(E)}{n(S)} = \frac{25}{50} = \frac{1}{2}$ (Ans)

4. Find the chance of throwing an even number with an ordinary 6 faced die.

Solution:

The outcomes when a die is thrown are: 1, 2, 3, 4, 5, 6

So, the total outcome = 6

Even numbers are 2, 4, 6

Therefore, no. of favorable outcomes = 3

The probability of getting a even number

= $\frac{3}{6} = \frac{1}{2}$ (Ans)

5. If $P(A) = \frac{7}{13}$, $P(B) = \frac{9}{13}$ and $P(A \cap B) = \frac{4}{13}$, evaluate $P(A|B)$.

Solution:

$P(A|B) = P(A \cap B) / P(B) = (4/13) / (9/13) = 4/9$.

6. One unbiased die is rolled, the probability of an even point or one point will appear top of the die is-

(a) $\frac{1}{2}$
 (b) $\frac{1}{3}$
 (c) $\frac{2}{3}$
 (d) none of these

Solution:

The no. of total outcomes = {1, 2, 3, 4, 5, 6} = 6

No. of favorable outcomes = {1, 2, 4, 6} = 4

∴ The required probability = $\frac{4}{6} = \frac{2}{3}$

Answer: (c)

7. Two unbiased coins are tossed. The probability of obtaining at least one tail is-

(a) $\frac{4}{3}$
 (b) $\frac{3}{4}$
 (c) $\frac{1}{3}$
 (d) none

Solution: If two unbiased coins are tossed simultaneously, then the total outcomes may be HH, HT, TH, TT.

No. of total outcomes = 4

Favorable outcome for at least one tail are are TH, HT, TT. No. of favorable outcomes = 3

$P(\text{at least one tail}) = \frac{3}{4}$

Answer: (b)

8. Two coins are tossed. The probability of getting at least one head is- (a) $\frac{3}{4}$ (b) $\frac{1}{2}$ (c) $\frac{1}{4}$ (d) none of these.

Solution:

No. of total outcome {HH, TT, TH, HT} = 4

No. of favorable outcome {HH, TH, HT} = 3

The probability of getting at least one head

= $\frac{\text{No. of favorable outcome}}{\text{No. of total outcome}} = \frac{3}{4}$

Answer: (a)

9. Three coins are tossed at random, then the probability that there will be at least one head is-

(a) $\frac{3}{8}$
 (b) $\frac{7}{8}$
 (c) $\frac{2}{9}$
 (d) none

Solution: The possible outcomes when three coins are tossed together are

{HHH, TTT, THT, TTH, HHT, HTH, THH, HTT}

Therefore, the number of possible outcomes when two coins are tossed is 8.

Now, the possible outcomes of getting at least one head are {HHH, THT, TTH, HHT, HTH, THH, HTT}, which means the number of favorable outcome is 7.

∴ Probability of getting at least one head is $P = \frac{7}{8}$

Answer: (b)

10. A coin is tossed two times in succession, write the sample space.

Solution:

A coin has two faces: head (H) and tail (T).

When a coin is tossed two times, the total number of possible outcomes is $2^2 = 4$

Thus, when a coin is tossed two times, the sample space is given by: $S = \{HH, HT, TH, TT\}$ (Ans)

11. A coin is tossed three times in succession, write the sample space.

Answer:

A coin has two faces: head (H) and tail (T).

When a coin is tossed three times, the total number of possible outcomes is $2^3 = 8$

Thus, when a coin is tossed three times, the sample space is given by:

$S = \{HHH, HHT, HTH, HTT, THH, THT, TTH, TTT\}$

12. The probability of any event A satisfies-

- (a) $-1 \leq P(A) \leq 1$
- (b) $-1 \leq P(A) \leq 0$
- (c) $0 \leq P(A) \leq 1$
- (d) none of these

Explanation: If any event is not possible then the probability for that event will be 0. If the event is certain then the probability will be 1. So, the range of probability for an event is $0 \leq P(A) \leq 1$.

Answer: (c)

13. One card is drawn from a standard pack of 52. The probability which is either king or queen is-

- (a) $1/13$
- (b) $2/13$
- (c) $3/13$
- (d) none of these

Solution: Since, Total cards = 52

No. of kings = 4

No. of queens = 4

No. of favorable cards that are king or queen = 8

$\therefore$ Required probability $\frac{8}{52} = \frac{2}{13}$

Answer: (b)

14. An urn contains 6 white balls and 12 black balls. A ball is drawn at random. What is the probability of being it to be a white?

- (a) $2/3$
- (b) $1/2$
- (c) $3/4$
- (d) $1/3$

Solution: Number of white balls in urn = 6

Number of black balls in urn = 12

Total number of balls = $6 + 12 = 18$

$\therefore$ Total outcomes = 18

Favorable outcomes = 6

Probability of drawing white ball

$$= \frac{\text{Favourable outcomes}}{\text{Total outcomes}} = \frac{6}{18} = \frac{1}{3}$$

Answer: (d)

15. Two events A and B are mutually exclusive if-

- (a) $P(A \cap B) = P(A)P(B)$
- (b) $P(A \cap B) = 1$
- (c) $P(A \cap B) = 0$
- (d) none of these

Explanation: Two exclusive events signify that they have nothing in common between them.

So, $P(A \cap B) = 0$

Answer: (c)

16. A is tossed. The events {H}, {T} are-

- (a) mutually exclusive
- (b) independent
- (c) not exhaustive
- (d) none of these

Explanation: Since $\{H\} \cap \{T\} = \emptyset$

$\therefore$ The events are mutually exclusive.

Answer: (a)

17. Let A and B be two events and $P(A) = 0.2$, $P(B) = 0.3$ and $P(A \cap B) = 0.1$ then the value of $P(A \cup B)$ is-

- (a) 0.4
- (b) 0.6
- (c) 0.06
- (d) none of these

Solution:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$= 0.2 + 0.3 - 0.1$$

$$= 0.4$$

Answer: (a)

18. Let 'A' be any event and $P(A) = 0.4$ then find $P(A^c)$.

Answer:

We know, for any events

$$P(A) + P(A^c) = 1$$

$$\Rightarrow 0.4 + P(A^c) = 1$$

$$\Rightarrow P(A^c) = 1 - 0.4 = 0.6 \quad (\text{Ans})$$

19. If $P(A) = 0.4$, $P(B) = 0.35$ find $P(\bar{A}) + P(\bar{B}) - 1$

Solution:

Given, $P(A) = 0.4$, $P(B) = 0.35$

$$\therefore P(\bar{A}) = 1 - 0.4 = 0.6$$

$$P(\bar{B}) = 1 - 0.35 = 0.65$$

$$\therefore P(\bar{A}) + P(\bar{B}) - 1$$

$$= 0.6 + 0.65 - 1 = 0.25 \quad (\text{Ans})$$

20. If $P(A) = p$, $P(B^c) = 0.7$, $P(A \cup B) = 0.8$ and $P(A \cap B) = 1 - p$, then find the value of p.

Solution:

Given, $P(A) = p$

$$P(B^c) = 0.7, \Rightarrow P(B) = 1 - 0.7 = 0.3$$

We know, $P(A \cup B) = P(A) + P(B) - P(A \cap B)$

$$\Rightarrow 0.8 = p + 0.3 - 1 + p$$

$$\Rightarrow 0.8 = 2p - 0.7$$

$$\Rightarrow 2p = 0.8 + 0.7 = 1.5$$

$$\Rightarrow p = \frac{1.5}{2} = 0.75 \quad (\text{Ans})$$

21. Find the probability of obtaining at least one '6' is game of throwing of two dice.

Solution:

Total number of possible cases $6^2 = 36$

Favourable cases of getting at least one six

$$= \{(1, 6), (2, 6), (3, 6), (4, 6), (5, 6), (6, 6), (6, 1), (6, 2), (6, 3), (6, 4), (6, 5)\}$$

Total number of favourable cases = 11

∴ The probability of getting at least one '6' in two dice is $\frac{11}{36}$ (Ans)

22. Find the probability of getting an even number from the numbers up to 50 (including 50).

Solution: Numbers 1 to 50

Choosing an Even number {2, 4, 6, ..., 50}

So, 25 even numbers out of 50 number

Probability of choosing an even number is

$$\frac{25}{50} = \frac{1}{2} \text{ (Ans)}$$

23. A number is chosen from 1 to 50, what is the probability of getting a prime number?

Solution:

Between 1 and 50, there are 15 prime numbers.

So the probability of getting a prime number is

$$\frac{15}{50} = \frac{3}{10} \text{ (Ans)}$$

24. A coin and a die are thrown simultaneously. Find the probability of occurrence of

(a) head and even face

(b) tail and multiple of three

Solution:

Let S be the sample space.

∴ $S = \{(H, 1), (H, 2), (H, 3), (H, 4), (H, 5), (H, 6), (T, 1), (T, 2), (T, 3), (T, 4), (T, 5), (T, 6)\}$

$n(S) = 12$

(a) Let A be the event of getting head and even face ∴ $A = \{(H, 2), (H, 4), (H, 6)\}$

$$\Rightarrow n(A) = 3$$

$$\therefore P(A) = \frac{n(A)}{n(S)} = \frac{3}{12} = \frac{1}{4} \text{ (Ans)}$$

(b) Let B be the event of getting tail and multiple of three

$$\therefore B = \{(T, 3), (T, 6)\}$$

$$\Rightarrow n(B) = 2$$

$$\therefore P(B) = \frac{n(B)}{n(S)} = \frac{2}{12} = \frac{1}{6} \text{ (Ans)}$$

25. Two dice are rolled. What is the probability that the sum total of the points on the dice be 9?

Solution:

There is 36 possible outcomes

(6 for the first die and 6 for the second, thus $6 \times 6 = 36$)

Out of the 36 outcomes, let's list the favorable outcomes: (6, 3), (3, 6), (4, 5), (5, 4)

There is 4 favorable outcomes out of 36.

Thus the probability is $\frac{4}{36} = \frac{1}{9}$ (Ans)

26. Two dice are rolled. What is the probability that the sum total of the points on the dice is 6?

Solution:

There is 36 possible outcomes

(6 for the first die and 6 for the second, thus $6 \times 6 = 36$)

Out of the 36 outcomes, let's list the favorable outcomes: (1, 5) (2, 4) (3, 3) (4, 2) (5, 1)

There is 5 favorable outcomes out of 36.

Thus the probability is $\frac{5}{36}$ (Ans)

27. Define complementary event.

Answer:

Complementary Events: If P(E) is the probability of an event and P(E^c) is the probability of an event not happening, then: $P(E) + P(E^c) = 1$

$$\Rightarrow P(E^c) = 1 - P(E)$$

28. What is the probability that the sum of outcomes is 13, when three dice are rolled?

Solution:

When three dice are rolled.

Total number of outcomes $= 6^3 = 216$

Sum of 13 can be achieved in the following ways:

From the digits 6, 4, 3. So, there are $3!$ ways = 6

From the digits 6, 2, 5. So, there are $3!$ ways = 6

From the digits 5, 4, 4. So, there are $\frac{3!}{2!}$ ways = 3

From the digits 6, 6, 1. So, there are $\frac{3!}{2!}$ ways = 3

From the digits 3, 5, 5. So, there are $\frac{3!}{2!}$ ways = 3

So, total numbers whose sum is 13

$$= 6 + 6 + 3 + 3 + 3 = 21$$

Therefore, the probability of getting sum as 21 on rolling three dice $= \frac{21}{216}$ (Ans)

29. A bag contains 6 red, 4 white and 8 blue balls. If three balls are drawn at random, find the probability that one is red, one is white and one is blue.

Solution:

The bag contains, 6 red, 4 white, 8 blue balls

As there balls are drawn at random.

$$\therefore n(S) = {}^{18}C_3$$

Let E the event that one red ball, one white ball and one blue ball was drawn.

$$\therefore n(E) = {}^6C_1 \times {}^4C_1 \times {}^8C_1$$

$$\therefore P(E) = \frac{{}^6C_1 \times {}^4C_1 \times {}^8C_1}{{}^{18}C_3} = \frac{6 \times 4 \times 8}{816} = \frac{4}{17} \text{ (Ans)}$$

30. There are two identical urns containing 3 white and 5 red balls; 4 white and 9 red balls. An urn is chosen at random and a ball is drawn from it. Find the probability that the ball is white.

Solution:

By the question, there are two urn. Say A and B.

A contains 3 white and 5 red balls. i.e. total 8 balls

B contains 4 white and 9 red balls. i.e. total 13 balls

Now, probability of getting a specific urn.

$$P(A) = P(B) = \frac{1}{2}$$

Let, W be the event that a white ball is drawn.

$$\therefore P\left(\frac{W}{A}\right) = \frac{3}{8} \text{ and } P\left(\frac{W}{B}\right) = \frac{4}{13}$$

$$\begin{aligned} \therefore P(W) &= P\left(\frac{W}{A}\right)P(A) + P\left(\frac{W}{B}\right)P(B) \\ &= \frac{3}{8} \times \frac{1}{2} + \frac{4}{13} \times \frac{1}{2} \\ &= \frac{1}{2} \left(\frac{39+32}{104} \right) \\ &= \frac{71}{208} \end{aligned}$$

$\therefore$ The probability of getting white ball is $\frac{71}{208}$ (Ans)

31. There are two identical urns containing 4 white and 3 red balls; 3 white and 7 red balls. An urn is chosen at random and a ball is drawn from it. Find the probability that the ball is white. What is the probability that it is from the first urn if the ball drawn is white?

Solution:

By the question, there are two urn. Say A and B.

A contains 4 white and 3 red balls. i.e. total 7 balls

B contains 3 white and 7 red balls. i.e. total 10 balls

Now, probability of getting a specific urn.

$$P(A) = P(B) = \frac{1}{2}$$

Let, W be the event that a white ball is drawn.

$$\therefore P\left(\frac{W}{A}\right) = \frac{4}{7} \text{ and } P\left(\frac{W}{B}\right) = \frac{3}{10}$$

$$\begin{aligned} \therefore P(W) &= P\left(\frac{W}{A}\right)P(A) + P\left(\frac{W}{B}\right)P(B) \\ &= \frac{4}{7} \times \frac{1}{2} + \frac{3}{10} \times \frac{1}{2} \\ &= \frac{1}{2} \left(\frac{40+21}{70} \right) \\ &= \frac{61}{140} \end{aligned}$$

$\therefore$ The probability of getting white ball is $\frac{61}{140}$ (Ans)

Compute the probability that the ball is from the first urn given that it is white.

$$\begin{aligned} P(A|W) &= \frac{P\left(\frac{W}{A}\right) \times P(A)}{P(W)} = \frac{\frac{4}{7} \times \frac{1}{2}}{\frac{61}{140}} \\ &= \frac{4 \times 140}{7 \times 2 \times 61} = \frac{40}{61} \text{ (Ans)} \end{aligned}$$

32. Define sample space.

Answer: The set of outcomes in a random experiment is known as sample space.

For example- say, rolling a die

Sample space = {1,2,3,4,5,6}

33. An urn contains 2 white and 4 red balls. Two balls are drawn randomly one after another without replacement. What is the probability that the drawn balls are red?

Solution:

Total number of balls = 2 white + 4 red = 6 balls

Since the draws are without replacement, the probability of each event depends on the previous draw.

Probability of drawing a red ball on the first draw:

$$= \frac{4}{6} = \frac{2}{3}$$

Since one red ball has already been removed, we now have 3 red balls left out of 5 remaining balls.

Probability of drawing a red ball on the second draw (given the first was red):

$$= \frac{3}{5}$$

Using the multiplication rule for dependent events:

The probability that both drawn balls are red is-

$$= \frac{2}{3} \times \frac{3}{5} = \frac{2}{5} \text{ (Ans)}$$

34. Five men in a company of twenty are graduates. If three men are picked out of 20 at random, what is the probability that at least one is graduates?

Solution:

Total number of people in the company = 20

Number of graduates = 5

Number of non-graduate = 15

$\therefore$ 3 men can be picked up in ${}^{20}C_3$ ways.

If 3 men picked up, no one of them is a graduate, can be picked up in ${}^{15}C_3$ ways.

$\therefore$ Probability that no one of them is a graduate is

$$= \frac{{}^{15}C_3}{{}^{20}C_3}$$

The probability that at least one is a graduate

= 1 - probability that no one of them is a graduate

$$= 1 - \frac{{}^{15}C_3}{{}^{20}C_3} = 1 - \frac{455}{1140} = \frac{685}{1140} = \frac{137}{228} = 0.6 \text{ (Ans)}$$

35. A batch of 50 fuses is known to have 10 defective fuses. If 10 fuses are selected at random, find the probability that all will be good.

Solution:

Given, total number of fuses = 50

Number of defective fuse = 10

$\therefore$ Number of effective fuse = 40

10 fuses are selected at random,

$\therefore$ Total number of outcomes = ${}^{50}C_{10}$

Now, the number of cases favorable to the event 'all the selected 10 fuses will be good' = ${}^{40}C_{10}$

$\therefore$ The probability that all selected 10 fuse are good

$$= \frac{{}^{40}C_{10}}{{}^{50}C_{10}} = 0.0825 \text{ (Ans)}$$

36. A problem in mechanics is given to three students A, B and C whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. What is the probability that the problem will be solved?

Solution:

The probability that A fails to solve the problem $= 1 - \frac{1}{2} = \frac{1}{2}$, similarly probability of failure of B and C are $1 - \frac{1}{3} = \frac{2}{3}$ and $1 - \frac{1}{4} = \frac{3}{4}$ respectively.

The events are independent. So probability that all fail to solve $= \frac{1}{2} \times \frac{2}{3} \times \frac{3}{4} = \frac{1}{4}$

Hence, the probability that the problem will be solved $= 1 - \frac{1}{4} = \frac{3}{4}$ (Ans)

37. How many outcomes are there for tossing 3 coins simultaneously?

- (a) 2
(b) 4
(c) 6
(d) 8

Solution:

There are 2 outcomes for each coin: Head or tail. Therefore, for tossing three coins, there are $2^3 = 8$ outcomes.

Answer: (d)

38. Two cards are drawn in succession from a pack of 52 cards. Find the chance that the first is a king and second a queen, if the first card is –

- i) replaced, ii) not replaced.

Solution:

In pack of cards, there are 4 kings and 4 queens.

When first card is drawn the chance of getting king $= \frac{4}{52} = \frac{1}{13}$

- i) When first card is replaced by 2nd card there also 52 cards. So, in this case chance of getting queen

$$\frac{4}{52} = \frac{1}{13}$$

So, the required chance of getting a king and a queen is $= \frac{1}{13} \times \frac{1}{13} = \frac{1}{169}$ (Ans)

- ii) When first card is not replaced by 2nd card there are 51 cards. So, chances of getting queen in 2nd draw $= \frac{4}{51}$

∴ The required chance of getting a king and a queen is $= \frac{1}{13} \times \frac{4}{51} = \frac{4}{663}$ (Ans)

39. P can hit a target 4 times in a 5 shots. Q 3 times in 4 shots and R twice in 3 shots. They fire a volley. What is the probability that two shots at least hit the target?

Solution: According to given condition P, Q, and R, with their respective probabilities of hitting a target: $P(A) = \frac{4}{5}$, $P(B) = \frac{3}{4}$, $P(C) = \frac{2}{3}$

They fire a volley, meaning they shoot simultaneously.

$$P(\text{miss by P}) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$P(\text{miss by Q}) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(\text{miss by R}) = 1 - \frac{2}{3} = \frac{1}{3}$$

Thus, the probability that all three miss:

$$P(0 \text{ hits}) = \frac{1}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{1}{60}$$

$$P(P \text{ hits, Q and R miss}) = \frac{4}{5} \times \frac{1}{4} \times \frac{1}{3} = \frac{1}{15}$$

$$P(Q \text{ hits, P and R miss}) = \frac{3}{4} \times \frac{1}{5} \times \frac{1}{3} = \frac{1}{20}$$

$$P(R \text{ hits, P and Q miss}) = \frac{2}{3} \times \frac{1}{5} \times \frac{1}{4} = \frac{1}{30}$$

Summing above three cases,

$$\frac{1}{15} + \frac{1}{20} + \frac{1}{30} = \frac{4+3+2}{60} = \frac{9}{60}$$

$$\therefore P(\text{at least 2 hits}) = 1 - P(0 \text{ hits}) - P(1 \text{ hit})$$

$$= 1 - \frac{1}{60} - \frac{9}{60}$$

$$= \frac{60-1-9}{60}$$

$$= \frac{50}{60} = \frac{5}{6} \text{ (Ans)}$$

40. A can hit a target 4 times out of 5, B 3 times out of 4 and C 2 times out of 3. Each is allowed to shot once. What is the probability that the target is hit?

Solution:

Probability that each of them misses the target:

$$P(A \text{ misses}) = 1 - P(A \text{ hits}) = 1 - \frac{4}{5} = \frac{1}{5}$$

$$P(B \text{ misses}) = 1 - P(B \text{ hits}) = 1 - \frac{3}{4} = \frac{1}{4}$$

$$P(C \text{ misses}) = 1 - P(C \text{ hits}) = 1 - \frac{2}{3} = \frac{1}{3}$$

Now, let's find the probability that all three of them miss the target:

$P(\text{all three miss})$

$$= P(A \text{ misses}) \times P(B \text{ misses}) \times P(C \text{ misses})$$

$$= \left(\frac{1}{5}\right) \times \left(\frac{1}{4}\right) \times \left(\frac{1}{3}\right)$$

$$= \frac{1}{60}$$

Now, we can find the probability that the target is hit: $P(\text{target is hit}) = 1 - P(\text{all three miss})$

$$= 1 - \frac{1}{60}$$

$$= \frac{59}{60} \text{ (Ans)}$$

- 41. A coin is tossed three times. Find the probability that the second throw will be different from the first and third.**

Solution:

As the tossing of a coin, there are 2 possible outcomes and the coin tossed 3 times

$$\therefore \text{No. of total outcomes } n = 2^3 = 8$$

All possible outcomes of three throws are:

$$\{(HHH), (HHT), (HTH), (THH), \\ \{ (HTT), (THT), (TTH), (TTT) \}$$

These are mutually exclusive, exhaustive and equally likely. Out of these 8 events only (HTH) and (THT) i.e. 2 cases are favorable to the required event. So favorable outcomes $m = 2$

$\therefore$ Probability that the second throw will be different from the first and third

$$= \frac{m}{n} = \frac{2}{8} = \frac{1}{4} \quad (\text{Ans})$$

- 42. A coin is tossed three times. Find the probability that the two head and one tail.**

Solution:

As the tossing of a coin, there are 2 possible outcomes and the coin tossed 3 times

$$\therefore \text{No. of total outcomes } n = 2^3 = 8$$

All possible outcomes of three throws are:

$$\{(HHH), (HHT), (HTH), (THH), \\ \{ (HTT), (THT), (TTH), (TTT) \}$$

These are mutually exclusive, exhaustive and equally likely. Out of these 8 events only (HTH), (HHT) and (THH) i.e. 3 cases are favorable to the required event. So favorable outcomes $m = 3$

$\therefore$ Probability that the second throw will be different from the first and third

$$= \frac{m}{n} = \frac{3}{8} \quad (\text{Ans})$$

- 43. Find the probability that getting at least two heads in a toss of unbiased coin 3 times.**

Solution: As the tossing of a coin, there are 2 possible outcomes and the coin tossed 3 times

$$\therefore \text{No. of total outcomes } n = 2^3 = 8$$

All possible outcomes of three throws are:

$$\{(HHH), (HHT), (HTH), (THH), \\ \{ (HTT), (THT), (TTH), (TTT) \}$$

These are mutually exclusive, exhaustive and equally likely. Out of these 8 events only (HTH), (HHT), (THH) and (HHH) i.e. 4 cases are favorable to the required event. So favorable outcomes $m = 4$

$\therefore$ Probability that getting at least two heads

$$= \frac{m}{n} = \frac{4}{8} = \frac{1}{2} \quad (\text{Ans})$$

- 44. Two dice are rolled. What is the probability that the sum total of the points on the dice will be 9?**

Solution: The total number of possible outcomes is
 $= 6^2 = 36$

The favorable outcomes where the sum of the two dice is 9 are { (3, 6), (4, 5), (6, 3), (5, 4) } = 4

$\therefore$ Probability of getting a sum of 9 is-

$$= \frac{4}{36} = \frac{1}{9} \quad (\text{Ans})$$

- 45. What is the chance that a leap year, selected at random will contain 53 Sunday?**

Solution: A leap year has 366 days, which means it consists of 52 full weeks and 2 extra days. The possible extra 2 days in a leap year can be:

{(Mon, Tue), (Tue, Wed), (Wed, Thu), (Thu, Fri), (Fri, Sat), (Sat, Sun), (Sun, Mon)}

So, number of total outcome = 7

Out of these 2 cases {(Sat, Sun), (Sun, Mon)} are favorable i. e. 2 events contain Sundays.

$\therefore$ The probability that a leap year, selected at random will contain 53 Sunday is $= \frac{2}{7}$ (Ans)

- 46. What is the probability of picking an odd number from 1 to 100?**

Solution: Total number = 100

Odd numbers are 1, 3, 5, ..., 99

$\therefore$ Total number of odd number

$$= \frac{99-1}{2} + 1 = 50$$

$$\therefore P(\text{odd number}) = \frac{50}{100} = \frac{1}{2} \quad (\text{Ans})$$

- 47. What is the probability of selecting a prime number from 1 to 30?**

Solution: Total numbers: 30

Prime numbers between 1 and 30:

2, 3, 5, 7, 11, 13, 17, 19, 23, 29

Number of prime numbers = 10

$$\therefore P(\text{prime number}) = \frac{10}{30} = \frac{1}{3} \quad (\text{Ans})$$

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